ELEC3104: Digital Signal Processing

Topic 1: DSP fundamentals

- Answers to Multiple Choice Questions

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Topic 1: Tutorial Questions (SOLO - Level 1)

1. An analogue signal contains frequencies up to 10 kHz. Suppose that we sample this signal at a sampling frequency f_s = 8kHz. Examine what will happen to the frequency f_1 = 5 kHz? (Tick the correct answer)

   a) 5kHz frequency will remain unchanged
   ✓b) 5kHz will alias to 3kHz
   c) 5kHz will alias to 0Hz
   d) None of the above

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f_1 (5 kHz) > $\frac{f_s}{2}$ ($\frac{8\,kHz}{2}$) ∴ Aliasing will occur

New frequency is $f_1' = f_1 - k f_s$ (↑ $k=1$) $= 5000 - 1 \times 8000$
$= -3000$ Hz

i.e 5 kHz will alias to 3 kHz

2. Which of the following is a not periodic signal ? (Tick the correct answer)

   a) x[n] = sin[πn/9] ← Period = 18
   b) x[n] = sin[nπ] ← Period = 2
   ✓c) x[n] = cos[πn²/15] ← non Periodic (because of n^2)
   d) x[n] = sin[πn/5 + π] + cos[πn/10 − π] ← Period = least common multiple of N_1 & N_2 = LCM(10, 20) = 20
      Period N_1 = 10; Period N_2 = 20

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3. Given a real-valued bandpass signal x_a(t) with a bandpass spectrum of X_a(f) nonzero only for $9\ kHz < |f| < 12kHz$, determine the minimum sampling frequency (f_s) for which no aliasing occurs? (Tick the correct answer)

   a) f_s = 18 kHz
   ✓b) f_s = 6 kHz
   c) f_s = 24 kHz
   d) None of the above

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Bandwidth (B) = 12 − 9 kHz = 3 kHz
f_h = 12 kHz which is integer multiple of the Bandwidth.
∴ f_s (minimum) = 2 × B = 2 × 3 kHz = 6 kH

4. A discrete-time signal x(n) = 6.35 cos(nπ/10) is quantised with a resolution Δv=0.2 $volts$. How many bits are required in the Analogue to Digital Converter? (Tick the correct answer)

   ✓a) 6 bits
   b) 8 bits
   c) 12 bits

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$\Delta V = \frac{2 \times 6.35}{2^B - 1}$ (↑ 0.2V)

B - number of bits

$2^B - 1 = 63.5$

∴ B = 6

Topic 1: Tutorial Questions (SOLO - Level 1)

5. Consider the amplitude modulated signal $x(t) = 5\,cos(3\pi t)\,cos(\pi t)$. Determine its spectral frequency components [you may assume that the frequencies are in kHz) (Tick the correct answer)
   a) ± 1.5kHz and ± 0.5kHz
   b) ± 3kHz and ± 1kHz
   c) ± 2kHz
   ✓d) ± 2kHz and ± 1kHz

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$x(t) = 5\cos(2\times 1.5\pi t)\cdot\cos(2\times 0.5\pi t)$

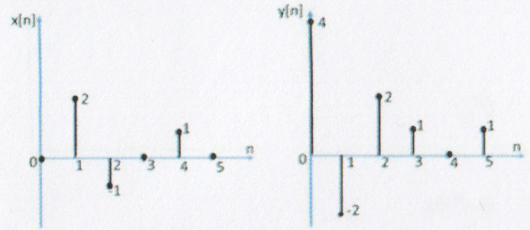
f_1 f_2

Product of two real sinusoids results in the sum and difference frequencies.

i.e. $\pm(f_1+f_2)$ & $\pm(f_1-f_2)$

$\pm$ 2kHz & $\pm$ 1kHz

6. Given the discrete-time sequences x[n] and y[n] shown, which difference formula best describes their relationship ? (Tick the correct answer)



   ✓a) y[n] = x[n-1] + 2x[n+1]
   b) y[n] = 2x[n] – x[n-1]
   c) y[n] = 2x[n-1] + x[n-2]
   d) y[n] = 2x[n-2] – x[n]

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$x[0]=0$; $y[0]\neq 0$ $(y[0]=4)$

$n=0$ $n=0$

∴ The signal x[n] must have been advanced by one sample

7. Determine the period of the discrete time signal $x[n] = \sin(0.2n + \pi)$. (Tick the correct answer)
   (a) 1 sample (b) 5 samples
   (c) π samples ✓(d) Non of the above

Topic 1 : Page 8 Example 4

Topic 1: Tutorial Questions (SOLO-Level 2)

1. A sinusoidal signal with peak-to-peak amplitude of 5V is sampled at 50kHz with uniform quantisation. Find the minimum number of bits for the analogue to digital converter to achieve a SQNR of at least 92dB. State any assumptions made.

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$SQNR = 6.02B + 1.76\,dB$ (B-number of bits)

$SQNR \geq 92\,dB$

$92 \leq 6.02B + 1.76$

$\therefore B \geq 14.9$

$\therefore \boxed{B = 15}$ (minimum number of bits)

2. Show that the signal to quantisation noise ratio (SQNR) of a linear B-bit analogue-to-digital converter is given by

$$SQNR = 6.02B + 4.77 - 20\log\left(A/\sigma_{sig}\right)$$

where the input range of the A/D converter is $\pm$A and the rms value of the input signal is σ_{sig}. Determine the SQNR if B is 16 bits and the input is a signal with a rms value of A/5.

Topic 1: Page 29 → $SQNR = 10\log\frac{\sigma_s^2}{\sigma_e^2} = 10\log\frac{\sigma_s^2}{(\Delta V^2/12)} = 10\log\frac{\sigma_s^2}{(2A/2^B)^2/12} = 10\log\sigma_s^2 \cdot \frac{2^{2B} \times 3}{A^2}$

$\therefore SQNR = 10\log\frac{\sigma_s^2}{A^2} + 10\log 2^{2B} + 10\log 3$

$= 20\log\frac{\sigma_s}{A} + B(10\log 2^2) + 4.77$

$= 6.02$

$\boxed{SQNR = 6.02B + 4.77 - 20\log\frac{A}{\sigma_s}}$

$\sigma_s = \sigma_{sig}$

$SQNR = 6.02 \times 16 + 4.77 - 20\log\frac{A}{(A/5)} = 6.02 \times 16 + 4.77 - 20\log(5)$

$\boxed{SQNR = 87.1\,dB}$

Topic 1: Tutorial Questions (SOLO-Level 2)

3. An analogue signal $x_a(t) = \cos(2\pi 500t)$ is sampled at a rate of f_s=4kHz. Determine the resulting analogue and digital magnitude spectra.

$FT[\cos\omega_1 t] = \pi\delta(\omega-\omega_1) + \pi\delta(\omega+\omega_1)$

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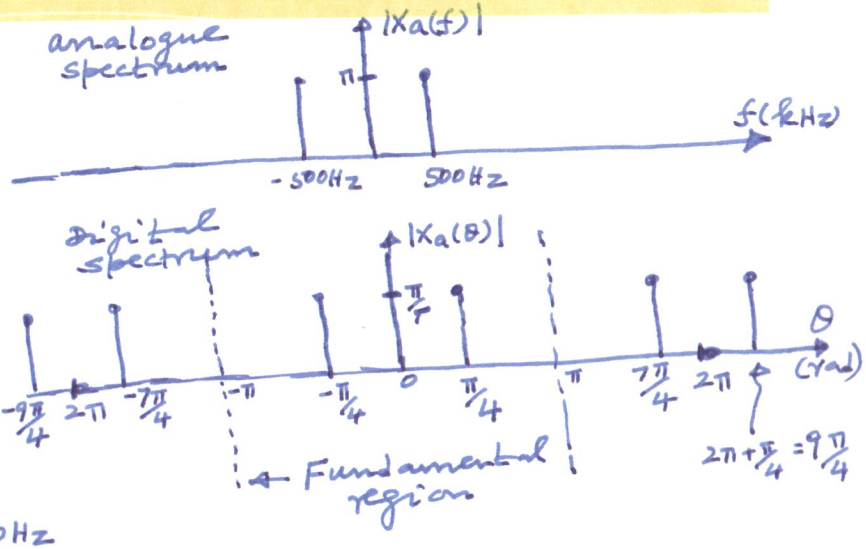
$x_a(t) = \cos(2\pi 500t)$

$\therefore f_1 = 500\,Hz$

$\therefore$ Corresponding digital frequency $\theta = 2\pi\frac{f_1}{f_s} = 2\pi\times\frac{500}{4000} = \frac{\pi}{4}$



4. An analogue signal $x(t) = \sin(480\pi t) + 3\sin(720\pi t)$ is sampled 600 times per sec:

(a) Determine the Nyquist sampling rate for x(t).

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$x(t) = \sin(2\pi\times240t) + 3\sin(2\pi\times360t)$

$f_1 = 240\,Hz$; $f_2 = 360\,Hz \Rightarrow f_{max} = 360\,Hz$ $\therefore F_{Nyquist} = 2\times f_{max} = 720\,Hz$

(b) Determine the folding frequency (or half the sampling frequency).

$f_s = 600\,Hz$ $\therefore f_{fold}$ or $\frac{f_s}{2} = 300\,Hz$

(c) What are the frequencies, in radians, in the resulting discrete time signal x[n]?

$f_2 > \frac{f_s}{2}$ ($\because$ it is affected by aliasing) $\therefore$ new frequency due to aliasing $f_2' = f_2 - k\cdot f_s = 360 - 1\cdot600 = -240\,Hz$ ($k=1$)

(d) If x[n] is passed through an ideal D/A converter what is the reconstructed signal y(t)?

We have $t = nT = \frac{n}{f_s} = \frac{n}{600}$

$y(t) = -2\sin(2\pi 240t)$

$\therefore y(t) = -2\sin(480\pi t)$

$\therefore x[n] = \sin\left(2\pi\frac{240}{600}n\right) + 3\sin\left(2\pi\frac{(-240)}{600}n\right)$

$x[n] = -2\sin\left(\frac{4\pi}{5}n\right)$

$x[n] = -\sin\left(2\pi\frac{240}{600}n\right)$

note: $\frac{n}{600} = t$

Topic 1: Tutorial Questions (SOLO-Level 2)

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5. A signal of $v(t) = 4.5\sin(1000\pi t)$ Volts is sampled at 2.5 kHz. An eight-bit A/D converter with a nominal input range of ± 5 Volts is used to convert sampled voltages to integer codes.

(a) Calculate the quantization interval of the A/D converter.

(b) Determine the maximum quantization error.

(a) Peak to Peak amplitude = 10 V

Quantisation interval or resolution (ΔV) = $\frac{10V}{2^B - 1} \approx \frac{10V}{2^B}$ (if B is Large)

$\therefore \Delta V = \frac{10V}{2^8} = 39.062\ mV$

↑ 8-bit A/D converter

(b) Maximum Quantisation error $e = \frac{\Delta V}{2} = 19.53\ mV$

Topic 1: Tutorial Questions (SOLO-Level 2)

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6. Consider the discrete-time sequence x[n] = cos[(nπ/8) + (π/4)] .
   (i) Determine the fundamental period of x[n].
   (ii) If the sampling frequency f_s = 10 kHz, write the equation of the analogue signal x(t).

(i) $x[n] = \cos[n\theta + \pi/4]$

digital frequency $\theta = \frac{\pi}{8}$; The fundamental period $N = \frac{2\pi k}{\theta}$

$N = \frac{2\pi \times 1}{\pi/8}$ ← k=1 $= 16$ samples

Period = 16 samples

(ii) $\theta = \frac{\pi}{8}$; $2\pi \frac{f_a}{f_s} = \frac{\pi}{8} \Rightarrow \frac{f_a}{f_s} = \frac{1}{16} \Rightarrow f_a = \frac{10000}{16}$

$= 625$ Hz

$x[n] = \cos(n\theta + \pi/4)$

$x(t) = \cos(2\pi(625)t + \pi/4)$

7. Sketch the following:
   (c) x[n] = δ[n+2] + 2 δ[n-1] - 0.5 δ[n-3]

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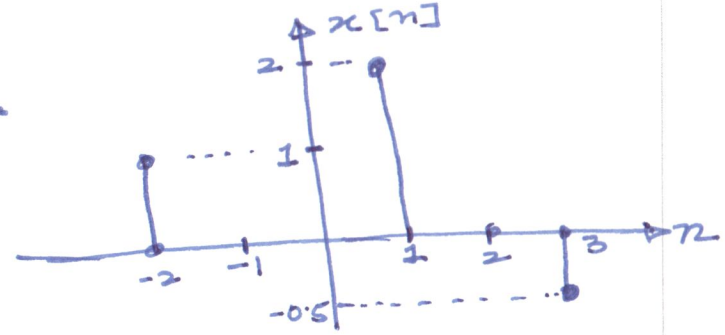
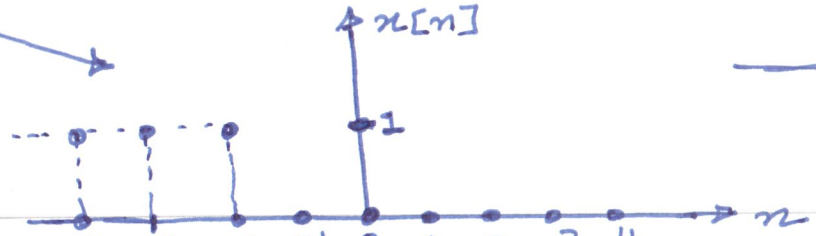
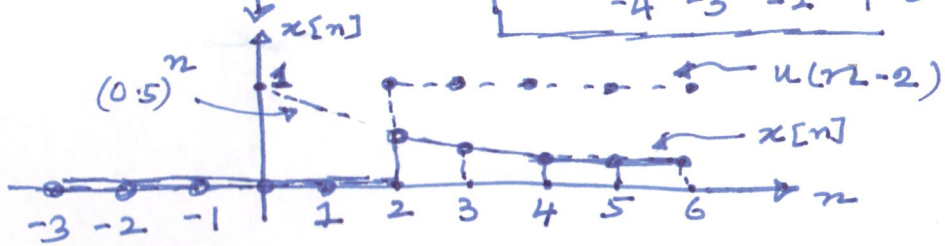


   (d) x[n] = u[-n-2]



   (e) x[n] = 0.5^n u[n-2]



Topic 1: Tutorial Questions (SOLO-Level 2)

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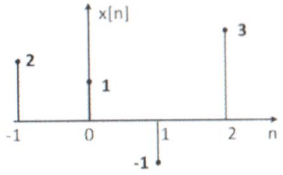
8. Write the equation for x[n] (see diagram) as a sum of scaled and delayed unit impulses

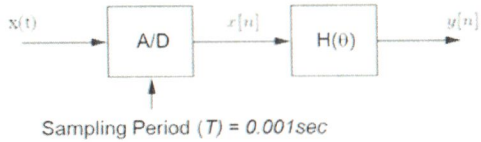


$x[n] = 2\delta(n+1) + \delta(n) - \delta(n-1) + 3\delta(n-2)$

9. For the system shown below , the input x(t) = cos(400πt) + cos(600πt). Find the output y[n] where

$f_1 = 200$ Hz, $f_2 = 300$ Hz

$$H(\theta) = \begin{cases} 1 & |\theta \le 0.5\pi| \\ 0 & otherwise \end{cases}$$



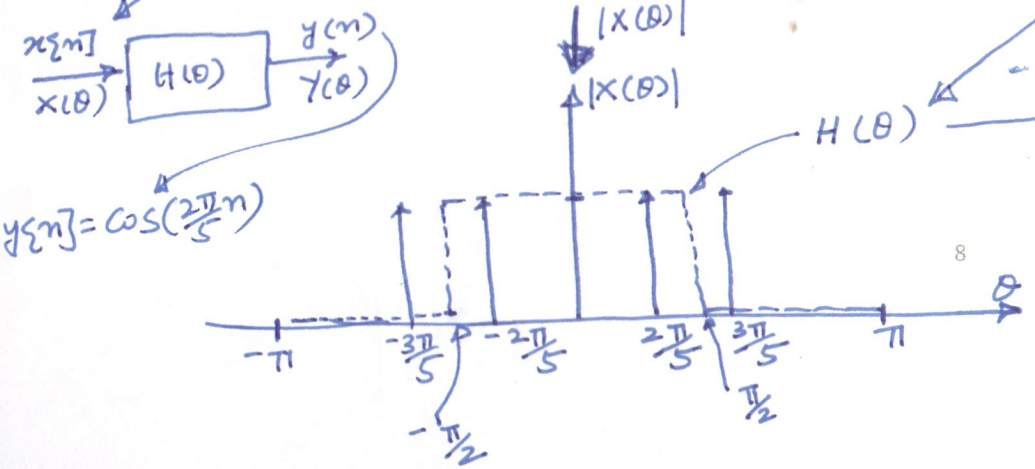
$f_s = \frac{1}{0.001} = 1000$ Hz

Sampling frequency

x(t) to x[n] transformation

$x[n] = \cos\left(2\pi \frac{200}{1000} n\right) + \cos\left(2\pi \frac{300}{1000} n\right)$

$x[n] = \cos\left(\frac{2\pi}{5} n\right) + \cos\left(\frac{3\pi}{5} n\right)$



$y[n] = \cos\left(\frac{2\pi}{5} n\right)$

The filter BW is $\pi/2 = 0.5\pi$ it removes the two frequencies $\pm \frac{3\pi}{5}$ and hence the term $\cos\left(\frac{3\pi}{5}\right)$ is removed.

$\therefore \boxed{y[n] = \cos\left(\frac{2\pi}{5} n\right)}$

Topic 1: Tutorial questions (SOLO - Level 3)

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1. A signal of $v(t) = 4.5\sin(1000\pi t)$ Volts is sampled at 2.5 kHz. An eight-bit A/D converter with a nominal input range of ±5 Volts is used to convert sampled voltages to integer codes. At what time (t>0) does the quantization error first reach its maximum value?

$\Delta V = \frac{10}{2^8} = 39.06\,mV$; Maximum quantisation error (e_m) = $\frac{\Delta V}{2}$

We will denote at $t = t_1$, the maximum quantisation error occurs

$\therefore \frac{\Delta V}{2} = 4.5\sin(1000\pi t_1) \Rightarrow t_1 = \left\{\sin^{-1}\left[\frac{\Delta V}{9}\right]\right\} \Big/ (1000\pi)$, $\therefore$ $t_1 = 1.38\,\mu s$

Note: $t_1 = 1.38\,\mu s$ suggests that the e_m will oscillate at a frequency ($\approx 362\,kHz$) much higher than the sampling frequency of 2.5 kHz ; $\therefore$ Quantization error appears as noise at the output.

2. A system samples a signal $x(t) = \cos(\omega_0 t)\cos(3\omega_0 t)$ at f_s = 7 kHz and reconstructs the signal using an ideal LPF with a cutoff frequency of 3.5 kHz. The value of ω_0 is $2\pi \times 10^3$ rad.

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(a) What is the highest frequency component present in the signal?

Hint : $\cos(\alpha)\cos(\beta) = \frac{1}{2}[\cos(\alpha+\beta) + \cos(\alpha-\beta)]$

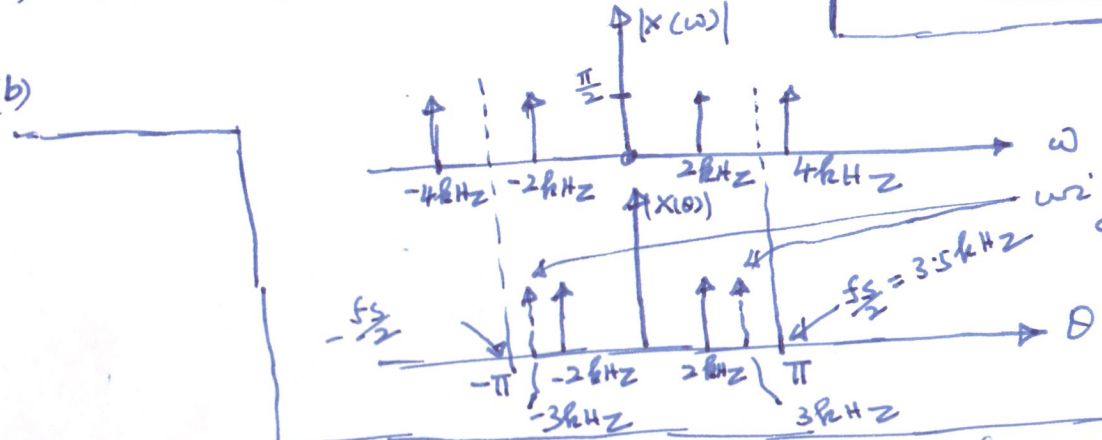
(b) Sketch $|X(\omega)|$ and $|X(\theta)|$ – the magnitude spectrum of the signal before and after sampling for (a) above.

(c) Can aliasing be prevented if the sampling frequency is increased to 8kHz? Explain.

(a) $x(t) = \cos(\omega_0 t)\cdot\cos(3\omega_0 t) = \frac{1}{2}[\cos(4\omega_0 t) + \cos(2\omega_0 t)]$

Highest frequency component $\omega = 4\omega_0 = 4 \times 2\pi \times 10^3$ ≡ 4 kHz

(b)



$f_s = 7\,kHz$ $\frac{f_s}{2} = 3.5\,kHz$

4 kHz will be aliased.

$\therefore$ New frequency $f_2' = f_2 - k \times f_s$

$= 4 - 1.7$

$= -3\,kHz$

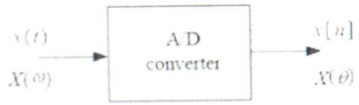
(c) NO. To avoid aliasing, we need $f_s > 2\,f_{highest}$ ($f_{highest} = 4\,kHz$)

$\therefore$ $f_s > 8\,kHz$

Topic 1: Tutorial questions (SOLO - Level 3)

3. If an analogue signal $x(t)$ is sampled every T seconds, then at the output of the Analogue to Digital Converter, as shown in a diagram below. Using Inverse Fourier Transforms in Continuous time and discrete time domains, show that

$$X(\theta) = \frac{1}{T}\sum_{k=-\infty}^{\infty} X(\omega + k\frac{2\pi}{T})$$



IFT (Inverse Fourier Transform) in continuous time is given by

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)\, e^{j\omega t}\, d\omega$$

given $x[n] = x(t)\Big|_{t=nT}$

$$x[n] = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)\, e^{j\omega(nT)}\, d\omega$$

$e^{j\omega nT}$ is Periodic function of period 2π

$$\therefore x[n] = \frac{1}{2\pi}\sum_{k=-\infty}^{\infty}\int_{2k\pi-\pi}^{2k\pi+\pi} X(\omega)\, e^{j\omega nT}\, d(\omega T)\times\frac{1}{T}$$

$$= \frac{1}{2\pi}\sum_{k=-\infty}^{\infty}\frac{1}{T}\int_{-\pi}^{\pi} X(\omega + k\tfrac{2\pi}{T})\, e^{j(\omega+k\frac{2\pi}{T})nT}\, d(\omega T)$$

$$= \frac{1}{2\pi}\cdot\frac{1}{T}\sum_{k=-\infty}^{\infty}\int_{-\pi}^{\pi} X(\omega + k\tfrac{2\pi}{T})\, e^{j\omega nT}\, d(\omega T)$$

let $\omega T = \theta$

$$x[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi}\underbrace{\frac{1}{T}\sum_{k=-\infty}^{\infty} X(\omega + k\tfrac{2\pi}{T})}_{=X(\theta)}\, e^{j\theta n}\cdot d\theta$$

$$\therefore X(\theta) = \frac{1}{T}\sum_{k=-\infty}^{\infty} X(\omega + k\tfrac{2\pi}{T})$$

IDFT (for Discrete signals)

$$x[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(\theta)\, e^{j n\theta}\, d\theta$$

Topic 1: Tutorial questions (SOLO - Levels 3)

4. Find the Inverse DTFT of $X(\theta)$ for the figure shown below and show that $x[n] = \frac{2\sin(n\pi/4)\cos(n\pi/2)}{n\pi}$

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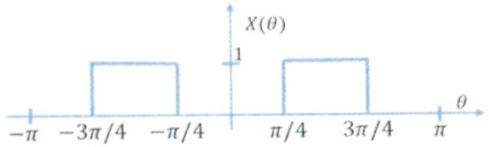


$$X(\theta) = \sum_{n=-\infty}^{\infty} x(n)e^{-jn\theta}$$

DTFT pair

$$x(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(\theta)e^{jn\theta}\,d\theta$$

θ = digital frequency; $-\pi \le \theta \le \pi$

Using the inverse DTFT, we have

$$x[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(\theta)e^{jn\theta}\,d\theta = \frac{1}{2\pi}\int_{\pi/4}^{3\pi/4} 1\cdot e^{jn\theta}\,d\theta + \frac{1}{2\pi}\int_{-3\pi/4}^{-\pi/4} 1\cdot e^{jn\theta}\,d\theta$$

$$= \frac{1}{2\pi jn}\left[e^{jn\theta}\right]_{\pi/4}^{3\pi/4} + \frac{1}{2\pi jn}\left[e^{jn\theta}\right]_{-3\pi/4}^{-\pi/4}$$

$$x[n] = \frac{1}{2\pi jn}\left[e^{j\frac{3\pi}{4}n} - e^{j\frac{\pi}{4}n}\right] + \frac{1}{2\pi jn}\left[e^{-j\frac{\pi}{4}n} - e^{-j\frac{3\pi}{4}n}\right]$$

Rearranging the terms, we have

$$x[n] = \frac{1}{2\pi jn}\left[e^{j\frac{3\pi}{4}n} - e^{-j\frac{3\pi}{4}n}\right] - \frac{1}{2\pi jn}\left[e^{jn\frac{\pi}{4}} - e^{-jn\frac{\pi}{4}}\right]$$

$$= \frac{1}{\pi n}\left[\frac{e^{j\frac{3\pi}{4}n} - e^{-j\frac{3\pi}{4}n}}{2j}\right] - \frac{1}{\pi n}\left[\frac{e^{jn\frac{\pi}{4}} - e^{-jn\frac{\pi}{4}}}{2j}\right]$$

$$x[n] = \frac{\sin\left(\frac{3\pi}{4}n\right)}{\pi n} - \frac{\sin\left(\frac{\pi}{4}n\right)}{\pi n}$$

$$\boxed{x[n] = 2\,\frac{\sin\left(\frac{n\pi}{4}\right)\cdot\cos\left(\frac{n\pi}{2}\right)}{n\pi}}$$

$$\left\{\sin A - \sin B = 2\cos\frac{A+B}{2}\cdot\sin\frac{A-B}{2}\right\}$$

Topic 1: Tutorial questions (SOLO - Level 3)

5. Find the Fourier Transform of x(t) = δ(t-a). Hence, show that $\int_{-\infty}^{\infty} e^{j\omega t}\,dt = 2\pi\delta(\omega)$.

$$X(\omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}\, dt$$

$\delta(t-a) = x(t)$

$$= \int_{-\infty}^{\infty} \delta(t-a)\, e^{-j\omega t}\, dt$$

$\delta(t-a) = 1$ at $t = a$

$$\boxed{X(\omega) = e^{-j\omega a}}$$

$$FT\{\delta(t-a)\} = e^{-j\omega a}$$

$$\therefore FT^{-1}\{e^{-j\omega a}\} = \delta(t-a)$$

(Inverse)

$$\frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-j\omega a}\cdot e^{j\omega t}\, d\omega = \delta(t-a)$$

Inverse Fourier Transform

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)\, e^{j\omega t}\, d\omega$$

$$\int_{-\infty}^{\infty} e^{j\omega(t-a)}\, d\omega = 2\pi\delta(t-a)$$

If $a = 0$

$$\int_{-\infty}^{\infty} e^{j\omega t}\, d\omega = 2\pi\delta(t)$$

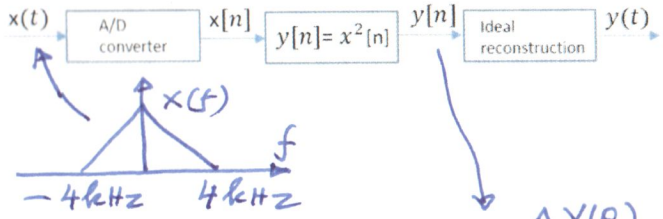
Interchanging the variables, we also have

$$\boxed{\int_{-\infty}^{\infty} e^{j\omega t}\, dt = 2\pi\delta(\omega)}$$

i.e. $t = \omega$ $\therefore dt = d\omega$ & the limits will be the same

Topic 1: Tutorial questions (SOLO - Level 4)

1. A speech signal is band limited to 4 kHz is ideally sampled at a sampling rate, f_s, and the sampled signal $x[n]$, is processed by squaring operations $y[n] = x^2[n]$, whose output is ideally reconstructed to obtain y(t) (see diagram on the right):



What minimum sampling rate f_s, will ensure $y(t) = x^2(t)$?

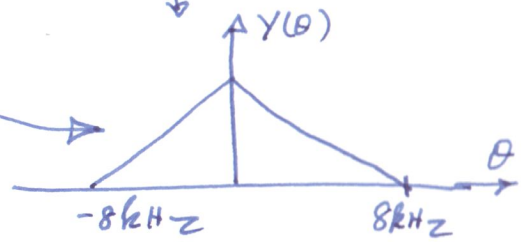
let $x(t) = A\sin(\omega_0 t)$

$y(t) = [x(t)]^2 = A^2\sin^2(\omega_0 t) = \frac{A^2}{2}(1 - \cos 2\omega_0 t)$

(squaring operation produces $2\omega_0$)

∴ The spectrum of $y[n]$ extends to twice the frequency of $x[n]$

∴ minimum sampling frequency should be 16kHz



2. The signal $x(t) = 2\cos(2000\pi t) - 4\sin(4000\pi t)$ is quantised by rounding, using a 12 bit quantiser. What is the rms quantisation error and the SQNR?

The signal Power $(P_s) = \left(\frac{2}{\sqrt{2}}\right)^2 + \left(\frac{4}{\sqrt{2}}\right)^2 = \sqrt{5}$

The peak value of $x(t)$ cannot exceed 6, so the peak to peak amplitude is ≈ 12.

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$\therefore \Delta V = \frac{12}{2^B - 1} = \frac{12}{2^{12} - 1}$ (using 12 bit quantiser) $\approx \frac{12}{2^{12}}$

RMS quantisation error $= \frac{\Delta V}{\sqrt{12}} = \frac{12/2^{12}}{\sqrt{12}} = 0.0008457$

$SQNR\,(dB) = 10\,Log\,\frac{P_s(\text{signal Power})}{P_n(\text{noise Power})} = 10\,Log\,\frac{\sqrt{5}}{\frac{\Delta V^2}{12}} = 10\,Log\,\frac{\sqrt{5}}{\frac{(12/2^{12})^2}{12}}$

$= 64.95$

Topic 1: Tutorial questions (SOLO-Level 5)

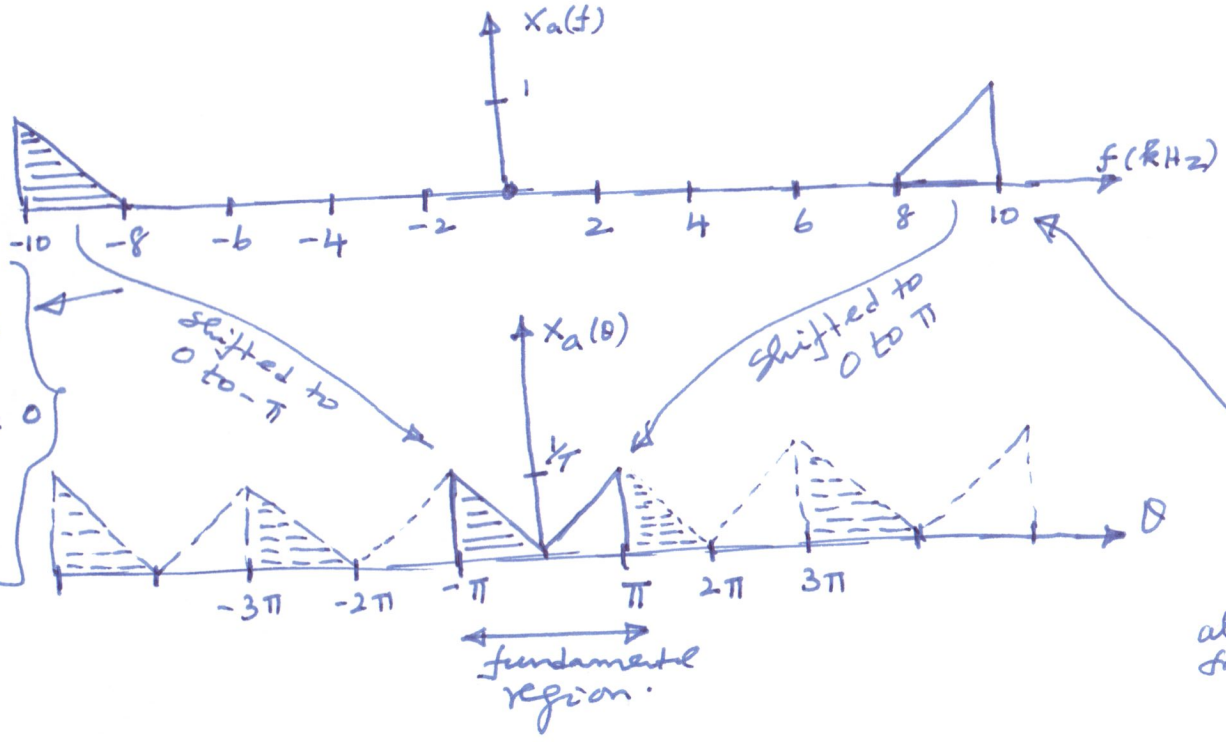
1. Given a real-valued bandpass signal $x_a(t)$ and $X_a(f) = 0$ for $|f| < f_1$ and $|f| > f_2$, the Nyquist sampling theorem says that the minimum sampling frequency is $f_s = 2f_2$. However , in some cases , the signal may be sampled at a lower rate.

   (a) Suppose that f_1 = 8 kHz and f_2 = 10 kHz. Draw a sketch of $X_a(\theta)$, if the sampling frequency f_s = 4 kHz.

Note

$$X(\theta) = \frac{1}{T}\sum_{k=-\infty}^{\infty} X\left(\omega + k\frac{2\pi}{T}\right)$$

$X(\theta)$ is formed by shifting $X_a(f)$ by integer multiples of the sampling frequency.



$f_3 = -8\,kHz$

$f_4 = -10\,kHz$

$f_3' = f_3 - k f_s$

$= -8 - (-2)4 = 0$

$f_4' = f_4 - (-2)4$

$= -2\,kHz$

$f_s = 4\,kHz$,

$f_1 = 8\,kHz$ & $f_2 = 10\,kHz$

$f_1' = f_1 - k f_s = 8 - 2\times4 = 0$ [$k=2$]

aliased frequency → $f_2' = f_2 - k f_s = 10 - 2\times4 = 2\,kHz$

Topic 1: Tutorial questions (SOLO-Level 5)

1. (b) If the bandwidth of the bandpass signal is defined as $B = f_2 - f_1$ and the centre frequency to be $f_c = \frac{(f_2+f_1)}{2}$, Show that if $f_c > B/2$ and f_2 is an integer multiple of the bandwidth B, no aliasing will occur if $x_a(t)$ is sampled at a sampling frequency $f_s = 2B$.

If f_2 is an integer multiple of B, we may express f_1 & f_2 as follows

$f_1 = (m-1)B$; $f_2 = mB$ m is an integer ; also $f_2 - f_1 = B$

With the sampling frequency $f_s = 2B$, the sampled signal has a spectrum

$$X_a(0) = \frac{1}{T}\sum_{-\infty}^{\infty} X_a(f - k\underbrace{2B}_{f_s})$$

If $X_a(f)$ is non zero only for $(m-1)B < |f| < mB$, there is only one term in the sum that contributes to $X_a(f)$ in the frequency range $0 < f < B$ and only one term that contributes to the frequency range $-B < f < 0$ (draw a sketch similar to Part a). Therefore, there is no aliasing and $x_a(t)$ may be sampled without aliasing if $f_s = 2B$.

1. (c) Repeat part (b) for the case in which f_2 is not an integer multiple of the bandwidth B.

If f_2 is not an integer multiple of B, we may always increase B until this is the case.

Let $k = \left[\frac{f_2}{B}\right]$ take only integer part

Now, if we simply increase B to B' where

$k = \frac{f_2}{B'}$ (or $B' = \frac{f_2}{k}$)

and f_2 now becomes integer multiple of B'.

$\therefore x_a(t)$ may be sampled without aliasing, a sampling frequency of

$$f_s = 2B' = \frac{2f_2}{\left[\frac{f_2}{B}\right]_{\text{integer part only}}}$$

Topic 1: Tutorial questions (SOLO-Level 5)

2. The D/A output approximates the analogue signal by a series of rectangular pulses, whose heights are equal to the corresponding value of the signal pulse as shown in Figure A.

a) Consider one pulse, $\hat{y}(t)$, with a height k and show that the corresponding frequency response $\hat{Y}(\omega)$ is a sinc function given by the following equation:

$$\hat{Y}(\omega) = k\,\frac{T}{2}\,e^{-\frac{j\omega T}{2}}\,\frac{\sin\frac{\omega T}{2}}{\frac{\omega T}{2}}$$

b) Explore the effect of $\hat{Y}(\omega)$ on the output signal spectrum $Y(\omega)$ and illustrate your understanding of this using a spectral plot. If compensation is required for the $sinx/x$, roll-off inherent in the D/A process, what function would you use to compensate? Briefly explain why.

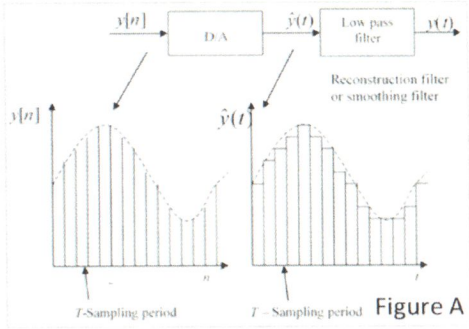
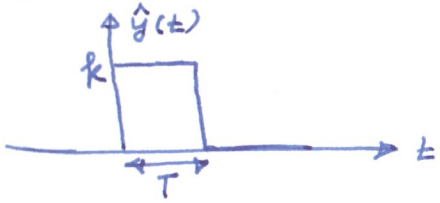


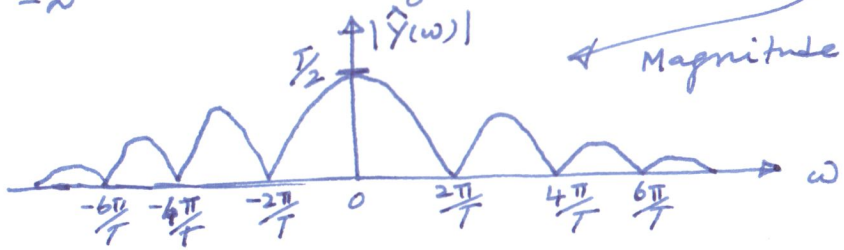
Figure A

(a) Just consider one pulse:



The corresponding frequency response is

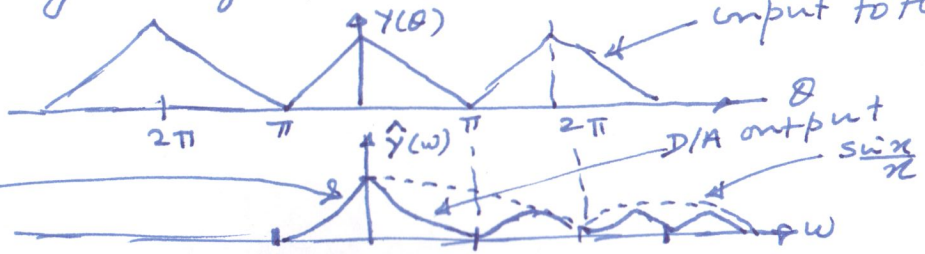
$$\hat{Y}(\omega) = \int_{-\infty}^{\infty} \hat{y}(t)\,e^{-j\omega t}\,dt = \int_{0}^{T} e^{-j\omega t}\,dt = \frac{T}{2}\,e^{\frac{j\omega T}{2}}\,\frac{\sin\frac{\omega T}{2}}{\frac{\omega T}{2}}$$



← Magnitude spectrum $\left(\frac{\sin x}{x}\right)$

(b) In the frequency domain, the staircase of the D/A converter introduces a type of distortion known as $\frac{\sin x}{x}$ where $x = \frac{\omega T}{2}$. The amplitude of the output signal spectrum $Y(\omega)$ is multiplied by the $\frac{\sin x}{x}$ function, which acts like a lowpass filter with the high frequency heavily attenuated.

The $\frac{\sin x}{x}$ effect is due to the holding action of the D/A and in signal recovery introduces an amplitude distortion



This distortion can be eliminated by equalisation. Before converting to analogue form the signal is passed through a digital filter whose amplitude response has a $\frac{x}{\sin x}$ shape.

ELEC3104: Digital Signal Processing

Topic 2: Discrete-Time Systems

Model answers to tutorial questions

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Term 3, 2021

Topic 2: Tutorial Questions (SOLO -Level 1)

SOLO – Level 1

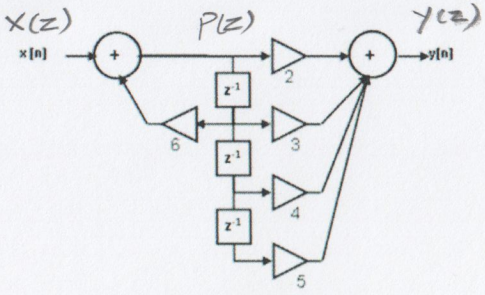
1. What is the equivalent transfer function of the block diagram below?
(Tick the correct answer)

(a) $H(z) = \frac{z(z^{-1}+3)(z^{-1}+4)(z^{-1}+5)}{6-z^{-1}}$

✓(b) $H(z) = \frac{2+3z^{-1}+4z^{-2}+5z^{-3}}{1-6z^{-1}}$

(c) $H(z) = \frac{1+6z^{-1}}{2+3z^{-1}+4z^{-2}+5z^{-3}}$

(d) $H(z) = \frac{2+3z^{-1}+4z^{-2}+5z^{-3}}{1+6z^{-1}}$



$$\frac{P(z)}{X(z)} = \frac{1}{1-6z^{-1}}$$

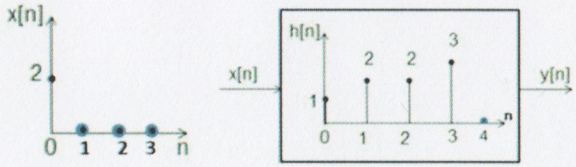
$$\frac{Y(z)}{P(z)} = 2+3z^{-1}+4z^{-2}+5z^{-3}$$

$$\therefore \frac{Y(z)}{X(z)} = \frac{Y(z)}{P(z)} \cdot \frac{P(z)}{X(z)}$$

$$= 2+3z^{-1}+4z^{-2}+5z^{-3} \times \frac{1}{1-6z^{-1}}$$

2. The impulse response ,h[n], of a discrete time system is given below. The output signal, y[n] , for the input signal, x[n] is given by: (Tick the correct answer)

(a) {2 0 0 0 0}
(b) {1 2 2 0 0}
(c) {3 4 5 2 0}
✓(d) {2 4 4 6 0}
(e) None of the above



$$y(n) = x(n) * h(n)$$

$$= 2\delta(n) * h[n]$$

↑ impulse (unit amplitude)

$$= 2\,h[n]$$

$$= \{2\ 4\ 4\ 6\ 0\}$$

Topic 2: Tutorial Questions (SOLO -Level 1)

SOLO – Level 1

3. The impulse response of the system with a transfer function $H(z) = \frac{\alpha+\alpha z^{-1}}{1-\alpha z^{-1}}$ is given by: (Tick the correct answer)

   (a) $\alpha^{n-1} + \alpha \quad n < 0$

   ✓(c) $\alpha^{n+1}\, u[n] + \alpha^{n}\, u[n-1]$

   (b) $\alpha^{n}\, u[n] + \alpha^{n-1}\, u[n-1]$

   (d) None of the above

$$H(z) = \frac{\alpha}{1-\alpha z^{-1}} + \frac{\alpha \cdot z^{-1}}{1-\alpha z^{-1}}$$ use inverse z-transform

inverse z-transform →

$$h[n] = \alpha \cdot (\alpha)^{n} u[n] + \alpha \cdot (\alpha)^{n-1} u[n-1]$$

$$h[n] = \alpha^{n+1} u[n] + \alpha^{n} u[n-1]$$

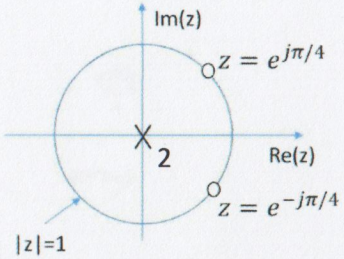
4. The pole-zero pattern of a system has zeros on the unit circle as show below. The system function is given by: (Tick the correct answer)

   (a) $H(z) = 1 + \frac{1}{\sqrt{2}} z^{-1} + z^{-2}$

   (b) $H(z) = z - \frac{1}{\sqrt{2}} + z^{-1}$

   (c) $H(z) = z^2 - 1 + z^{-2}$

   ✓(d) $H(z) = 1 - \sqrt{2}\, z^{-1} + z^{-2}$



$$H(z) = \frac{(z - 1 \cdot e^{j\frac{\pi}{4}})(z - 1 \cdot e^{-j\frac{\pi}{4}})}{z^2}$$

$$= z^2 - \underbrace{(e^{j\frac{\pi}{4}} + e^{-j\frac{\pi}{4}})}_{=2\cos\frac{\pi}{4}} z + 1 \Big/ z^2$$

$$H(z) = \frac{z^2 - 2\cos\frac{\pi}{4} z + 1}{z^2} = \frac{z^2 - \sqrt{2} z + 1}{z^2}$$

$$H(z) = 1 - \sqrt{2} z^{-1} + z^{-2}$$

5. The DFT of a real signal is given below. Therefore, the signal energy is equal to: (Tick the correct answer)

   (a) 0

   (b) 5

   (c) 7

   ✓(d) None of the above

DFT: [1, $\sqrt{2}$, -1, B, -7, -j2, C, $\sqrt{2}$]

k=0 k=7

use conjugate symmetry of DFT to find B & C

$$X[k] = [1 \quad \sqrt{2} \quad -1 \quad j2 \quad -7 \quad -j2 \quad -1 \quad \sqrt{2}]$$

$$[1 \quad \sqrt{2} \quad -1 \quad B \quad -7 \quad -j2 \quad C \quad \sqrt{2}]$$

$\therefore B = j2$

$C = -1$

$$\text{signal energy} = \frac{1}{8} \sum |X[k]|^2$$

$$= \frac{1}{8} [1^2 + \sqrt{2}^2 + (-1)^2 + 2^2 + 7^2 + 2^2 + 1^2 + (\sqrt{2})^2]$$

$$= 8$$

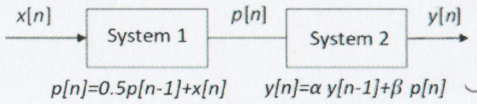
Topic 2: Tutorial Questions (Level 2)

SOLO – Level 2

1. Consider the cascade of the following two systems as depicted below. The difference equation relating $x[n]$ of $y[n]$ is given by:

$$y[n] = x[n] + \frac{3}{4}y[n-1] - \frac{1}{8}y[n-2]$$

Determine α and β.



$P[n] = 0.5P[n-1] + x[n] \;\therefore\; \frac{P(z)}{X(z)} = \frac{1}{1-0.5z^{-1}}$

$y[n] = \alpha y[n-1] + \beta P[n] \rightarrow \frac{Y(z)}{P(z)} = \frac{\beta}{1-\alpha z^{-1}}$

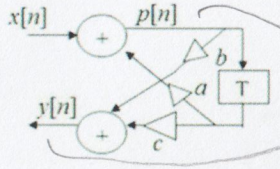
$\therefore \frac{Y(z)}{X(z)} = \frac{Y(z)}{P(z)} \cdot \frac{P(z)}{X(z)} = \frac{\beta}{1-\alpha z^{-1}} \cdot \frac{1}{1-0.5z^{-1}}$

$\therefore \boxed{\frac{Y(z)}{X(z)} = \frac{\beta}{1-(\alpha+0.5)z^{-1}+0.5\alpha z^{-2}}} \quad -(1)$

$\frac{Y(z)}{X(z)} = \frac{1}{1-\frac{3}{4}z^{-1}+\frac{1}{8}z^{-2}} \quad -(2)$

Equate (1) & (2) $\boxed{\beta = 1}$; $0.5\alpha = \frac{1}{8} \;\therefore\; \boxed{\alpha = \frac{1}{4}}$

2. For the block diagram realisations given below, develop the relation between y[n] and x[n] in each case.



(a)

$P(z) = X(z) + az^{-1}P(z) \Rightarrow \frac{P(z)}{X(z)} = \frac{1}{1-az^{-1}}$

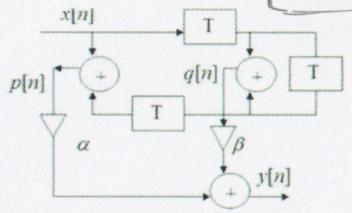
$Y(z) = bP(z) + cz^{-1}P(z) \Rightarrow \frac{Y(z)}{P(z)} = b+cz^{-1}$

$\Rightarrow \frac{Y(z)}{X(z)} = \frac{b+cz^{-1}}{1-az^{-1}}$

$Y(z)[1-az^{-1}] = X[z](b+cz^{-1})$

$Y(z) = bX(z) + cz^{-1}X(z) + az^{-1}Y(z)$

$\boxed{y[n] = bx[n] + cx[n-1] + ay[n-1]}$



(b)

$q[n] = x[n-1] + x[n-2]$

$p[n] = x[n] + x[n-3]$

$y[n] = \alpha p[n] + \beta q[n] = \alpha\{x[n] + x[n-3]\} + \beta\{x[n-1] + x[n-2]\}$

$\therefore \boxed{y[n] = \alpha x[n] + \beta x[n-1] + \beta x[n-2] + \alpha x[n-3]}$

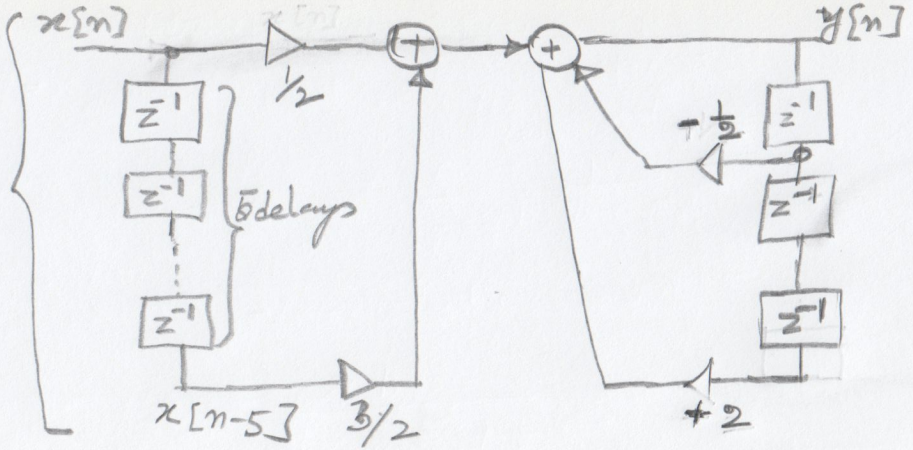
Topic 2: Tutorial Questions (Level 2)

SOLO – Level 2

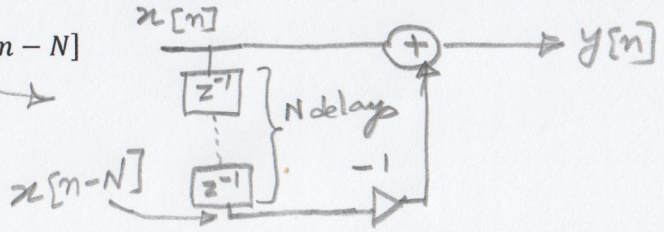
3. Draw a system implementation for each of the following equations:

a) $2y[n] + y[n-1] - 4y[n-3] = x[n] + 3x[n-5]$

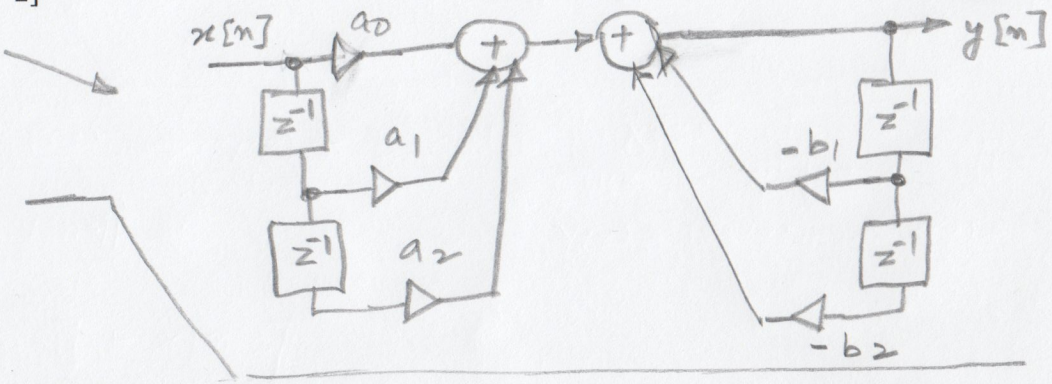
$y[n] = \frac{1}{2}x[n] + \frac{3}{2}x[n-5] - \frac{1}{2}y[n-1] + 2y[n-3]$



b) $y[n] = x[n] - x[n-N]$



c) $y[n] = a_0x[n] + a_1x[n-1] + a_2x[n-2] - b_1y[n-1] - b_2y[n-2]$



4. A difference equation for a particular filter is given by

$y[n] = 0.1x[n] - 0.1x[n-2] + 0.8x[n-3] + 0.1x[n-4] + 0.6x[n-6]$

Find the impulse response of the above filter.

This is a FIR filter.

$\therefore h[n] = \{0.1, 0, -0.1, 0.8, 0.1, 0, 0.6\}$ & $h[n] = 0,\ n \geq 7$

$n = 0$

Topic 2: Tutorial Questions (Level 2)

SOLO – Level 2

5. Consider the system described by the difference equation

$$y[n] = ay[n-1] + bx[n]$$

Determine '*b*' in terms of '*a*' so that $\sum_{n=-\infty}^{\infty} h[n] = 1$.

Let $x[n] = \delta[n]$ then $y[n] = h[n]$

$\delta[0] = 1$

$n=0 \quad y[0] = a\,y[-1] + b\,x[0] = b$

$n=1 \quad y[1] = a\,y[0] + b\,x[1] = ab$

$n=2 \quad y[2] = a\,y[1] + b\,x[2] = a^2 b$

$n = n-1 \quad y[n-1] = a^{n-1} b$

$n = n \quad \boxed{y[n] = a^n b \quad n \geq 0}$

$$\sum_{n=-\infty}^{\infty} h[n] = \sum_{n=0}^{\infty} a^n b = b\left[1 + a + a^2 + \cdots\right] = b \cdot \frac{1}{1-a}$$

$$\sum_{n=-\infty}^{\infty} h[n] = 1 \text{ (given)} \quad \therefore b \cdot \frac{1}{1-a} = 1 \Rightarrow \boxed{b = 1-a}$$

6. Let $x[n] = \{1\ 4\ 0\ 2\}$ and $h[n] = \{1\ 2\ 1\}$. Find their convolution (Both sequences start at n=0).

$$X(z) = 1 + 4z^{-1} + 2z^{-3}; \quad H(z) = 1 + 2z^{-1} + z^{-2}$$

$$Y(z) = X(z) \cdot H(z) = 1 + 2z^{-1} + z^{-2} + 4z^{-1} + 8z^{-2} + 4z^{-3} + 2z^{-3} + 4z^{-4} + 2z^{-5}$$

$$Y(z) = 1 + 6z^{-1} + 9z^{-2} + 6z^{-3} + 4z^{-4} + 2z^{-5}$$

Take inverse z-transform $\rightarrow y[n] = x(n) * h[n] = \{1, 6, 9, 6, 4, 2\}$

7. Find the impulse response h[n] for the following system:

$$H(z) = \frac{1 - \frac{1}{2}z^{-1}}{\left[1 - \frac{1}{4}z^{-1}\right]\left[1 + \frac{1}{4}z^{-1}\right]}$$

$$\frac{1 - \frac{1}{2}z^{-1}}{\left[1 - \frac{1}{4}z^{-1}\right]\left[1 + \frac{1}{4}z^{-1}\right]} = \frac{A}{\left[1 - \frac{1}{4}z^{-1}\right]} + \frac{B}{1 + \frac{1}{4}z^{-1}}$$

To find A, substitute $z^{-1} = 4$ $\{$i.e. $1 - \frac{1}{4}z^{-1} = 0\}$ $\rightarrow A = \left.\frac{1 - \frac{1}{2}z^{-1}}{1 + \frac{1}{4}z^{-1}}\right|_{z^{-1}=4} = \frac{1 - \frac{1}{2} \times 4}{1 + \frac{1}{4} \times 4} = -\frac{1}{2}$

similarly $B = \left.\frac{1 - \frac{1}{2}z^{-1}}{1 - \frac{1}{4}z^{-1}}\right|_{z^{-1}=-4} = \frac{1 - \frac{1}{2}(-4)}{1 - \frac{1}{4}(-4)} = \frac{3}{2}$

$$\therefore H(z) = \frac{-\frac{1}{2}}{1 - \frac{1}{4}z^{-1}} + \frac{\frac{3}{2}}{1 + \frac{1}{4}z^{-1}}$$

$$\therefore h[n] = -\frac{1}{2}\left(\frac{1}{4}\right)^n u[n] + \frac{3}{2}\left(-\frac{1}{4}\right)^n u[n]$$

Topic 2: Tutorial Questions (Level 2)

SOLO – Level 2

8. Consider the following system:

$$H(z) = \frac{1 - 2z^{-1} + 2z^{-2} - z^{-3}}{(1 - z^{-1})(1 - 0.5z^{-1})(1 - 0.2z^{-1})}$$

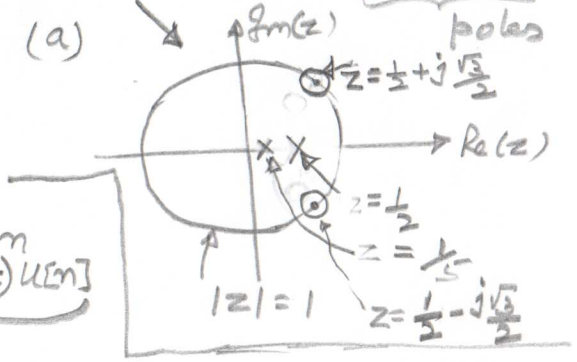
a) Sketch the pole-zero pattern. Is this system stable?

b) Determine the impulse response of the system.

$= \frac{(1 - z^{-1} + z^{-2})(1 - z^{-1})}{(1 - z^{-1})(1 - 0.5z^{-1})(1 - 0.2z^{-1})} = \frac{1 - z^{-1} + z^{-2}}{(1 - \frac{1}{2}z^{-1})(1 - \frac{1}{5}z^{-1})} = \frac{z^2 - z + 1}{(z - \frac{1}{2})(z - \frac{1}{5})}$

zeros: $z = \frac{1 \pm j\sqrt{3}}{2}$

(a)



(b) $H(z) = 1 + \left[\frac{A}{1 - \frac{1}{2}z^{-1}} + \frac{B}{1 - \frac{1}{5}z^{-1}}\right] z^{-1}$ ← similar to Q7, we can find $A = \frac{5}{2}$, $B = \frac{-14}{5}$

$H[z] = 1 + \frac{(5/2)z^{-1}}{1 - \frac{1}{2}z^{-1}} - \frac{(14/5)z^{-1}}{1 - \frac{1}{5}z^{-1}}$

Inverse z-transform

$\therefore h[n] = \delta[n] + \frac{5}{2}\left(\frac{1}{2}\right)^{n-1}u[n-1] - \left(\frac{14}{5}\right)\left(\frac{1}{5}\right)^{n-1}u[n-1] = \underbrace{\delta[n] + 5\left(\frac{1}{2}\right)^{n}u[n] - 14\left(\frac{1}{5}\right)^{n}u[n]}_{= h[n]}$

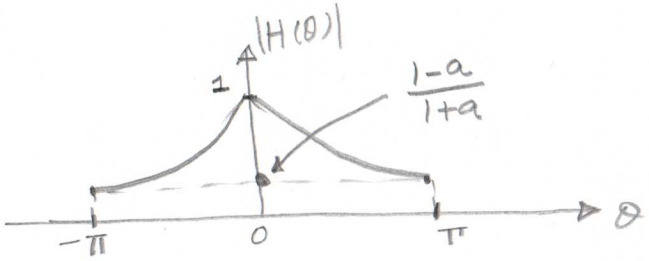
9. Determine and sketch the approximate magnitude response for the following filter: $H(z) = \frac{1-a}{1-az^{-1}}$ $0<a<1$

$H(\theta) = H(z)\Big|_{z = e^{j\theta}} = \frac{1-a}{1-ae^{-j\theta}}$

$|H(\theta)|^2 = H(\theta) \cdot H^*(\theta) = \frac{1-a}{1-ae^{-j\theta}} \cdot \frac{1-a}{1-ae^{j\theta}} = \frac{(1-a)^2}{1-a(e^{j\theta}+e^{-j\theta})+a^2} = \frac{(1-a)^2}{1-2a\cos\theta+a^2}$

$H^*(\theta)$: complex conjugate

$\therefore |H(\theta)| = \frac{1-a}{\sqrt{1-2a\cos\theta+a^2}}$



Topic 2: Tutorial Questions (Level 2)

SOLO – Level 2

10. A speech signal is sampled with a sampling period of 125ms. A frame of 256 samples is selected and a 256 point DFT is computed. What is the spacing between the DFT values in Hz.

$f_s = \frac{1}{125\times10^{-3}}$ $\quad N = 256$

$\Delta f = \frac{f_s}{N} = \frac{1}{125\times10^{-3}} \cdot \frac{1}{256} = 0.03125\,Hz$

10. Let x[n]={1, 2, 1, 0}. With N=4, compute the 4-point DFT.

$X[k] = \sum_{n=0}^{N-1} x[n]\, e^{-j\frac{2\pi}{N} n\cdot k}$; $N=4$, $\therefore X[k] = \sum_{n=0}^{3} x[n]\, e^{-j\frac{2\pi}{4} n k}$

$X[k] = x[0] + x[1]e^{-j\frac{\pi}{2}k} + x[2]e^{-j\pi k} + x[3]e^{-j\frac{3\pi}{2}k}$

$\quad\quad = 1 \quad\quad 2 \quad\quad 1 \quad\quad 0$

$\boxed{X[k] = 1 + 2e^{-j\frac{\pi}{2}k} + e^{-j\pi k}}$

$k=0$ $X[0] = 4$; $k=1$ $X[1] = 1 + 2e^{-j\frac{\pi}{2}} + e^{-j\pi} = -j2$

$k=2$ $X[2] = 1 + 2e^{-j\pi} + e^{-j2\pi} = 0$

$k=3$ $X[3] = 1 + 2e^{-j\frac{3\pi}{2}} + e^{-j3\pi} = j2$

$\boxed{X[k] = [4, -j2, 0, j2]}$

12. Find the 8-point DFT of $x[n]$ ={1, 1, 0, 0, 0, 0, 0, 0}, using conjugate symmetry property of $X[k]$.

Since x[2] to x[7] are all zeros, the upper index in the DFT summation will be n=1

$X[k] = \sum_{n=0}^{1} x[n]\, e^{-j\frac{2\pi}{8} n k}$ {N=8} ; $X[k] = x[0] + x[1]e^{-j\frac{2\pi}{8}k} = 1 + e^{-j\frac{\pi}{4}k}$ for $k = 0, 1, 2, \cdots 7$

N=8;

$X[k] = 1 + e^{-j\frac{\pi}{4}k}$

$k=0$ $X[0] = 1+1 = 2$

$k=1$ $X[1] = 1 + e^{-j\frac{\pi}{4}} = 1.707 - j0.707$

$k=2$ $X[2] = 1 + e^{-j\frac{\pi}{2}} = 1 - j$

$k=3$ $X[3] = 1 + e^{-j\frac{3\pi}{4}} = 0.293 - j0.707$

$k=4$ $X[0] = 1 + e^{-j\pi} = 0$

Using conjugate symmetry property

$X[5] = X[3] = 0.293 - j0.707$

$X[6] = X[2] = 1 - j$

$X[7] = X[1] = 1.707 - j0.707$

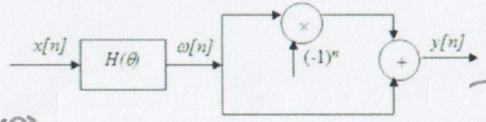
$X[0]$ $X[1]$ $X[2]$ $X[3]$ $X[4]$ $X[5]$ $X[6]$ $X[7]$

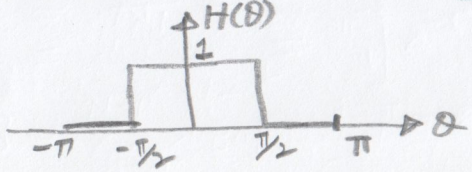
Topic 2: Tutorial questions (Levels 2)

SOLO – Level 2

13. For the system in the figure below, sketch the output *y[n]* when the input *x[n]* is *δ[n]* and *H(θ)* is an ideal lowpass filter as follows:

$$H(\theta) = \begin{cases} 1 & 0 \le |\theta| \le \frac{\pi}{2} \\ 0 & \frac{\pi}{2} < |\theta| \le \pi \end{cases}$$





$w[n] = h[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} H(\theta)\, e^{j\theta n}\, d\theta = \frac{1}{2\pi}\int_{-\pi/2}^{\pi/2} 1\, e^{j\theta n}\, d\theta$

$h[n] = \frac{1}{2\pi}\,\frac{e^{j\frac{\pi}{2}n} - e^{-j\frac{\pi}{2}n}}{jn} = \frac{1}{\pi}\,\frac{\sin n\frac{\pi}{2}}{n}$

$w[n]$ becomes $h[n]$ when $x[n] = \delta[n]$

$y[n] = (-1)^n h[n] + h[n] = h[n]\{1 + (-1)^n\}$

$y[n] = \frac{1}{2}\,\frac{\sin\frac{n\pi}{2}}{\frac{n\pi}{2}}\{1 + (-1)^n\}$

$$y[n] = \frac{1}{2}\{1 + (-1)^n\} \cdot \frac{\sin\frac{n\pi}{2}}{\frac{n\pi}{2}}$$

14. A 3- second signal is sampled at f_s = 100 Hz. The maximum spectral spacing is to be $\Delta f = 0.25\ Hz$. How many samples are needed for the DFT and FFT ?

Number of samples $N = \frac{f_s}{\Delta f} = \frac{100}{0.25} = 400$

3s signal gives 300 samples. To make 400 samples, we must pad the signal with 100 zeros.

DFT : 400 samples

FFT : 512 samples (2^9) ie another 112 zeros must be added. ie spacing $\Delta f = \frac{100}{512} = 0.1953\ Hz$ ← better than required, resolution of 0.25 Hz

15. Classify the following system in terms of its linearity, time invariance, memory and causality: $y[n] = |x[-n]|$

answer : non-linear, time invariant, dynamic, non-causal

Linearity : $x[n] \xrightarrow{H} y[n] = |x[-n]|$

$x_1[n] \xrightarrow{H} y_1[n] = |x_1[-n]|$

$x_2[n] \xrightarrow{H} y_2[n] = |x_2[-n]|$

$a_1 y_1[n] + a_2 y_2[n] = a_1|x_1[-n]| + a_2|x_2[-n]|$ — ①

$a_1 x_1[n] + a_2 x_2[n] \xrightarrow{H} |a_1 x_1[-n] + a_2 x_2[-n]|$ — ②

① & ② are not same

Non-linear

Time invariant

$x(n-n_0) \xrightarrow{H} |x[-(n-n_0)]|$ — ③

$y(n-n_0) \xrightarrow{H} |x[-(n-n_0)]|$ — ④

③ & ④ are same

So time invariant

Memory

If $n = 2$, $y[2] = |x[-2]|$

output relies on the past input

∴ dynamic

causal

If $n = -2$, $y[-2] = |x[2]|$

output relies on future input

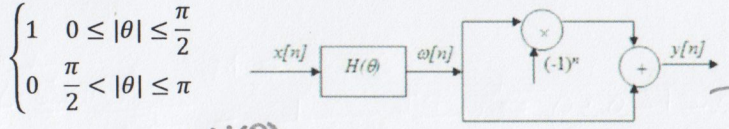
∴ non-causal

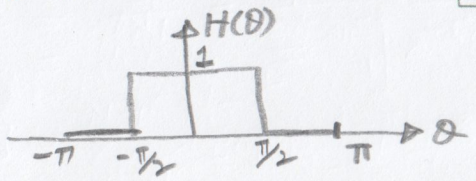
Topic 2: Tutorial questions (Levels 2)

SOLO – Level 2

13. For the system in the figure below, sketch the output *y[n]* when the input *x[n]* is $\delta[n]$ and $H(\theta)$ is an ideal lowpass filter as follows:

$$H(\theta) = \begin{cases} 1 & 0 \le |\theta| \le \frac{\pi}{2} \\ 0 & \frac{\pi}{2} < |\theta| \le \pi \end{cases}$$





$w[n] = h[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} H(\theta)\, e^{j\theta n}\, d\theta = \frac{1}{2\pi}\int_{-\pi/2}^{\pi/2} 1\, e^{j\theta n}\, d\theta$

$h[n] = \frac{1}{2\pi}\,\frac{e^{j\frac{\pi}{2}n} - e^{-j\frac{\pi}{2}n}}{jn} = \frac{1}{\pi}\,\frac{\sin n\frac{\pi}{2}}{n}$

$y[n] = (-1)^n h[n] + h[n] = h[n]\left\{1 + (-1)^n\right\}$

$y[n] = \frac{1}{2}\,\frac{\sin\frac{n\pi}{2}}{\frac{n\pi}{2}}\left\{1 + (-1)^n\right\}$

$$y[n] = \frac{1}{2}\left\{1 + (-1)^n\right\}\cdot\frac{\sin\frac{n\pi}{2}}{\frac{n\pi}{2}}$$

w[n] becomes h[n] when x[n] = δ[n]

14. A 3- second signal is sampled at f_s = 100 Hz. The maximum spectral spacing is to be $\Delta f = 0.25\ Hz$. How many samples are needed for the DFT and FFT ?

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15. Classify the following system in terms of its linearity, time invariance, memory and causality: $y[n] = |x[-n]|$

Answer : non-linear, time invariant, dynamic, non-causal

Linearity : $x[n] \xrightarrow{H} y[n] = |x[-n]|$

$x_1[n] \xrightarrow{H} y_1[n] = |x_1[-n]|$

$x_2[n] \xrightarrow{H} y_2[n] = |x_2[-n]|$

$a_1 y_1[n] + a_2 y_2[n] = a_1\,|x_1[-n]| + a_2\,|x_2[-n]|$ — ①

$a_1 x_1[n] + a_2 x_2[n] \xrightarrow{H} |a_1 x_1[-n] + a_2 x_2[-n]|$ — ②

① & ② are not same

Non-linear

Time invariant

$x(n-n_0) \xrightarrow{H} |x[-(n-n_0)]|$ — ③

$y(n-n_0) \xrightarrow{H} |x[-(n-n_0)]|$ — ④

③ & ④ are same

So time invariant

Memory

If $n=2$, $y[2] = |x[-2]|$

output relies on the past input

∴ Dynamic

Causal

If $n=-2$, $y[-2] = |x[2]|$

output relies on future input

∴ non-causal

Topic 2: Tutorial questions (Level 3)

SOLO – Level 3

1. If $y[n] = x[n] * h[n]$, then show that $Y(z) = X(z) \cdot H(z)$.

$$y[n] = \sum_{k=-\infty}^{\infty} h[k]\, x[n-k] \quad ; \quad Y(z) = \sum_{n=-\infty}^{\infty} y[n]\, z^{-n}$$

← Definition of z-transform

$$\therefore Y(z) = \sum_{n=-\infty}^{\infty} \left[\sum_{k=-\infty}^{\infty} h[k]\, x[n-k] \right] z^{-n} = \sum_{n=-\infty}^{\infty} \left[h[k] \cdot z^{-k} \cdot \sum_{k=-\infty}^{\infty} x[n-k]\, z^{k} \cdot z^{-n} \right]$$

{Multiplying by $z^{k} \cdot z^{-k}$ and changing the order of the summation}

$$Y(z) = \sum_{k=-\infty}^{\infty} \left\{ h[k] z^{-k} \cdot \sum_{n=-\infty}^{\infty} x[n-k]\, z^{-(n-k)} \right\}$$

By substituting $m = n - k$

$$Y(z) = \sum_{k=-\infty}^{\infty} h[k] z^{-k} \cdot \sum_{m=-\infty}^{\infty} x[m]\, z^{-m}$$

$= H(z)$ $\quad$ $= X(z)$

$$\Rightarrow Y(z) = H(z) \cdot X(z)$$

$$Y(z) = X(z) \cdot H(z)$$

2. Find the z-transform, G(z), if $g[n] = \begin{cases} na^n & n \geq 0 \\ 0 & n < 0 \end{cases}$ where a is a constant

Let $x[n] = a^n \Rightarrow X(z) = \frac{z}{z-a} \quad \therefore \frac{dX(z)}{dz} = \frac{-a}{(z-a)^2}$

$g[n] = n a^n$

$G(z) = -z \frac{dX(z)}{dz}$ ←

$$G(z) = -z \cdot \frac{-a}{(z-a)^2} = \frac{a z}{z^2 (1 - a z^{-1})^2} = \frac{a z^{-1}}{(1 - a z^{-1})^2}$$

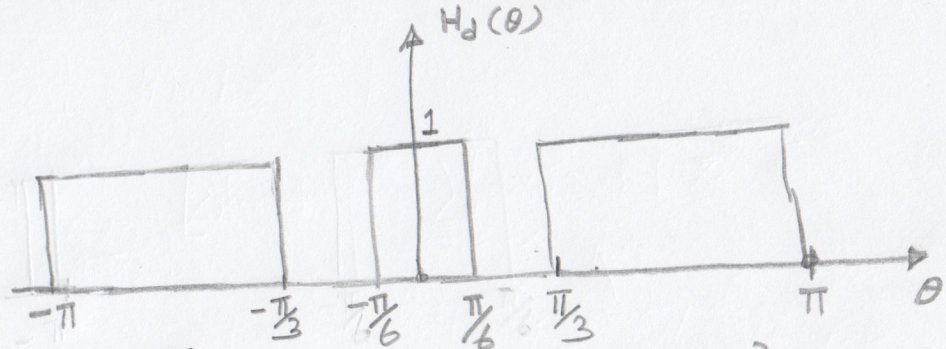
$$G(z) = \frac{a z^{-1}}{(1 - a z^{-1})^2}$$

Topic 2: Tutorial questions (Level 3)

SOLO – Level 3

3. The frequency response of a band stop filter is given by

$$H_d(\theta) = \begin{cases} 1 & |\theta| \le \frac{\pi}{6} \\ 0 & \frac{\pi}{6} < |\theta| < \frac{\pi}{3} \\ 1 & \frac{\pi}{3} \le |\theta| \le \pi \end{cases}$$



Show that the impulse response $h_d[n]$ of the band stop filter is given by

$$h_d[n] = \delta[n] - \frac{\sin\left(\frac{\pi}{3}n\right)}{\pi n} + \frac{\sin\left(\frac{\pi}{6}n\right)}{\pi n}$$

$$h_d[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} H_d(\theta)e^{jn\theta} = \frac{1}{2\pi}\left\{\int_{-\pi}^{-\pi/3} 1\cdot e^{jn\theta}d\theta + \int_{-\pi/6}^{\pi/6} 1\cdot e^{jn\theta}d\theta + \int_{\pi/3}^{\pi} 1\cdot e^{jn\theta}d\theta\right\}$$

$$= \frac{1}{2\pi}\left\{\left[\frac{e^{jn\theta}}{jn}\right]_{-\pi}^{-\pi/3} + \left[\frac{e^{jn\theta}}{jn}\right]_{-\pi/6}^{\pi/6} + \left[\frac{e^{jn\theta}}{jn}\right]_{\pi/3}^{\pi}\right\}$$

$$= \frac{1}{2\pi}\left\{\frac{e^{-j\frac{\pi}{3}n} - e^{-j\pi n}}{jn} + \frac{e^{j\frac{\pi}{6}n} - e^{-j\frac{\pi}{6}n}}{jn} + \frac{e^{j\pi n} - e^{j\frac{\pi}{3}n}}{jn}\right\}$$

$$= \frac{1}{\pi n}\left\{-\underbrace{\frac{(e^{j\frac{\pi}{3}n} - e^{-j\frac{\pi}{3}n})}{2j}}_{-\sin\frac{\pi}{3}n} + \underbrace{\frac{e^{j\frac{\pi}{6}n} - e^{-j\frac{\pi}{6}n}}{2j}}_{=\sin\frac{\pi}{6}n} + \underbrace{\frac{e^{j\pi n} - e^{-j\pi n}}{2j}}_{=\sin\pi n}\right\}$$

$$= \frac{\sin\pi n}{\pi n} - \frac{\sin\frac{\pi}{3}n}{\pi n} + \frac{\sin\frac{\pi}{6}n}{\pi n}$$

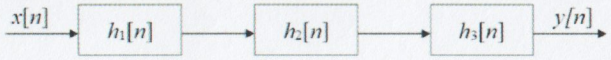
note:
$\frac{\sin n\pi}{\pi n} = 1$ when $n=0$
& zero elsewhere
$\therefore \delta[n]$

$$h_d[n] = \delta[n] - \frac{\sin\frac{\pi}{3}n}{\pi n} + \frac{\sin\frac{\pi}{6}n}{\pi n}$$

Topic 2: Tutorial questions (Level 3)

SOLO – Level 3

4. Three systems with the following impulse responses are connected in cascade: $h_1[n] = \delta(n) - \delta(n-1)$; $h_2[n] = g[n]$; $h_3[n] = u[n]$; where $u[n]$ is the unit step sequence and $\delta[n]$ is the unit sample sequence (i.e an impulse).



What is the impulse response $h_c(n)$ of the overall system?

$h_1[n] = \delta[n] - \delta[n-1] \Rightarrow H_1(z) = 1 - z^{-1}$

$h_3[n] = u[n] \Rightarrow H_3(z) = \frac{1}{1-z^{-1}}$

$y[n] = x[n] * h_1[n] * h_2[n] * h_3[n]$

$\frac{Y(z)}{X(z)} = H_1(z) \cdot H_2(z) \cdot H_3(z)$

$H_c(z) = (1 - z^{-1}) \cdot G(z) \cdot \frac{1}{1-z^{-1}}$

$H_c(z) = G(z)$

$\therefore h_c[n] = g[n]$

5. Show that the convolution of the two infinite duration sequences

$$p[n] = a^n u[n] \quad \text{and} \quad q[n] = b^n u[n]$$

for all n, where $u[n]$ is the unit step function and $a \neq b$, is given by

$$y[n] = \frac{a^{n+1} - b^{n+1}}{a - b}$$

$y[n] = p[n] * q[n] = \sum_{k=-\infty}^{\infty} a^k u[k] \cdot b^{n-k} u[n-k]$

$u[k] = 1$ when $k \geq 0$; $u[n-k] = 1$ when $n - k \geq 0$, i.e. $k \leq n$

$= \sum_{k=0}^{n} a^k b^{n-k}$

$= \sum_{k=0}^{n} \left(\frac{a}{b}\right)^k \cdot b^n$

$= b^n \sum_{k=0}^{n} \left(\frac{a}{b}\right)^k$

$y[n] = b^n \sum_{k=0}^{n} \left(\frac{a}{b}\right)^k$

$= b^n \left[1 + \left(\frac{a}{b}\right) + \left(\frac{a}{b}\right)^2 + \cdots + \left(\frac{a}{b}\right)^n\right]$

$= b^n \frac{1 - \left(\frac{a}{b}\right)^{n+1}}{1 - \frac{a}{b}}$

$= \frac{b^n}{b^{n+1}} \frac{b^{n+1} - a^{n+1}}{b - a} = \frac{b^n \times b}{b^{n+1}} \cdot \frac{a^{n+1} - b^{n+1}}{a - b}$

$$y[n] = \frac{a^{n+1} - b^{n+1}}{a - b}$$

Topic 2: Tutorial questions (Level 3)

SOLO – Level 3

6. Consider a system with an input x[n] and an output y[n] which satisfies the following difference equation:

$$y[n] = n\,y[n-1] + x[n]$$

If $x[n] = \delta[n]$, determine $y[n]$ for all values of n.

$x[n] = \delta[n]$ $\quad = 1$ at $n=0$

$n=0$ $\quad y[0] = 0\,y[-1] + x[0] = 1$

$n=1$ $\quad y[1] = 1\,y[0] + x[1]_{=0} = 1 \cdot 1$

$n=2$ $\quad y[2] = 2[1] + x[2]_{=0} = 2 \cdot 1$

$n=3$ $\quad y[3] = 3[2\times1] + x[3]_{=0} = 3 \cdot 2 \cdot 1$

$n=4$ $\quad y[4] = 4[3\times2\times1] + x[4]_{=0} = 4 \cdot 3 \cdot 2 \cdot 1$

$n=n$ $\quad y[n] = n \cdot (n-1)(n-2)(n-3)\cdots 1 = n!$

$$\boxed{y[n] = n! \quad n \geq 1}$$

Topic 2: Tutorial questions (Level 4)

SOLO – Level 4

1. A system L is known to be linear. The responses of L to inputs $x_1[n]$ and $x_2[n]$ are given by $y_1[n]$ and $y_2[n]$ respectively (see Figure 1).

   (a) Determine whether the system L could be time-invariant.

   (b) If the input x[n] to the system is $\delta[n]$, what is the system response y[n]?

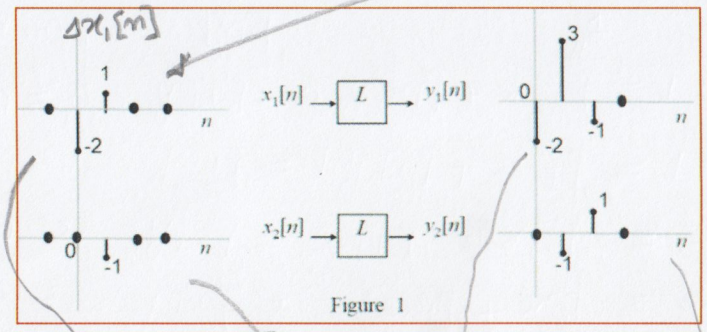


Figure 1

(a) $x_1[n]$ can be formulated from the graph as follows:

$x_1[n] = -x_2[n] + 2x_2[n+1]$

If the system is time invariant, then

$y_1[n] = -y_2[n] + 2y_2[n+1]$

$[0, 1, -1]$ $[-2, 2, 0]$

$y_1[n] = \{(0-2), (1+2), (-1+0)\}$

$= \{-2, 3, -1\}$

Same as the output of $y_1[n]$ shown in the figure

$\therefore$ L is time-invariant

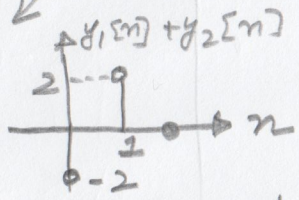
(b) add $x_1[n] + x_2[n] = -2\delta[n]$

$-2\delta[n]$ → L → $y_1[n] + y_2[n] = -2h[n]$

$h[n]$

$-2\delta[n] \xrightarrow{L} -2h[n]$

$\delta[n] \longrightarrow h[n]$



Divide it by -2, then you get $h[n]$

$\therefore h[n] = \{1, -1\}$

Topic 2: Tutorial questions (Level 4)

SOLO – Level 4

2. Consider the interconnection of LTI systems as shown in Figure 2:

   (a) Express the overall impulse response in terms of $h_1[n]$, $h_2[n]$, $h_3[n]$ and $h_4[n]$.

   (b) Determine *h[n]* when

   $h_1[n] = \{\frac{1}{2}\ \underset{\uparrow}{\frac{1}{4}}\ \frac{1}{2}\}$; $h_2[n] = h_3[n] = (n+1)\,u[n]$; $h_4[n] = \delta[n-2]$

   (c) Determine the response of the system in part(b) if

   $x[n] = \delta[n+2] + 3\delta[n-1] - 4\delta[n-3]$

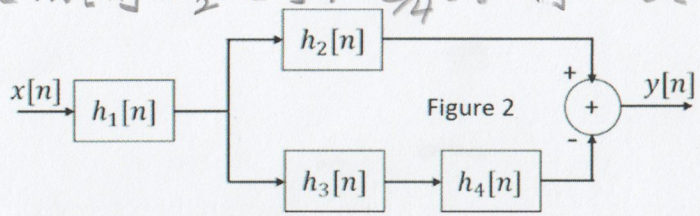


Figure 2

(a) $h[n] = h_1[n] * [h_2[n] - h_3[n] * h_4[n]]$

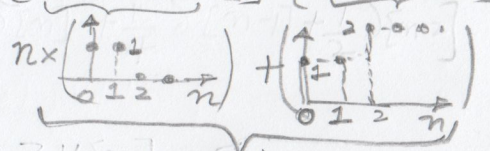
(b) $h_3[n] * h_4[n] = (n+1)u[n] * \delta[n-2]$

$= (n-1)u[n-2]$

$h_2[n] - h_3[n] * h_4[n]$

$= (n+1)u[n] - (n-1)u[n-2]$

$= n\{u[n] - u[n-2]\} + \{u[n] + u[n-2]\}$

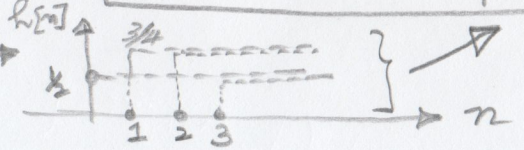


$= 2u[n] - \delta[n]$

Given (see (b)) $h_1[n] = \frac{1}{2}\delta[n] + \frac{1}{4}\delta[n-1] + \frac{1}{2}\delta[n-2]$

Hence $h[n] = \{\frac{1}{2}\delta[n] + \frac{1}{4}\delta[n-1] + \frac{1}{2}\delta[n-2]\} * \{2u[n] - \delta[n]\}$

$h[n] = \frac{1}{2}\delta[n] + \frac{5}{4}\delta[n-1] + 2\delta[n-2] + \frac{5}{2}u[n-3]$



$x[n] = \{1, 0, \underset{\uparrow n=0}{0}, 3, 0, -4\}$

$h[n] = \frac{1}{2}\delta[n] + \frac{5}{4}\delta[n-1] + 2\delta[n-2] + \frac{5}{2}u[n-3]$

$y[n] = x[n] * h[n]$

$= \{\frac{1}{2}, \frac{5}{4}, \underset{\uparrow n=0}{2}, \frac{25}{4}, \frac{13}{2}, 5, 2, 0, 0 \ldots\}$

Note

$H(z) = \left[\frac{1}{2} + \frac{1}{4}z^{-1} + \frac{1}{2}z^{-2}\right] \cdot \left[\frac{2}{1-z^{-1}} - 1\right]$

$= \frac{\frac{1}{2} + \frac{3}{4}z^{-1} + \frac{3}{4}z^{-2} + \frac{1}{2}z^{-3}}{1 - z^{-1}}$

$h[n] = \frac{1}{2}u[n] + \frac{3}{4}u[n-1] + \frac{3}{4}u[n-2] + \frac{1}{2}u[n-3]$

Topic 2: Tutorial questions (Level 4)

SOLO – Level 4

3. Find a difference equation to implement a system that has a unit impulse response:

$$h[n] = (\tfrac{1}{4})^n \cos(n\tfrac{\pi}{3})\, u[n]$$

Note: To find the difference equation, we must first find the z-transform response $H(\theta)$

Using the z-transform table

$$H(z) = \frac{1 - z^{-1}(\frac{1}{4})\cos(\frac{\pi}{3})}{1 - z^{-1}2(\frac{1}{4})\cos\frac{\pi}{3} + (\frac{1}{4})^2 z^{-2}} = \frac{1 - \frac{1}{8}z^{-1}}{1 - \frac{1}{4}z^{-1} + \frac{1}{16}z^{-2}} = \frac{Y(z)}{X(z)}$$

$$Y(z)\left[1 - \frac{1}{4}z^{-1} + \frac{1}{16}z^{-2}\right] = X(z)\left[1 - \frac{1}{8}z^{-1}\right]$$

$$\therefore y[n] = x[n] - \frac{1}{8}x[n-1] + \frac{1}{4}y[n-1] - \frac{1}{16}y[n-2]$$

4. An L[th] order moving average filter is a linear time invariant system that, for an input $x[n]$, produces the output $y[n] = \frac{1}{L+1}\sum_{k=0}^{L} x[n-k]$. Show that the frequency response of this system is $H(\theta) = \frac{1}{L+1} e^{-jL\theta/2} \frac{\sin[(L+1)\theta/2]}{\sin(\frac{\theta}{2})}$

If the input $x[n] = \delta[n]$, the response, by definition will be the impulse response $h[n]$

$$\therefore h[n] = \frac{1}{L+1}\sum_{k=0}^{L}\delta[n-k] \Rightarrow H(\theta) = \frac{1}{L+1}\sum_{k=0}^{L} e^{-jk\theta}$$

$$H(\theta) = \frac{1}{L+1}\left[1 + e^{-j\theta} + e^{-j2\theta} + e^{-j3\theta} + \cdots + e^{-jL\theta}\right] = \frac{1}{L+1}\cdot\frac{1 - e^{-j(L+1)\theta}}{1 - e^{-j\theta}}$$

(using Geometric series)

Factoring out $e^{-j\frac{(L+1)\theta}{2}}$ from the numerator and a term $e^{-j\frac{\theta}{2}}$ from the denominator, we have

$$H(\theta) = \frac{1}{L+1} e^{-jL\theta/2} \cdot \frac{e^{j\frac{(L+1)\theta}{2}} - e^{-j\frac{(L+1)\theta}{2}}}{e^{+j\theta/2} - e^{-j\theta/2}} = \frac{1}{L+1}\cdot e^{-\frac{jL\theta}{2}} \cdot \frac{\sin(L+1)\theta/2}{\sin\theta/2}$$

Topic 2: Tutorial questions (Level 5)

SOLO – Level 5

1. Consider a bandpass filter with a spectrum X(θ) as shown in Figure A. Show that x[n] can be formulated as:

$$x[n] = (k_p + k_q)\cos(n\theta_0) - k_p\cos(n\theta_u) - k_q\cos(n\theta_l)$$

$$k_p = \frac{1}{\pi n^2(\theta_u - \theta_0)}; \quad k_q = \frac{1}{\pi n^2(\theta_0 - \theta_l)}; \quad \theta = 2\pi\frac{f}{f_s}$$

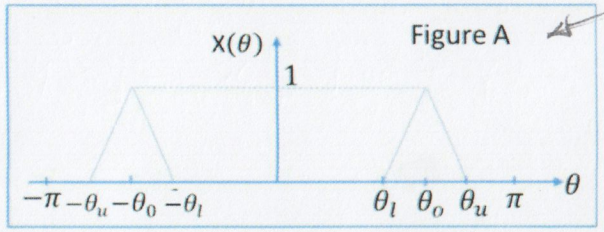
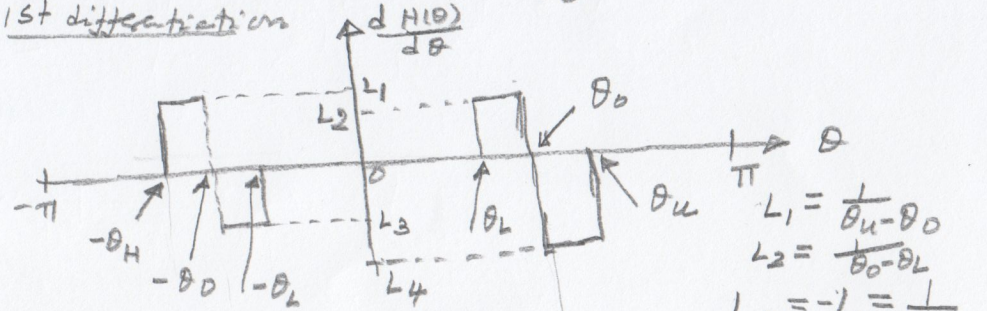


Figure A

note: $\theta_u \neq \theta_0$, $\theta_0 \neq \theta_L$

To find impulse response, we take IDTFT of $X(\theta)$. Use differentiation property of DTFT. $n\,h[n] \Leftrightarrow j\frac{dH(\theta)}{d\theta}$

1st differentiation $\frac{d\,H(\theta)}{d\theta}$



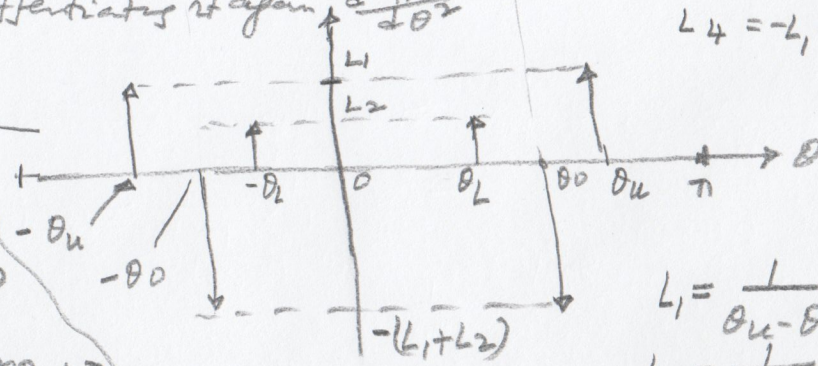
$L_1 = \frac{1}{\theta_u - \theta_0}$

$L_2 = \frac{1}{\theta_0 - \theta_L}$

$L_3 = -L_2 = \frac{1}{\theta_L - \theta_0}$

$L_4 = -L_1 = \frac{1}{\theta_0 - \theta_u}$

Differentiating it again $\frac{d^2H(\theta)}{d\theta^2}$



$L_1 = \frac{1}{\theta_u - \theta_0}$

$L_2 = \frac{1}{\theta_0 - \theta_L}$

Taking IDFT of $\frac{d^2H(\theta)}{d\theta^2} = \frac{1}{2\pi}\int_{-\pi}^{\pi}\frac{d^2H(\theta)}{d\theta^2}e^{jn\theta}\,d\theta$

$$= \frac{1}{2\pi}\int_{-\pi}^{\pi}\Big[L_2\{\delta(\theta-\theta_L)+\delta(\theta+\theta_L)\} + (L_1+L_2)\{\delta(\theta-\theta_0)+\delta(\theta+\theta_0)\} + L_1\{\delta(\theta-\theta_u)+\delta(\theta+\theta_u)\}\Big]e^{jn\theta}\,d\theta$$

$$= \frac{1}{2\pi}\Big\{L_2(e^{jn\theta_L}+e^{-jn\theta_L}) - (L_1+L_2)(e^{jn\theta_0}+e^{-jn\theta_0}) + L_1(e^{jn\theta_u}+e^{-jn\theta_u})\Big\}$$

$$= \frac{1}{\pi}\Big\{\underbrace{\frac{1}{\theta_u-\theta_0}}\cos n\theta_u + \frac{1}{\theta_0-\theta_L}\cos n\theta_L - \Big(\frac{1}{\theta_u-\theta_0}+\frac{1}{\theta_0-\theta_L}\Big)\cos n\theta_0\Big\}$$

To get $h[n]$ multiply the above by $\big(\frac{1}{n}\cdot\frac{1}{n} = \frac{1}{n^2}\big)$ ← [note $n\,h[n] \Leftrightarrow j\frac{dH(\theta)}{d\theta}$]

$$\therefore h[n] = \Big(\frac{1}{\theta_u-\theta_0}+\frac{1}{\theta_0-\theta_L}\Big)\cdot\frac{1}{n^2\pi}\cos n\theta_0 - \frac{1}{\theta_u-\theta_0}\cdot\frac{1}{n^2\pi}\cos n\theta_u - \frac{1}{\theta_0-\theta_L}\cdot\frac{1}{n^2\pi}\cos n\theta_L$$

(k_p, k_2, k_p, k_2)

Topic 2: Tutorial questions (Level 5)

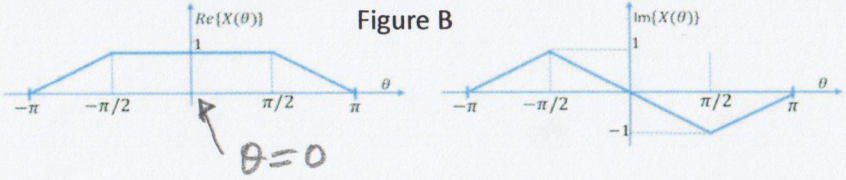
SOLO – Level 5

2. The centre of gravity of a sequence is defined by $c = \frac{\sum_{n=-\infty}^{\infty} n\,x[n]}{\sum_{n=-\infty}^{\infty} x[n]}$ and is used as a measure of the *time delay* of a sequence: Find (using the properties of DTFT)

   (a) an expression for *c* in terms of the DTFT of x[n].

   (b) the value of *c* for the sequence x[n] that has a DTFT as shown in Figure B.

Figure B



To find the value of c in terms of X(θ), first note that the denominator is simply the value of $X(\theta)\big|_{\theta=0}$ (ie DC value)

The Numerator $\sum_{n=-\infty}^{\infty} n\,x[n] = j\frac{d}{d\theta}X(\theta)\Big|_{\theta=0}$ $\quad \left[nx[n] \overset{DTFT}{\Leftrightarrow} j\frac{d}{d\theta}X(\theta)\right]$ see table in the notes

$$\therefore c = \frac{j\frac{d}{d\theta}X(\theta)\Big|_{\theta=0}}{X(\theta)\Big|_{\theta=0}}$$

From Figure B, we have $X(\theta)\big|_{\theta=0} = 1$

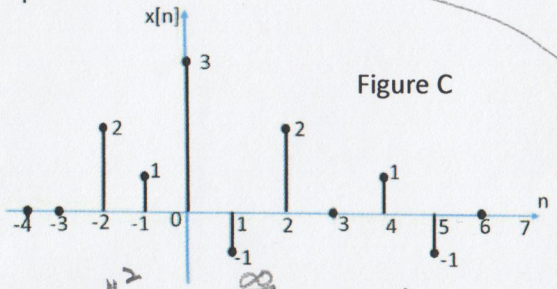
$$\frac{d}{d\theta}X(\theta) = \frac{d}{d\theta}Re[X(\theta)]\Big|_{\theta=0} + j\frac{d}{d\theta}Im(X(\theta))\Big|_{\theta=0}$$

$= 0$ (see Figure B gradient) $+ j\left(\frac{-1}{\pi/2}\right)$ see Figure B gradient

$$c = j\frac{\left(-\frac{2j}{\pi}\right)}{1} = \frac{2}{\pi}$$

3. For the sequence x[n] plotted in Figure C, evaluate the integral $\int_{-\pi}^{\pi}\left|\frac{d}{d\theta}X(\theta)\right|^2 d\theta$.

Figure C



$$\frac{1}{2\pi}\int_{-\pi}^{\pi}|X(\theta)|^2 d\theta = \sum_{n=-\infty}^{\infty}|x(n)|^2$$

Topic 2 Page 35

Using Parseval's Theorem

we have $n\,x[n] \overset{DTFT}{\Leftrightarrow} j\frac{d}{d\theta}X(\theta)$ $\therefore \frac{d}{d\theta}X(\theta) \Leftrightarrow -j\,n\,x[n]$

$$\therefore \int_{-\pi}^{\pi}\left|\frac{d}{d\theta}X(\theta)\right|^2 = 2\pi\sum_{n=-\infty}^{\infty}|-j\,n\,x[n]|^2 = 2\pi\sum_{n=-\infty}^{\infty}|n\,x[n]|^2 = (75)\times 2\pi$$

for x[n] see Figure C

ELEC3104: Digital Signal Processing

Topic 3: Digital Filter Fundamentals

Model Answers to Tutorial Questions

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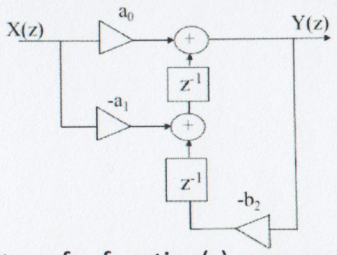
Term 3, 2021

Topic 3: Tutorial Questions (SOLO -Level 1)

SOLO – Level 1

1. A digital filter structure is shown below. Calculate the dc gain of the filter, in terms of the filter coefficients. (Tick the correct answer)



   (a) $1 + a_0 - a_1$

   ✓ (b) $\frac{a_0 - a_1}{1 + b_2}$

   (c) $\frac{a_0 - a_1}{1 - b_2}$

   (d) Non of the above

[see Page 13, Topic 3]

$H(z) = \frac{a_0 + a_1 z^{-1}}{1 + b_2 z^{-2}}$, $\therefore H(\theta) = \frac{a_0 - a_1 e^{-j\theta}}{1 + b_2 e^{-j2\theta}}$

DC gain $= H(\theta)\Big|_{\theta=0} = \frac{a_0 - a_1}{1 + b_2}$

2. Tick which of the following transfer function(s) represent high pass filters, where $0 < \alpha < 1$.

   ✓ (a) $H(z) = 1 - \alpha z^{-1}$ (b) $H(z) = \frac{1}{1 - \alpha z^{-1}}$

   (c) $H(z) = 1 + \alpha z^{-1}$ ✓ (d) $H(z) = \frac{1}{1 + \alpha z^{-1}}$

(b) & (c) are Lowpass filters (Topic 2: Page 34)

See Example 17: Topic 2 (Page 35)
(a) & (d) are Highpass filters

3. The Digital Filter $H(z) = \frac{2 + 2z^{-1}}{3 - z^{-1}}$; (Tick the correct answer)

   ✓ a) is a low-pass filter
   b) is a high-pass filter
   c) is an all-pass filter

$H(z) = \frac{2}{3} \frac{(1 + z^{-1})}{1 - \frac{1}{3} z^{-1}}$

$1 + z^{-1}$ is a Lowpass filter
& $\frac{1}{1 - \alpha z^{-1}}$ is a Lowpass filter
(Topic 2: Page 34)

4. A maximum phase filter has: (Tick the correct answer)

   a) All zeros and poles outside the unit circles
   b) All poles outside the unit circle
   ✓ c) All zeros outside the unit circle
   d) All poles and zeros inside the unit circle

See Page 32, Topic 3

Topic 3: Tutorial Questions (SOLO -Level 1)

SOLO – Level 1

5. Which of the following is/are linear phase filters: (Tick the correct answer)

✓(a) $H(z) = 0.25 + 0.5\,z^{-1} + 0.25\,z^{-2}$ ✓(b) $H(z) = z^{-4}$

✓(c) $H(z) = 1 - z^{-2}$ (d) $H(z) = 1 + 2z^{-1} + 2\,z^{-2} - z^{-3}$

(b) See Topic 3, Page 26
(a) impulse response are symmetric about
(c) its mid Point (Page 8, Topic 3)

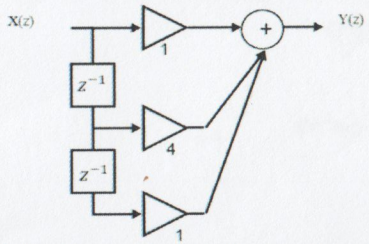
6. What can be said of the following filter ? (Tick the correct answer)

a) It has minimum phase and a finite impulse response.

b) It has linear phase and an infinite impulse response.

c) It has an infinite impulse response.

✓d) It has linear phase and a finite impulse response



$* H(z) = 1 + 4z^{-1} + z^{-2} = \frac{z^2 + 4z + 1}{z^2}$

$z = \frac{-4 \pm \sqrt{16-4}}{2}$

$z = -2 \pm \sqrt{3}$

one zero inside the unit circle & one outside the unit circle

(a) is wrong

* FIR filter
∴ (b) & (c) are wrong

* Symmetrical filter coefficients — Linear Phase

7. Which of the following system is linear, time invariant, causal and stable? (Tick the correct answer)

a) y[n] = |x[n]| ← non linear

b) y[n] = x[n] + n x[n+1] ← Time Variant

c) y[n] = x[-n] ← non causal

✓ *d)* $y[n] = \frac{1}{N}\sum_{k=0}^{N-1} x[n-k]$ (an FIR filter)

Topic 3: Tutorial Questions (Level 2)

SOLO – Level 2

1. Compute the 3dB bandwidth of the following filters:

$$H_1(z) = \frac{1-a}{1-az^{-1}};\quad H_2(z) = \frac{1-a}{2}\frac{1+z^{-1}}{1-az^{-1}},\qquad 0 < a < 1$$

Which filter has a smaller 3dB bandwidth?

$H_1(\theta) = \frac{1-a}{1-ae^{-j\theta}}$; $|H_1(\theta)|^2 = H_1(\theta)\cdot H_1^*(\theta)$

$= \frac{1-a}{1-ae^{-j\theta}}\cdot\frac{1-a}{1-ae^{j\theta}}$

$= \frac{(1-a)^2}{1+a^2-2a\cos\theta}$

$|H_1(\theta)|^2_{\theta=\theta_{c1}} = \frac{1}{2}$

$\frac{(1-a)^2}{1+a^2-2a\cos\theta_{c1}} = \frac{1}{2} \Rightarrow \cos\theta_{c1} = \frac{4a-a^2-1}{2a}$

$H_2(\theta) = \frac{1-a}{2}\frac{1+e^{-j\theta}}{1-ae^{-j\theta}}$

$|H_2(\theta)|^2 = \left(\frac{1-a}{2}\right)^2\frac{(1+\cos\theta)^2+\sin^2\theta}{(1-a\cos\theta)^2+a^2\sin^2\theta} = \left(\frac{1-a}{2}\right)^2\frac{2+2\cos\theta}{1+a^2-2a\cos\theta}$

$|H_2(\theta)|^2_{\theta=\theta_{c2}} = \frac{1}{2} \Rightarrow \frac{1}{2} = \left(\frac{1-a}{2}\right)^2\frac{2+2\cos\theta_{c2}}{1+a^2-2a\cos\theta_{c2}}$

$\therefore \cos\theta_{c2} = \frac{2a}{1+a^2}$

$\cos\theta_{c2} > \cos\theta_{c1}$ for $0<a<1$

$\therefore \theta_{c2} < \theta_{c1}$

2. Find the magnitude response of the filter *H(z)* and comment on your result.

$$H(z) = \frac{1+3z^2}{3+z^2}$$

Draw the canonic realization of the filter *H(z)*.

$H(\theta) = \frac{1+3e^{j2\theta}}{3+e^{j2\theta}}$ $\quad |H(\theta)|^2 = H(\theta)\cdot H^*(\theta)$

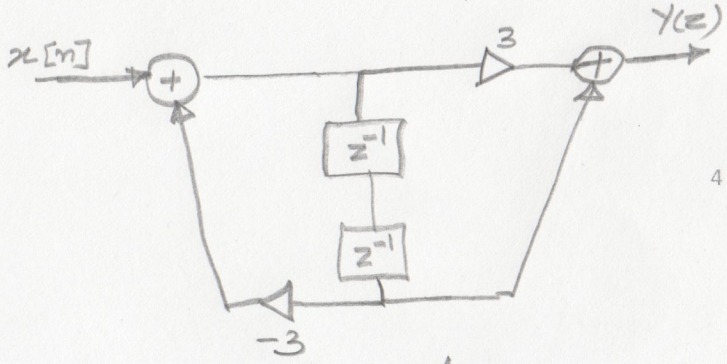
$|H(\theta)|^2 = \left(\frac{1+3e^{j2\theta}}{3+e^{j2\theta}}\right)\left(\frac{1+3e^{-j2\theta}}{3+e^{-j2\theta}}\right)$

$= \frac{(1+3\cos2\theta)^2+(3\sin2\theta)^2}{(3+\cos2\theta)^2+(\sin2\theta)^2} = \frac{10+6\cos2\theta}{10+6\cos2\theta}$

$= 1$

$\therefore |H(\theta)| = 1$

$H(z) = \frac{1+3z^2}{3+z^2} = \frac{3+z^{-2}}{1+3z^{-2}}$

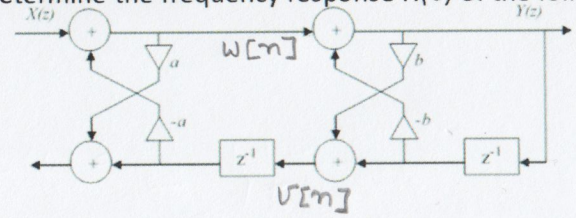


canonic form.

Topic 3: Tutorial Questions (Level 2)

SOLO – Level 2

3. Determine the frequency response $H(\theta)$ of the following filter:



$y[n] = w[n] - by[n-1]$ — ① Substitute ③ in ① to eliminate $w[n]$

$v[n] = y[n-1] + by[n]$ — ②

$\therefore y[n] = x[n] - av[n-1] - by[n-1]$

$w[n] = x[n] - av[n-1]$ — ③

$v[n-1] = y[n-2] + by[n-1]$

$y[n] = x[n] - ay[n-2] - ab\,y[n-1] - by[n-1]$

$\frac{Y(z)}{X(z)} = \frac{1}{1 + b(a+1)z^{-1} + az^{-2}}$

$\therefore H(\theta) = \frac{1}{1 + b(a+1)e^{-j\theta} + ae^{-j2\theta}}$

4. A two-pole lowpass filter has the transfer function

$$H(z) = \frac{k_0}{(1 - b_1 z^{-1})^2}$$

Determine the values of k_0 and b_1 such that the frequency response $H(\theta)$ satisfy the conditions

$$H(0) = 1; \quad |H(\theta)|^2_{\theta=\frac{\pi}{4}} = \frac{1}{2}$$

$H(\theta) = \frac{k_0}{(1 - b_1 e^{-j\theta})^2}$ · $|H(0)| = 1 \Rightarrow 1 = \frac{k_0}{(1-b_1)^2}$

$\therefore k_0 = (1-b_1)^2$

$|H(\theta)|^2 = \frac{k_0}{(1-b_1 e^{j\theta})^2} \cdot \frac{k_0}{(1-b e^{j\theta})^2} = k_0^2 \frac{1}{(1+b_1^2 - 2b_1\cos\theta)^2}$

$|H(\theta)|^2_{\theta=\pi/4} = \frac{1}{2} = \frac{k_0^2}{\left(1+b_1^2 - 2b_1\frac{1}{\sqrt{2}}\right)^2} \Rightarrow (1+b_1^2 - b_1\sqrt{2})^2 = 2k_0^2$

$\therefore (1+b_1^2 - b_1\sqrt{2}) = \sqrt{2}(1-b_1)$

$\therefore b_1 = 0.32$ & $k_0 = (1-0.32)^2 = 0.46$

5. Implement a parallel structure for the following $H(z)$:

$$H(z) = \frac{z^2 - 4z + 3}{z(z+0.5)^2}$$

$\frac{z^2-4z+3}{z(z+0.5)^2} = \frac{A}{z} + \frac{Bz+C}{(z^2+z+\frac{1}{4})}$

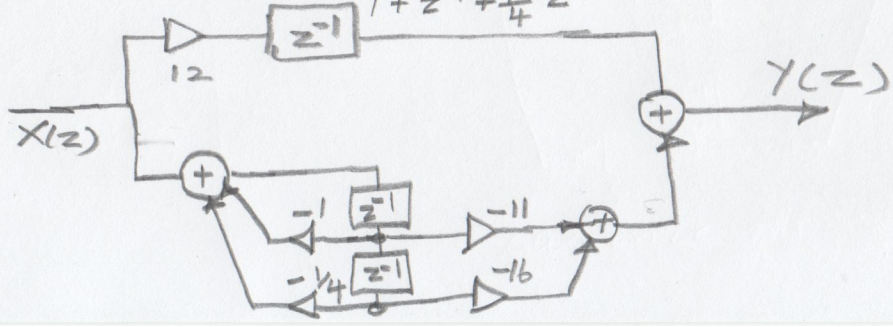
$\therefore z^2 - 4z + 3 = Az^2 + Az + \frac{A}{4} + Bz^2 + Cz$

By equating the coefficients, we obtain

$A+B=1$, $A+C=-4$, $\frac{A}{4} = 3$

$\therefore B = -11$, $C = -16$ & $A = 12$

$H(z) = 12 + \frac{-11z^{-1} - 16z^{-2}}{1 + z^{-1} + \frac{1}{4}z^{-2}}$



Topic 3: Tutorial Questions (Level 2)

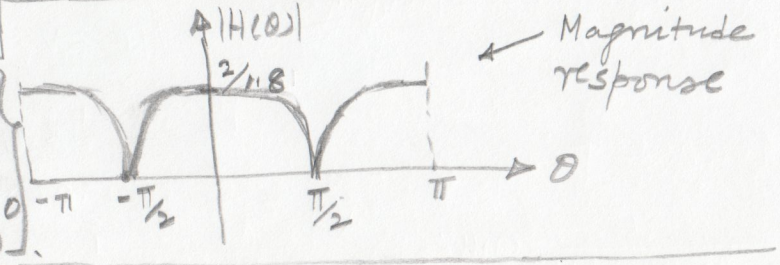
SOLO – Level 2

6. A notch filter is given by $H(z)=\frac{1+z^{-2}}{1+0.8z^{-2}}$. Determine the magnitude response at $\theta=0,\frac{\pi}{2}$, *and* π. Sketch the magnitude response in the interval $-\pi\le\theta\le\pi$.

$H(\theta)=\frac{1+e^{-j2\theta}}{1+0.8e^{-j2\theta}}$ at $\theta=\pi/2$ $\quad H(\theta)\big|_{\theta=\pi/2}=\frac{1+e^{-j\frac{\pi}{2}\times2}}{1+0.8e^{-j\frac{\pi}{2}\times2}}=\frac{1+\cos\pi-j\sin\pi}{1+0.8(\cos\pi-j\sin\pi)}=0$

DC gain $H(\theta)\big|_{\theta=0}=\frac{2}{1.8}$

at $\theta=\pi$, $H(\theta)\big|_{\theta=\pi}=\frac{1+e^{-j2\pi}}{1+0.8(\cos\pi-j\sin\pi)}=\frac{2}{1.8}$



Magnitude response

7. Determine and sketch the magnitude and phase response of the following filters

(a) $y[n]=\frac{1}{2}(x[n]-x[n-1])$ (b) $y[n]=x[n]-x[n-8]$

(c) $y[n]=x[n-4]$

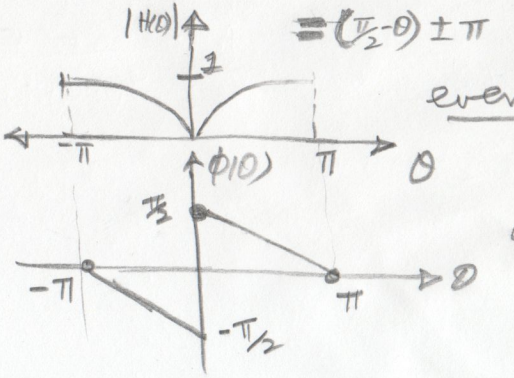
(a) $Y(z)=\frac{1}{2}X(z)-\frac{1}{2}z^{-1}X(z)\Rightarrow H(z)=\frac{1}{2}(1-z^{-1})$

$H(\theta)=\frac{1}{2}(1-e^{-j\theta})=\frac{1}{2}e^{-j\frac{\theta}{2}}\frac{e^{j\frac{\theta}{2}}-e^{-j\frac{\theta}{2}}}{(2j)}(2j)$

$=je^{-j\frac{\theta}{2}}\sin\theta/2=e^{j\frac{\pi}{2}}e^{-j\frac{\theta}{2}}\sin\theta/2=e^{j(\frac{\pi}{2}-\theta)}\underbrace{\sin\theta/2}_{B(\theta)}$

$|H(\theta)|=|\sin\theta/2|$; $\phi(\theta)=(\pi/2-\theta)$ if $B(\theta)\ge0$

$=(\pi/2-\theta)\pm\pi$ $B(\theta)<0$



even function

odd function

$y[n]=x[n]-x[n-8]$

$H(z)=\frac{Y(z)}{X(z)}=1-z^{-8}$

$H(\theta)=1-e^{-j8\theta}=e^{-j4\theta}[e^{+j4\theta}-e^{-j4\theta}]$

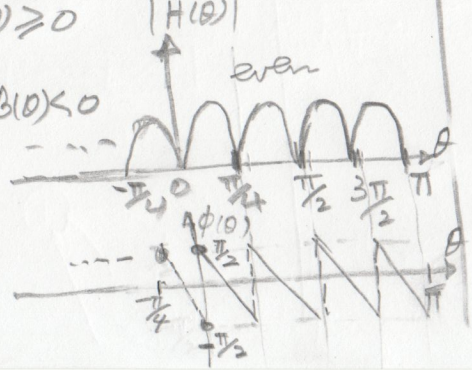
$H(\theta)=e^{-j4\theta}(2j)\sin4\theta$

$H(\theta)=e^{j(\frac{\pi}{2}-4\theta)}\underbrace{2\sin4\theta}_{=B(\theta)}$

$|H(\theta)|=|2\sin4\theta|$

$\phi(\theta)=\pi/2-4\theta$ $B(\theta)\ge0$

$=(\pi/2-4\theta)\pm\pi$ $B(\theta)<0$

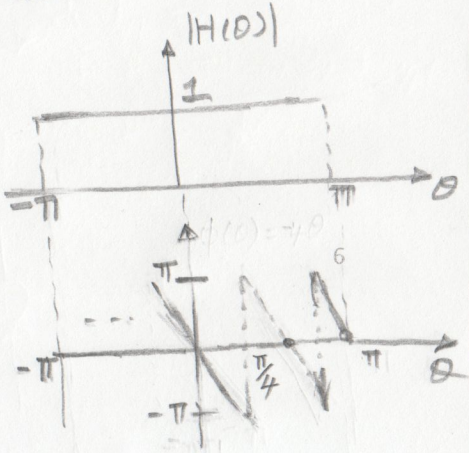


(c) $Y(z)=z^{-4}X(z)$

$H(z)=z^{-4}$; $H(\theta)=e^{-j4\theta}$

$|H(\theta)|=1$

$\phi(\theta)=-4\theta$

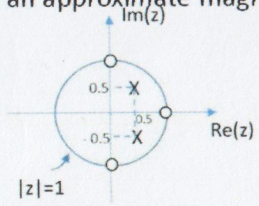


Topic 3: Tutorial Questions (Level 2)

SOLO – Level 2

8 (a) Determine the transfer function H(z) of a discrete - time system with the pole-zero map shown below:

(b) Using the pole-zero map , sketch an approximate magnitude response of the H(z) .



radius of the pole $(r) = \sqrt{(0.5)^2+(0.5)^2} = \frac{1}{\sqrt{2}}$

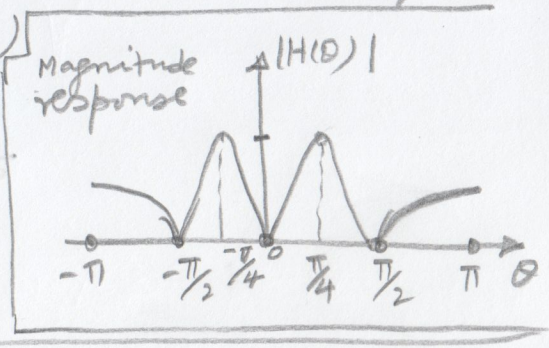
angle of the pole $= \theta$; $\tan\theta = 0.5/0.5 = 1$

$\theta = \pi/4$

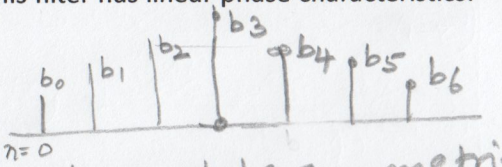
$$H(z) = \frac{(z-1)(z+j)(z-j)}{(z-re^{-j\frac{\pi}{4}})(z-re^{j\frac{\pi}{4}})}$$

$$= \frac{(z-1)(z^2+1)}{z^2 - \frac{2}{\sqrt{2}}\cos\frac{\pi}{4}z + \frac{1}{2}}$$

$$H(z) = \frac{(z-1)(z^2+1)}{z^2 - z + \frac{1}{2}}$$



9. A Finite impulse filter filter has seven filter coefficients b_0 to b_6, with $b_0 = 1$, $b_1 = 2$, $b_2 = 3$ and $b_3 = 4$.Determine b_4 to b_6 such t hat this filter has linear phase characteristics.



The coefficients must be symmetrial about b_3 for a linear phase filter $\Rightarrow$

$b_4 = b_2 = 3$ ✓

$b_5 = b_1 = 2$

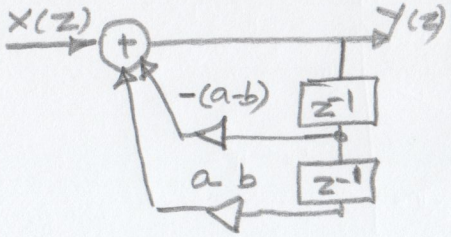
$b_6 = b_0 = 1$

10. Implement the following transfer function in cascade structure , in direct form II and in parallel structure. All coefficients are real.

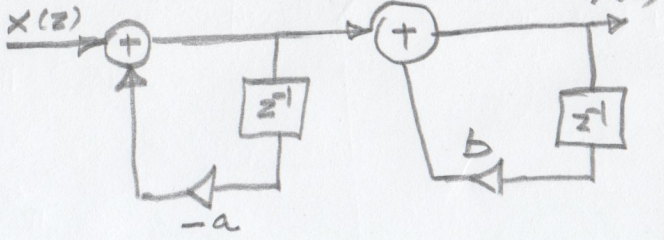
$$H(z) = \frac{1}{(1+az^{-1})(1-bz^{-1})}$$

Direct form II

$$H(z) = \frac{1}{1+(a-b)z^{-1}-abz^{-2}}$$



Cascade

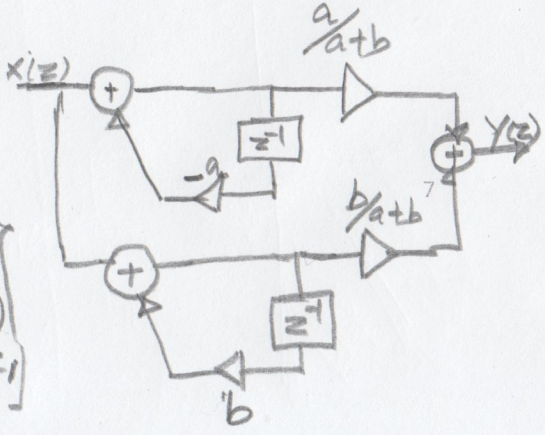


Parallel structure

$$H(z) = \frac{1}{(1+az^{-1})(1-bz^{-1})}$$

$$= \frac{A}{1+az^{-1}} + \frac{B}{1-bz^{-1}}$$

$$H(z) = \frac{(a/a+b)}{1+az^{-1}} + \frac{(b/a+b)}{1-bz^{-1}}$$



Topic 3: Tutorial questions (Level 3)

SOLO – Level 3

1. Show that both digital filters given below have the same magnitude response:

   (a) $y[n] = \sum_{i=-m}^{m} c_i\, x[n-i]$ (b) $y[n] = \sum_{i=-m}^{m} c_i\, x[n-m-i]$

   where $y[n]$ is the output, $x[n]$ is the input and c_i are the coefficients.

$$Y(z) = \sum_{i=-m}^{m} c_i\, X(z)\cdot z^{-i}$$

$$H_1(z) = \frac{Y(z)}{X(z)} = \sum_{i=-m}^{m} c_i\, z^{-i} \Rightarrow \boxed{|H_1(\theta)| = \left|\sum_{i=-m}^{m} c_i\, e^{-j\theta i}\right|}$$

$$Y(z) = \sum_{i=-m}^{m} c_i\, X(z)\cdot z^{-m}\cdot z^{-i}$$

$$H_2(z) = \frac{Y(z)}{X(z)} = \sum_{i=-m}^{m} c_i\, z^{-m}\cdot z^{-i}$$

$$H_2(\theta) = \sum_{i=-m}^{m} c_i\, e^{-j\theta i}\cdot e^{-j\theta m}$$

$$\boxed{|H_2(\theta)| = \left|\sum_{i=-m}^{m} c_i\, e^{-j\theta i}\right|\cdot\left|e^{-j\theta m}\right|}, \quad \left|e^{-j\theta m}\right| = 1$$

$$\therefore \underline{|H_1(\theta)| = |H_2(\theta)|}$$

2. Consider the following causal IIR transfer function:

$$H(z) = \frac{2z^3 - 4z^2 + 9}{(z-3)(z^2+z+0.5)}$$

Is *H(z)* a stable function? If it is not stable, find a stable transfer function *G(z)* such that $|G(\theta)| = |H(\theta)|$. Is there any transfer function having the same magnitude response as *H(z)*?

Since H(z) has a pole at z=3 (outside the unit circle), the given transfer function is unstable. To construct a stable transfer function having the same magnitude response consider another transfer function:

$$G(z) = H(z)\cdot A(z) \quad \leftarrow \text{Make } A(z) \text{ as an all-pass filter}$$

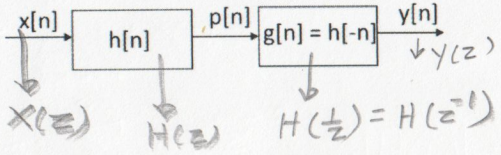
$$= \frac{2z^3-4z^2+9}{(z^2+z+0.5)(z-3)}\cdot\frac{z-3}{1-3z} \quad \leftarrow \text{all-pass filter (Magnitude response is constant or 1)}$$

$$\therefore |G(\theta)| = |H(\theta)||A(\theta)| \Rightarrow |G(\theta)| = |H(\theta)|$$

Topic 3: Tutorial questions (Level 3)

SOLO – Level 3

3. Show that if $x[n]$ is a real valued signal, the structure shown below has a zero-phase response; where $h[n]$ is the impulse response of the filter H(z) given by; $H(z) = (\frac{1}{2} + z^{-1} + \frac{1}{2}z^{-2})$.



4. The difference equation of a digital filter is given by

$$y[n] = (1/8)\{x[n] - x(n-8)\} - y(n-2)$$

(a) Find the transfer function for the above filter.

(b) Using the transfer function obtained in part (a) above, draw the pole zero map for the filter and sketch the corresponding magnitude response.

$\frac{Y(z)}{X(z)} = H(z) \cdot H(z^{-1})$ or $\frac{Y(\theta)}{X(\theta)} = H(\theta) \cdot H^*(\theta)$

$H(\theta) = \frac{1}{2} + e^{-j\theta} + \frac{1}{2}e^{-j2\theta}$; $H^*(\theta) = \frac{1}{2} + e^{j\theta} + \frac{1}{2}e^{j2\theta}$

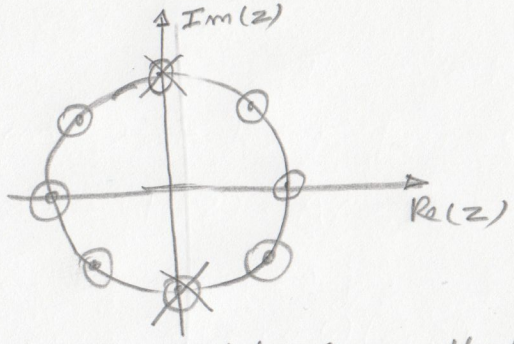
$\frac{Y(\theta)}{X(\theta)} = \left(\frac{1}{2} + e^{-j\theta} + \frac{1}{2}e^{-j2\theta}\right)\left(\frac{1}{2} + e^{j\theta} + \frac{1}{2}e^{j2\theta}\right)$

$= \frac{3}{2} + (e^{j\theta} + e^{-j\theta}) + \frac{1}{4}(e^{j2\theta} + e^{-j2\theta})$

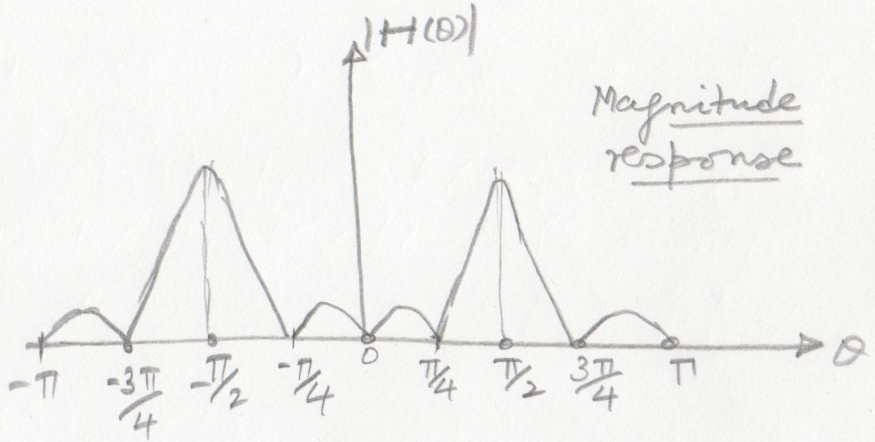
$\frac{Y(\theta)}{X(\theta)} = \frac{3}{2} + 2\cos\theta + \frac{1}{2}\cos 2\theta$ ← Zero phase

↑ Real Part only

(a) $H(z) = \frac{Y(z)}{X(z)} = \frac{1}{8}\frac{(1 - z^{-8})}{1 + z^{-2}}$ — 8 zeros; Two poles



Two zeros will be cancelled by TWO poles



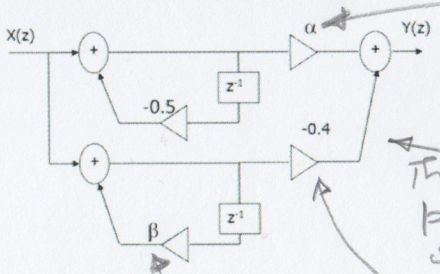
Topic 3: Tutorial questions (Level 3)

SOLO – Level 3

5. The following figure shows an IIR filter structure, and its transfer function given by

$$H(z) = \frac{Y(z)}{X(z)} = \frac{3z(5z-2)}{(z+0.5)(3z-1)}$$

Determine the values of the Multiplier Coefficients α and β.



This is a parallel structure

$$H(z) = \frac{Y(z)}{X(z)} = \frac{3z(5z-2)}{(z+0.5)(3z-1)} = \frac{(5-2z^{-1})}{(1+0.5z^{-1})(1-\frac{1}{3}z^{-1})}$$

$$\frac{5-2z^{-1}}{(1+\frac{1}{2}z^{-1})(1-\frac{1}{3}z^{-1})} = \frac{A}{1+\frac{1}{2}z^{-1}} + \frac{B}{1-\frac{1}{3}z^{-1}}$$

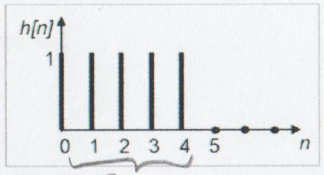
$\beta = \frac{1}{3}$

$\alpha = \frac{27}{5}$

$$A = \frac{5-2(-2)}{1-\frac{1}{3}(-2)} = \frac{27}{5}$$

$$B = \frac{5-2(3)}{1+\frac{1}{2}(3)} = -\frac{2}{5}$$

6. A FIR filter has an impulse response as shown below:



(a) Show that the frequency response of this filter is given by

$$H(\theta) = e^{-j(2\theta)} \frac{\sin\left(\frac{5}{2}\theta\right)}{\sin\left(\frac{\theta}{2}\right)}$$

(b) Draw an implementation of this filter as a cascade structure.

(a) $H(z) = 1 + z^{-1} + z^{-2} + z^{-3} + z^{-4}$

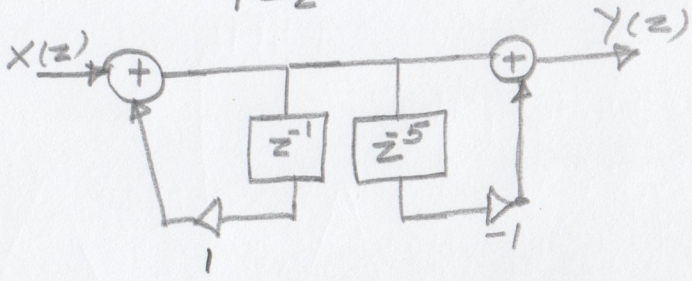
$$H(z) = \frac{1-z^{-5}}{1-z^{-1}}$$

$$H(\theta) = \frac{1-e^{-5\theta j}}{1-e^{-j\theta}} = \frac{e^{-j\frac{5}{2}\theta}}{e^{-j\frac{1}{2}\theta}} \frac{(e^{j\frac{5}{2}\theta} - e^{-j\frac{5}{2}\theta})}{(e^{-j\frac{1}{2}\theta} - e^{-j\frac{1}{2}\theta})}$$

$$H(\theta) = e^{-j2\theta} \frac{\sin\frac{5\theta}{2}}{\sin\frac{\theta}{2}}$$

(b)

$$H(z) = \frac{1-z^{-5}}{1-z^{-1}}$$

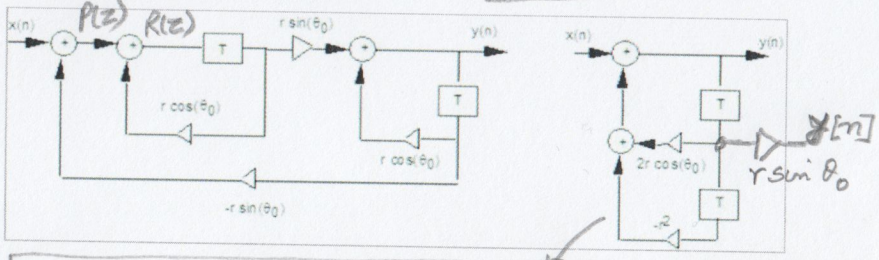


Cascade structure

Topic 3: Tutorial questions (Level 3)

SOLO – Level 3

7. Show that the systems (i) and (ii) shown below are equivalent:



$$R(z) = X(z) + z^{-1}Y(z)(-r\sin\theta_0) + (r\cos\theta_0)R(z)z^{-1}$$

$$R(z)(1 - r\cos\theta_0 z^{-1}) = X(z) - r\cos\theta_0 z^{-1}Y(z) \quad \text{— (1)}$$

$$Y(z) = R(z)(r\sin\theta_0)z^{-1} + Y(z)\,r\cos\theta_0\,z^{-1}$$

$$R(z) = \frac{Y(z)[1 - r\cos\theta_0 z^{-1}]}{r\sin\theta_0 z^{-1}} \quad \text{---- (2)}$$

From (1) & (2)

$$\frac{X(z) - r\sin\theta_0 z^{-1}Y(z)}{1 - r\cos\theta_0 z^{-1}} = \frac{Y(z)[1 - r\cos\theta_0 z^{-1}]}{r\sin\theta_0 z^{-1}}$$

$$\frac{Y(z)}{X(z)} = \frac{r\sin\theta_0 z^{-1}}{1 - 2r\cos\theta_0 z^{-1} + r^2 z^{-2}}$$

Both are equivalent!

$$\therefore \frac{Y(z)}{X(z)} = \frac{r\sin\theta_0 z^{-1}}{1 - 2r\cos\theta_0 z^{-1} + r^2 z^{-2}}$$

8. A third order FIR filter has a transfer function G(z) given by

$$G(z) = (9 + 6z^{-1} - 8z^{-2})(2 + 5z^{-1}).$$

From G(z), determine the transfer function of a FIR filter whose magnitude response is identical to that of G(z) and has

(a) a minimum phase response
(b) a maximum phase response

$$G(z) = (3 - 2z^{-1})(3 + 4z^{-1})(2 + 5z^{-1})$$

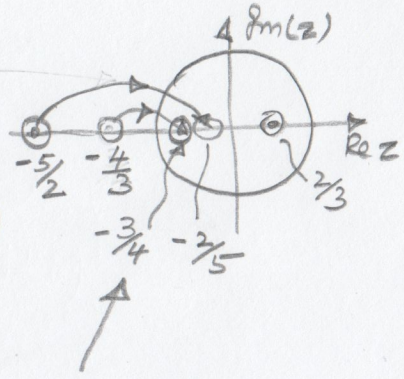
(a) $$G(z) = 18\left(1 - \tfrac{2}{3}z^{-1}\right)\left(1 + \tfrac{4}{3}z^{-1}\right)\left(1 + \tfrac{5}{2}z^{-1}\right)$$

($|z| < 1$, $|z| > 1$, $|z| > 1$)

$$G(z) = 60\left(1 - \tfrac{2}{3}z^{-1}\right)\left(1 + \tfrac{3}{4}z^{-1}\right)\left(1 + \tfrac{2}{5}z^{-1}\right)$$

($|z| < 1$, $|z| < 1$, $|z| < 1$)

For minimum phase reflect the zeros inside the unit circle

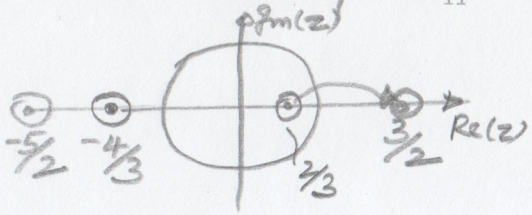


(b) $$G(z) = 12\left(1 - \tfrac{3}{2}z^{-1}\right)\left(1 + \tfrac{4}{3}z^{-1}\right)\left(1 + \tfrac{5}{2}z^{-1}\right)$$

($|z| > 1$, $|z| > 1$, $|z| > 1$)

all the zeros are outside the unit circle

∴ G(z) has maximum phase response



Topic 3: Tutorial questions (Level 3)

SOLO – Level 3

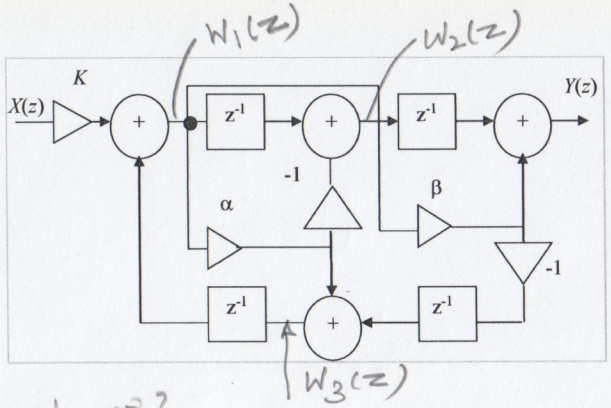
9. Analyse the digital structure shown on the right:

(a) Determine its transfer function $H(z)$ =(Y(z))/(X(z))

(b) What should be the value of the multiplier coefficient K so that the transfer function *H(z)* that you obtained in (a) above has a unity gain at $\theta = 0$?

(c) What should be the value of the multiplier coefficient *K* so that *H*(*z*) has a unity gain at $\theta = \pi$?

(d) Is there a difference between these two values of *K* obtained in Question (b) and (c)? If not, why not?



(a) $W_1(z) = K X(z) + W_3(z) z^{-1}$ — ①

$W_2(z) = W_1(z) z^{-1} + (-\alpha) W_1(z)$ — ②

$W_3(z) = \alpha W_1(z) - \beta W_1(z) z^{-1}$ — ③

$Y(z) = W_2(z) z^{-1} + \beta W_1(z)$ — ④

From ① & ③ $W_1(z) = K X(z) + z^{-1}(\alpha - \beta z^{-1}) W_1(z)$

$\therefore W_1(z) = \dfrac{K X(z)}{1 - \alpha z^{-1} + \beta z^{-2}}$

From ② & ④ $Y(z) = [W_1(z) z^{-1} - \alpha W_1(z)] z^{-1} + \beta W_1(z)$

$Y(z) = \{[z^{-1} - \alpha] z^{-1} + \beta\} W_1(z)$

$Y(z) = K X(z) \dfrac{z^2 - \alpha z^{-1} + \beta}{1 - \alpha z^{-1} + \beta z^{-2}}$

$\therefore \dfrac{Y(z)}{X(z)} = \dfrac{K[\beta - \alpha z^{-1} + z^{-2}]}{1 - \alpha z^{-1} + \beta z^{-2}}$

(b) $H(z) = K \dfrac{[\beta - \alpha z^{-1} + z^{-2}]}{1 - \alpha z^{-1} + \beta z^{-2}}$

$\therefore H(\theta)\big|_{\theta=0} = K \dfrac{\beta - \alpha + 1}{1 - \alpha + \beta} \Rightarrow$ $\boxed{K = \dfrac{1 - \alpha + \beta}{\beta - \alpha + 1} = 1}$

$H(\theta)\big|_{\theta=0} = 1$ (given)

(c) $H(\theta) = K \cdot \dfrac{\beta - \alpha e^{-j\theta} + e^{-j2\theta}}{1 - \alpha e^{-j\theta} + \beta e^{-j2\theta}}$; $H(\theta)\big|_{\theta=\pi} = 1$ (given)

$1 = K \dfrac{\beta - \alpha e^{-j\pi} + e^{-j2\pi}}{1 - \alpha e^{-j\pi} + \beta e^{-j2\pi}} = K\left(\dfrac{\beta - \alpha + 1}{1 - \alpha + \beta}\right)$

$\therefore \boxed{K = 1}$

(d) $H(0) = 1$

$H(\pi) = 1$

$H(z)$ is an all-pass function with a constant magnitude at all values of θ.

Topic 3: Tutorial questions (Level 3)

SOLO – Level 3

10. (a) Calculate the magnitude and phase responses of the following FIR filter:

$$y[n] = \frac{1}{4}[x[n] - 2x[n-1] + x[n-2]]$$

(b) Sketch the magnitude and phase responses.

(c) Show that the above FIR filter has a constant group delay.

(a) $H(z) = \frac{1}{4}(1 - 2z^{-1} + z^{-2})$

$H(\theta) = \frac{1}{4}(1 - 2e^{-j\theta} + e^{-j2\theta})$

$= \frac{1}{4} e^{-j\theta}(e^{j\theta} - 2 + e^{-j\theta})$

$= \frac{1}{4} e^{-j\theta}(2\cos\theta - 2)$

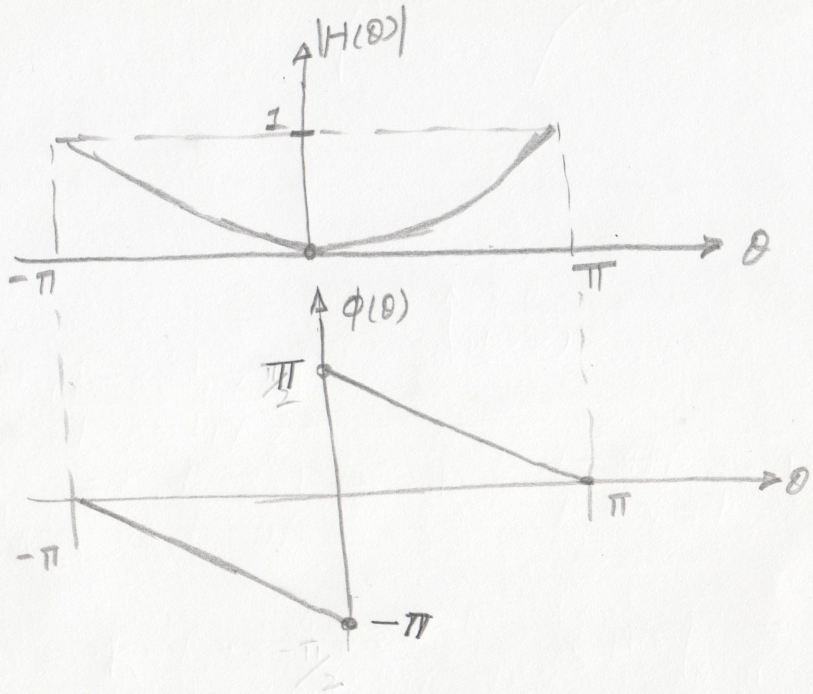
$= e^{-j\theta} \cdot \frac{1}{2}(\cos\theta - 1)$

$B(\theta)$

$|H(\theta)| = |B(\theta)| = \left|\frac{1}{2}(\cos\theta - 1)\right|$ is always negative

$\phi(\theta) = -\theta \pm \pi \quad B(\theta) \leq 0$

(b)



(c) group delay $\tau_g = -\frac{d\phi(\theta)}{d\theta}$

$= -\frac{1}{d\theta}(-\theta \pm \pi)$

$= 1$

Constant

Note: $\int \theta e^{jk\theta} d\theta = \frac{e^{jk\theta}}{k^2}(1-jk\theta)$ ← This will be given for Q1

Topic 3: Tutorial questions (Level 4)

SOLO – Level 4

1. The frequency response of an ideal differentiator is given by

$$H(\theta) = j\theta e^{-j\theta\tau} \quad -\pi \le \theta \le \pi$$

This response is periodic with period 2π. The quantity τ is the delay of the system in samples.

(a) Sketch the magnitude and phase responses of the system for $-\pi \le \theta \le \pi$.

(b) Find the impulse response *h[n]* of the system as a sum of a sine and cosine functions.

$H(\theta) = j\theta e^{-j\theta\tau} \quad -\pi \le \theta \le \pi$

$|H(\theta)| = \theta \quad 0 \le \theta \le \pi$

$= -\theta \quad -\pi < \theta < 0$

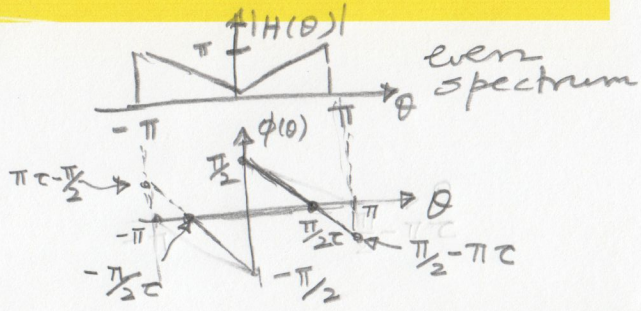
Magnitude response

$H(\theta) = e^{j\pi/2} \cdot \theta \cdot e^{-j\theta\tau}$, $e^{j\pi/2} = j$

$H(\theta) = \theta e^{-j(\theta\tau - \pi/2)}$

$\therefore \phi(\theta) = -\theta\tau + \pi/2 \quad \theta \ge 0$

$= -\theta\tau - \pi/2 \quad \theta < 0$



(b)

$h[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} H(\theta) e^{j\theta n} d\theta$

$= \frac{1}{2\pi}\int_{-\pi}^{\pi} j\theta e^{-j\theta\tau} e^{j\theta n} d\theta$

$h[n] = \frac{j}{2\pi}\int_{-\pi}^{\pi} \theta e^{j(n-\tau)\theta} d\theta$

$h[n] = \frac{j}{2\pi}\left[\frac{e^{j(n-\tau)\theta}}{(n-\tau)^2}(1-j(n-\tau)\theta)\right]_{-\pi}^{\pi}$

$= \frac{j}{2\pi(n-\tau)^2}\left[\{e^{j(n-\tau)\pi}(1-j(n-\tau)\pi\} - \{e^{j(n-\tau)(-\pi)}(1-j(n-\tau)(-\pi)\}\right]$

$= \frac{1}{2\pi(n-\tau)^2}\left[\{e^{j(n-\tau)\pi} - e^{-j(n-\tau)\pi}\} - \{j(n-\tau)\pi e^{j(n-\tau)\pi} - j(n-\tau)\pi e^{-j(n-\tau)\pi}\}\right]$

$h[n] = -\frac{\sin[\pi(n-\tau)]}{\pi(n-\tau)^2} + \frac{\cos[\pi(n-\tau)]}{n-\tau} \quad n \ne \tau$

2. Determine the magnitude and phase response of the multipath channel *y[n] = x[n] + x[n-M]*. At what frequencies $|H(\theta)| = 0$?

$y[n] = x[n] + x[n-M] \Rightarrow Y(z) = X(z) + X(z)z^{-M}$

$H(z) = 1 + z^{-M} \Rightarrow H(\theta) = 1 + e^{-j\theta M} = (1+\cos M\theta) + j\sin(M\theta)$

$|H(\theta)| = \sqrt{(1+\cos M\theta)^2 + (\sin M\theta)^2} = \sqrt{2+2\cos(M\theta)}$

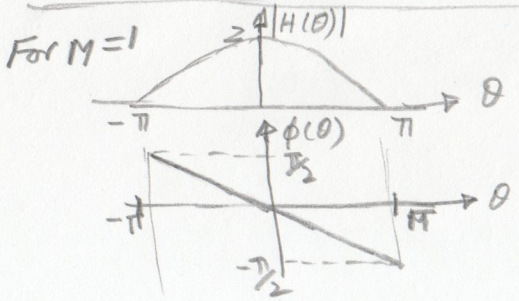
$\phi(\theta) = \tan^{-1}\left\{\frac{-\sin M\theta}{1+\cos M\theta}\right\}$; let $M = 1$

$\phi(\theta) = \tan^{-1}\frac{-\sin\theta}{1+\cos\theta}$

$= \tan^{-1}\left\{\frac{-2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{2+2\cos^2\frac{\theta}{2}-1}\right\}$

$\phi(\theta) = -\theta/2$

For $M=1$



$|H(\theta)| = 0 \Rightarrow \cos M\theta = -1$

$M\theta = \pi$

$\therefore \theta = \pi/M$

Topic 3: Tutorial questions (Level 4)

SOLO – Level 4

3. Let a causal linear time invariant discrete-time system be characterised by a real impulse response $h[n]$ with a DTFT of $H(\theta)$. Consider the system shown in Figure A, where $x[n]$ is a finite length sequence. Determine the frequency response of the overall system $G(\theta)$ $\{G(z) = {}^{Y(z)}/_{X(z)}\}$ in terms of $H(\theta)$ and show that it has a zero-phase response.

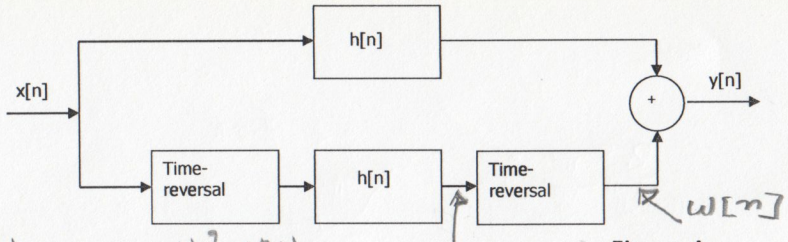


Figure A

$h[n] \xrightarrow{DTFT} H(\theta)$

$v[n] = x[-n] * h[n]$

$w[n] = v[-n] = x[n] * h[n]$ $\quad = w[n]$

$y[n] = h[n] * x[n] + x[n] * h[-n]$

$y[n] = \{h[n] + h[-n]\} * x[n]$

$Y(z) = [H(z) + H(z^{-1})] X(z)$

$\therefore G(z) = \frac{Y(z)}{X(z)} = H(z) + H(z^{-1})$

$G(\theta) = H(\theta) + H^*(\theta)$

$= (a + jb) + (a - jb) = 2a$

$G(\theta)$ is real

$\therefore G(\theta)$ has zero phase.

4. An analogue system may be implemented as shown in Figure B. Using the transformation $\frac{1}{s} \rightarrow k\frac{1+z^{-1}}{1-z^{-1}}$ a discrete time system may be obtained; where k is a constant.

   (a) Draw the digital structure of the system, using the minimum number of multipliers and delay units.

   (b) Write the difference equation that governs the behaviour of the system in the form $y[n] = 2\cos[\theta_0]\, y[n-1] - y[n-2]$ and find the value of $\cos\theta_0$.

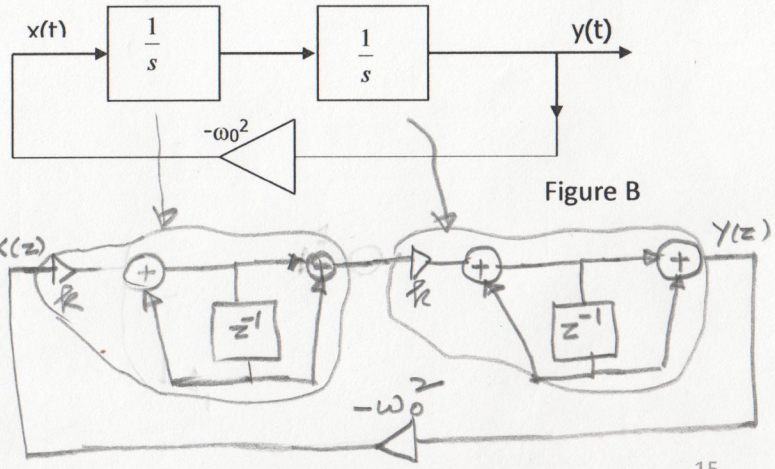


Figure B

(a)

Number of delays = 2

(b) $\frac{Y(z)}{X(z)} = k\frac{1+z^{-1}}{1-z^{-1}} \cdot k\frac{1+z^{-1}}{1-z^{-1}} = k^2\frac{1+2z^{-1}+z^{-2}}{1-2z^{-1}+z^{-2}}$ — ①

$X(z) = -\omega_0^2 Y(z) \Rightarrow x[n] = -\omega_0^2 y[n]$ — ②

$y[n] = k^2[x[n] + 2x[n-1] + x[n-2]] + 2y[n-1] - y[n-2]$

$y[n] = k^2\{-\omega_0^2 y[n] - 2\omega_0^2 y[n-1] - \omega_0^2 y[n-2]\} + 2y[n-1] - y[n-2]$

$y[n]\{1 + k^2\omega_0^2\} = 2(1 - k^2\omega_0^2)\, y[n-1] - (1 + k^2\omega_0^2)\, y[n-2]$

$\therefore y[n] = 2\frac{1 - k^2\omega_0^2}{1 + k^2\omega_0^2} y[n-1] - y[n-2] \Rightarrow$ $\frac{1 - k^2\omega_0^2}{1 + k^2\omega_0^2} = \cos\theta_0$

$y[n] = 2\cos\theta_0\, y[n-1] - y[n-2]$

$\left[\frac{d}{dx}\tan^{-1}u = \frac{1}{1+u^2}\frac{du}{dx}\right]$

Topic 3: Tutorial questions (Level 5)

SOLO – Level 5

1. (a) Find the group delay for each of the following system, where $|\alpha| < 1$

(i) $H_1(\theta) = 1 - \alpha e^{-j\theta}$ (ii) $H_2(\theta) = \frac{1}{1-\alpha e^{-j\theta}}$

(b) using the results obtained in (a) above, show that the group delay of an all-pass filter with a transfer function $H_0(z) = \frac{z^{-1}-\alpha}{1-\alpha z^{-1}}$ is $\frac{1-\alpha^2}{1+\alpha^2-2\alpha\cos(\theta)}$

(i) $H_1(\theta) = 1-\alpha e^{-j\theta}$; $H(\theta) = (1-\alpha\cos\theta) + j\alpha\sin\theta$

$\phi_1(\theta) = \tan^{-1}\frac{\alpha\sin\theta}{1-\alpha\cos\theta}$

The group delay $\tau_1(\theta) = -\frac{d}{d\theta}\phi_1(\theta) = -\frac{1}{1+\left(\frac{\alpha\sin\theta}{1-\alpha\cos\theta}\right)^2}\frac{d}{d\theta}\left(\frac{\alpha\sin\theta}{1-\alpha\cos\theta}\right)$

$\tau_1(\theta) = \frac{-1}{1+\left(\frac{\alpha\sin\theta}{1-\alpha\cos\theta}\right)^2}\cdot\frac{(1-\alpha\cos\theta)\alpha\cos\theta - (\alpha\sin\theta)^2}{(1-\alpha\cos\theta)^2}$

which after simplification

$$\tau_1(\theta) = \frac{\alpha^2-\alpha\cos\theta}{1+\alpha^2-2\alpha\cos\theta}$$

$H_0(z) = \frac{-\alpha\left[1-\frac{1}{\alpha}z^{-1}\right]}{1-\alpha z^{-1}} = -\alpha\cdot\underbrace{\left(1-\frac{1}{\alpha}z^{-1}\right)}_{\text{Part (i) above}}\cdot\underbrace{\frac{1}{1-\alpha z^{-1}}}_{\text{Part (ii) above}}$

The group delay of $H_0(z)$ is $\tau_3(\theta) = -\alpha\cdot\frac{\left(\frac{1}{\alpha}\right)^2-\frac{1}{\alpha}\cos\theta}{1+\left(\frac{1}{\alpha}\right)^2-2\left(\frac{1}{\alpha}\right)\cos\theta} + -\frac{(\alpha^2-\alpha\cos\theta)}{1+\alpha^2-2\alpha\cos\theta}$

$\leftarrow$ simplify

$$\tau_3(\theta) = \frac{1-\alpha^2}{1+\alpha^2-2\alpha\cos\theta}$$

(ii) Having found the group delay for $H_1(\theta) = 1-\alpha e^{-j\theta}$, the group delay for $H_2(\theta)$ is the inverse of $H_1(\theta)$

$\therefore \phi_2(\theta) = -\phi_1(\theta)$

$\therefore \tau_2(\theta) = -\tau_1(\theta) = -\frac{(\alpha^2-\alpha\cos\theta)}{1+\alpha^2-2\alpha\cos\theta}$

2. Find the minimum phase system that has a magnitude response given by

$$|H(\theta)|^2 = \frac{\frac{5}{4}-\cos\theta}{\frac{10}{9}-\frac{2}{3}\cos\theta}$$

To solve this problem consider $|H(\theta)|^2$ in terms of complex exponentials:

$$|H(\theta)|^2 = \frac{\frac{5}{4}-\frac{1}{2}e^{j\theta}-\frac{1}{2}e^{-j\theta}}{\frac{10}{9}-\frac{1}{3}e^{j\theta}-\frac{1}{3}e^{-j\theta}} \quad \left\{\cos\theta = \frac{e^{j\theta}+e^{-j\theta}}{2}\right\}$$

Replacing $e^{j\theta}$ by z and $e^{-j\theta}$ by z^{-1}, this becomes

$$H(z)H(z^{-1}) = \frac{\frac{5}{4}-\frac{1}{2}z-\frac{1}{2}z^{-1}}{\frac{10}{9}-\frac{1}{3}z-\frac{1}{3}z^{-1}} = \frac{\left(1-\frac{1}{2}z^{-1}\right)\left(1-\frac{1}{2}z\right)}{\left(1-\frac{1}{3}z^{-1}\right)\left(1-\frac{1}{3}z\right)}$$

The minimum phase system is then formed by extracting the poles and zeros that are inside the unit circle

$$H_{min}(z) = \frac{1-\frac{1}{2}z^{-1}}{1-\frac{1}{3}z^{-1}}$$

Topic 3: Tutorial questions (Level 5)

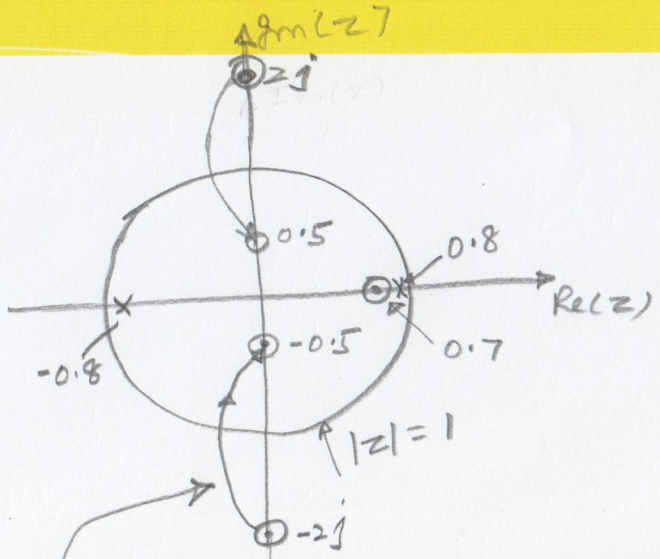
SOLO – Level 5

3. A stable and causal system (H(z)) may always be factored into a product of a minimum phase system with an all-pass system: $H(z) = H_{min}(z) \cdot H_{ap}(z)$ **or** $H(z) = H_{min}(z) \cdot H_{lp}(z)$; where $H_{lp}(z)$ is a linear phase system.

$$H(z) = \frac{(1 - 0.7z^{-1})(1 - j2z^{-1})(1 + j2z^{-1})}{(1 - 0.8z^{-1})(1 + 0.8z^{-1})}$$

(a) Find a minimum phase system function $H_{min}(z)$ and an all-pass system $H_{ap}(z)$, for the above H(z).

(b) Find a minimum phase system function $H_{min}(z)$ and a linear phase system $H_{lp}(z)$, for the above H(z).



(a) $H(z) = H_{min}(z) \cdot H_{ap}(z)$

The H(z) has non minimum phase factors $(1 - j2z^{-1})(1 + j2z^{-1})$ which may be written as the product of a minimum phase term and all-pass factors:

$$H(z) = \underbrace{\frac{(1 - 0.7z^{-1})\ (1 - \frac{1}{2}jz^{-1})(1 + \frac{1}{2}jz^{-1})}{(1 - 0.8z^{-1})(1 + 0.8z^{-1})}}_{H_{min}} \times \underbrace{\frac{(1 - j2z^{-1})(1 + j2z^{-1})}{(1 - \frac{1}{2}jz^{-1})(1 + \frac{1}{2}jz^{-1})}}_{H_{ap}}$$

Reflect the zeros outside the unit circle to inside the unit circle to obtain min phase filter.

$H_{LP} = (1 - \frac{1}{2}jz^{-1})(1 + \frac{1}{2}jz^{-1})(1 - 2jz^{-1})(1 + 2jz^{-1})$

(b) $H(z) = H_{min}(z) \cdot H_{lp}(z)$

The system is not minimum phase because the factors $(1 - j2z^{-1})\ (1 + j2z^{-1})$ are out side the unit circle. First we reflect the zeros about the unit circle and replace it with poles: $\frac{1}{(1 - \frac{1}{2}jz^{-1})}$ & $\frac{1}{1 + \frac{1}{2}jz^{-1}}$. Then we multiply each term with a linear phase factor that has zeros at $(z = \frac{1}{2}j$ & $z = 2j)$ and $z = -\frac{1}{2}j$ & $z = -2j$

$$\therefore H(z) = \underbrace{\frac{(1 - 0.7z^{-1})}{(1 - 0.8z^{-1})(1 + 0.8z^{-1})} \cdot \frac{1}{(1 - \frac{1}{2}jz^{-1})(1 + \frac{1}{2}jz^{-1})}}_{H_{min}} \times H_{lp}$$

Topic 3: Tutorial questions (Level 5)

SOLO – Level 5

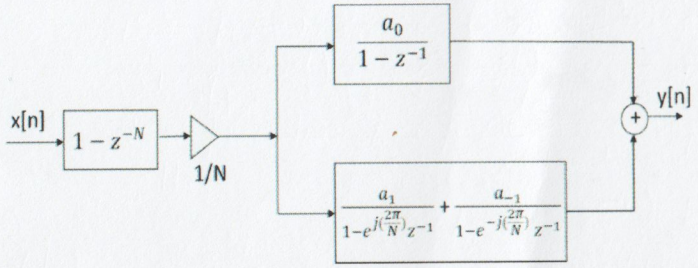
4. A filter structure with a transfer function $\{H(z) = \frac{Y(z)}{X(z)}\}$ is shown below. You may assume that $N = 3$, $a_0 = 3$ and $H(\theta)|_{\theta=2\pi/3} = 6+j3\sqrt{3}$.

   Also $H(e^{j\frac{2\pi k}{N}}) = a_k$ and $H(e^{-j\frac{2\pi k}{N}}) = a_{-k}$

   (a) Determine a_1 & a_{-1} such that this filter has a real valued impulse response *h[n];*

   (b) Find H(z) and draw a filter structure using delays elements, multipliers and adders.

   (c) Show that this filter structure can be realised as an FIR filter.



(a) $a_1 = 6+j3\sqrt{3}$ $\quad N=3$

$a_{-1} = 6-j3\sqrt{3}$

$$H(z) = \frac{1-z^{-3}}{3}\left[\frac{3}{1-z^{-1}} + \frac{6+j3\sqrt{3}}{1-e^{j\frac{2\pi}{3}}z^{-1}} + \frac{6-j3\sqrt{3}}{1-e^{-j\frac{2\pi}{3}}z^{-1}}\right]$$

$$= \frac{1-z^{-3}}{3}\left[\frac{3}{1-z^{-1}} + \frac{(6+j3\sqrt{3})(1-z^{-1}e^{-j\frac{2\pi}{3}}) + (6-j3\sqrt{3})(1-z^{-1}e^{-j\frac{2\pi}{3}})}{1+z^{-1}+z^{-2}}\right]$$

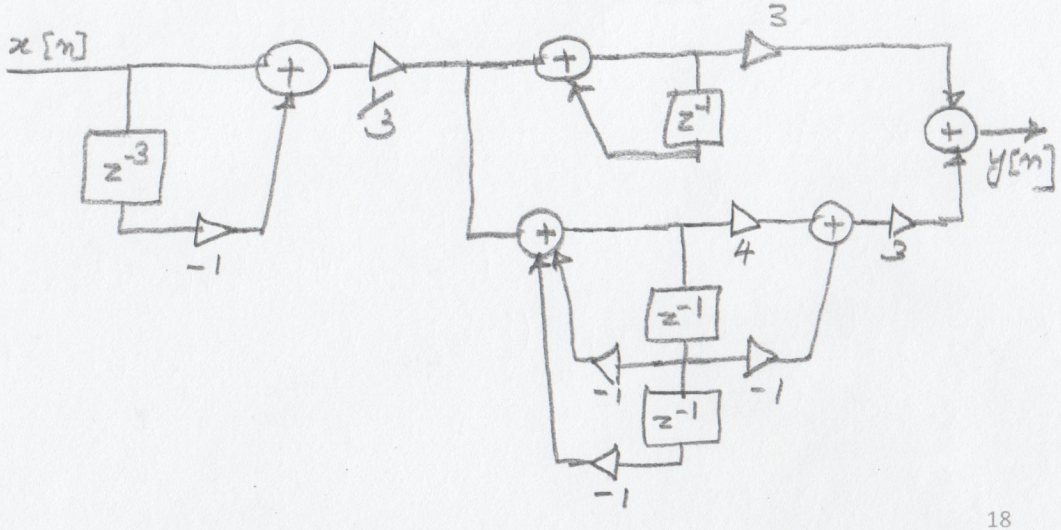
$$= \frac{1-z^{-3}}{3}\left[\frac{3}{1-z^{-1}} + \frac{3(4-z^{-1})}{1+z^{-1}+z^{-2}}\right]$$

$$H(z) = \frac{1-z^{-3}}{3} \cdot 3\left[\frac{5-4z^{-1}+2z^{-2}}{(1-z^{-1})(1+z^{-1}+z^{-2})}\right]$$

$$H(z) = 5-4z^{-1}+2z^{-2}$$

FIR filter.

(b)



ELEC3104: Digital Signal Processing

Topic 4: Digital Resonator Fundamentals

Model Answers to Tutorial questions

Prof. Eliathamby Ambikairajah, School of EE&T

Term 3, 2021

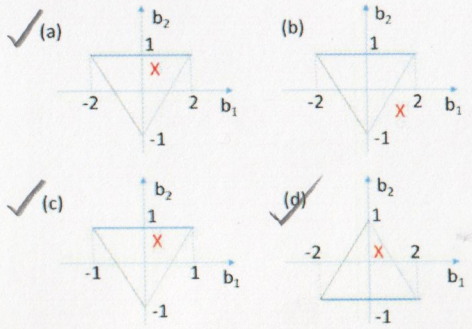
Topic 4: Tutorial Questions (SOLO -Level 1)

SOLO – Level 1

1. A second-order all-pole filter is described by $H(z) = \frac{1}{1+b_1 z^{-1}+b_2 z^{-2}}$

   Which of the following diagram(s) is used to conclude that the system is stable ? (Tick the correct answer)

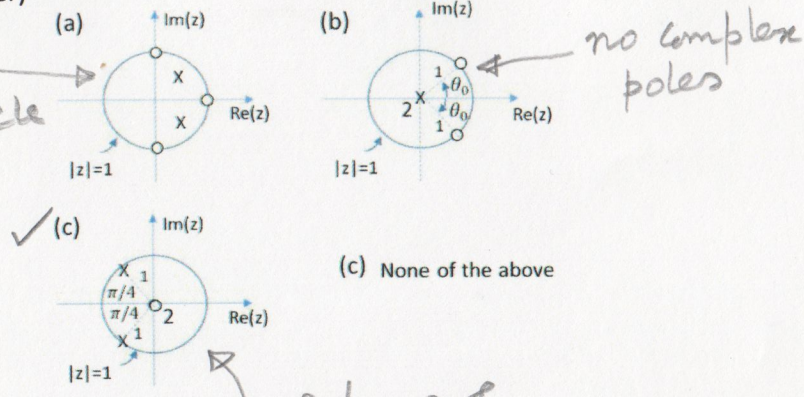
The poles may be real or complex, but for a stable system, they must be inside the stability triangle.



outside the stability triangle

2. Which of the following pole-zero plot represents a digital oscillator? (Tick the correct answer)

poles inside the unit circle



no complex poles

(c) None of the above

Poles are on the unit circle

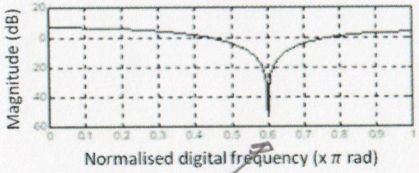
Topic 4: Tutorial Questions (SOLO -Level 1)

SOLO – Level 1

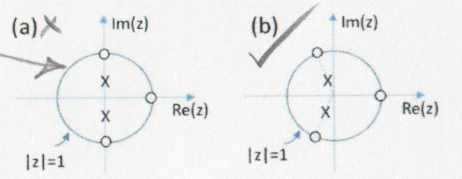
3. Which of the following pole-zero plot is most likely to produce the notch filter response shown below ?(Tick the correct answer)

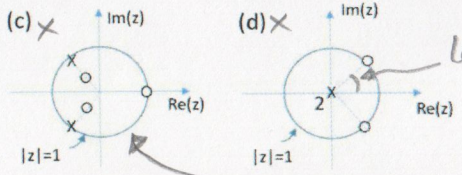
Notch frequency $\pi/2$

Notch filter magnitude response



notch frequency is > $\pi/2$





less than $\pi/2$

Zeros must be on the unit circle or closer to unit circle

4. An instrumentation and recording system require that the power-line frequency of 50Hz and harmonics to be eliminated. What type of filter would you choose? (tick the correct answer)

(a) All-pass filter
(b) Band-pass filter
✓(c) Notch filter
(d) High-pass filter
(e) Low- pass filter
(f) An oscillator
(g) None of the above

To remove a particular frequency use notch filter.

Topic 4: Tutorial Questions (Level 2)

SOLO – Level 2

1. A digital oscillator has a unit impulse response given by:

$$h[n] = \sin([n+1]\theta_0)\,u[n];$$

where θ_0 is the oscillator frequency;

(a) Find the transfer function of this oscillator

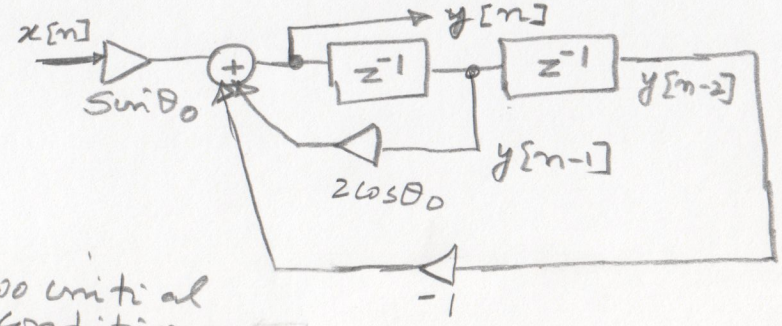
(b) Draw the structure for this transfer function using the transfer function obtained in part (a) above

(c) By setting the input in part (b) to zero and under certain initial conditions, sinusoidal oscillation can be obtained. Find these initial conditions.

(a)

From the z-transform table

$$H(z) = \frac{\sin\theta_0}{1 - 2\cos\theta_0 z^{-1} + z^{-2}}$$



$x[n] = 0$; $y[n] = \sin[(n+1)\theta_0]$

$n=0$ $\quad y[0] = \sin[(0+1)\theta_0] = \sin\theta_0$

$n=-1$ $\quad y[-1] = \sin[(-1+1)\theta_0] = 0$

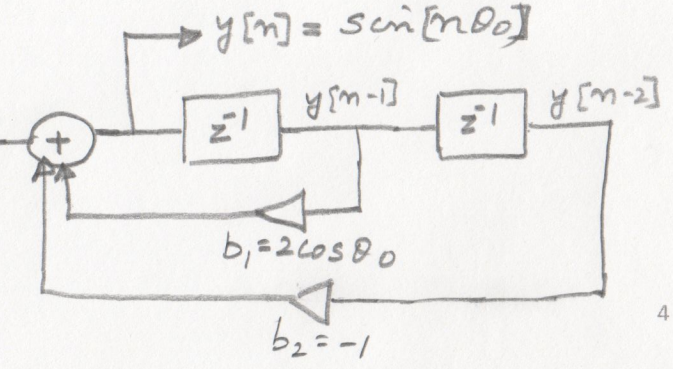
Two initial conditions

or

$n=-2$ $\quad y[-2] = \sin[(-2+1)\theta_0] = -\sin\theta_0$ ← Two initial conditions

2. Using the following difference equation, you are required to implement (i.e. draw a structure) a digital oscillator that generates y[n]=sin(θ_0n) for n ≥0. Choose appropriate initial conditions. Note that: $x[n]$ is the input; $y[n]$ is the output and r is a constant. You are required to choose an appropriate value for r in order to sustain oscillation.

$$y[n] = 2r\cos\theta_0\, y[n-1] - r^2 y[n-2] + x[n] - r\cos\theta_0\, x[n-1]$$



Let $x[n] = 0$, $\quad y[n] = 2r\cos\theta_0\, y[n-1] - r^2 y[n-2]$

For an oscillator $r = 1$ (Poles on the unit circle)

$\therefore y[n] = 2\cos\theta_0\, y[n-1] - y[n-2]$

output : $y[n] = \sin(\theta_0 n)$ ← given

$\therefore y[0] = \sin(0) = 0$ & $y[-1] = \sin(-\theta_0) = -\sin\theta_0$ ⇒

$y[0] = 0$
$y[-1] = -\sin\theta_0$

initial conditions

Topic 4: Tutorial Questions (Level 2)

SOLO – Level 2

3. Design a notch filter with notch frequency 60 Hz, stopband 5 Hz (bandwidth) and sampling frequency 300Hz.

$\Delta\theta_0 = 2\pi \frac{f_a}{f_s} = 2\pi \frac{5}{300} = \frac{\pi}{30}$ ($\Delta\theta_0$: bandwidth)

notch frequency $\theta_0 = 2\pi \times \frac{60}{300} = \frac{2\pi}{5}$

$r = 1 - 0.5\,\Delta\theta_0 = 1 - 0.5 \times \frac{\pi}{30} = 0.9476$

$\left(r = 1 - \pi \frac{\Delta f}{f_s}\right)$

→ stopband: Place zeros at the notch frequency

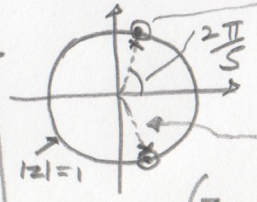
$z_1 = e^{+j\theta_0}$; $z_2 = e^{-j\theta_0}$ } zeros

→ $z_1 = e^{j\frac{2\pi}{5}}$; $z_2 = e^{-j\frac{2\pi}{5}}$

Passband: We place poles along the orientation of the zeros at

$P_{1,2} = r e^{\pm j\theta_0} \Rightarrow P_1 = 0.9476\, e^{j\frac{2\pi}{5}}$

$P_2 = 0.9476\, e^{-j\frac{2\pi}{5}}$



$\therefore H(z) = \frac{(z - e^{j\frac{2\pi}{5}})(z - e^{-j\frac{2\pi}{5}})}{(z - 0.9476\, e^{j\frac{2\pi}{5}})(z - 0.9476\, e^{-j\frac{2\pi}{5}})}$

$$H(z) = \frac{z^2 - 0.618z + 1}{z^2 - 0.5857z + 0.898}$$

4. Design a bandpass filter with centre frequency of 100Hz, passband 10Hz, stopband edges at 50 Hz and 150 Hz , and sampling frequency 400Hz.

Note: If the designed filter becomes non causal, then use double poles at each pole locations to get a causal filter.

$\theta_0 = 2\pi \frac{100}{400} = \frac{\pi}{2}$; $\Delta\theta_0 = 2\pi \frac{10}{400} = \frac{\pi}{20}$ ($\Delta\theta_0$: BW)

Stoband edges: $\theta_l = 2\pi \frac{50}{400} = \frac{\pi}{4}$; $\theta_u = 2\pi \frac{150}{400} = \frac{3\pi}{4}$

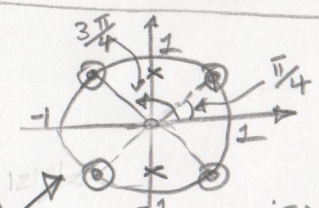
$r = 1 - 0.5\,\Delta\theta_0 = 1 - 0.5 \times \frac{\pi}{20} = 0.9125$

Passband: Place Poles at $P_{1,2} = r e^{\pm j\theta_0} = 0.9125\, e^{\pm j\frac{\pi}{2}}$

$= \pm j0.9125$

stopband: Place conjugate zeros at:

$z_{1,2} = e^{\pm j\frac{\pi}{4}}$ and $z_{3,4} = e^{\pm j\frac{3\pi}{4}}$



$H(z) = \frac{(z - e^{-j\frac{\pi}{4}})(z - e^{j\frac{\pi}{4}})(z - e^{-j\frac{3\pi}{4}})(z - e^{j\frac{3\pi}{4}})}{(z - 1\cdot0.9215)(z + j0.9215)}$

$$H(z) = \frac{z^4 + 1}{z^2 + 0.8941}$$

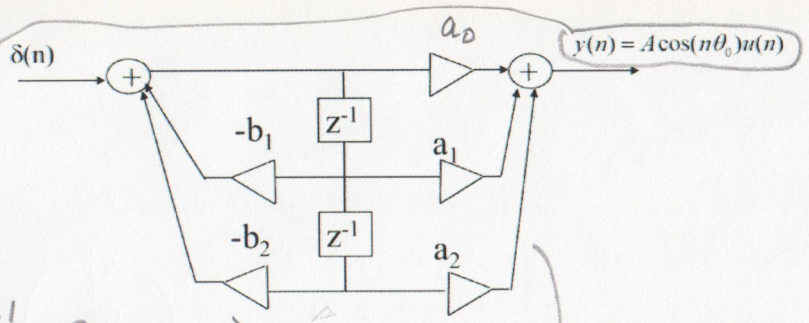
← This is a non-causal filter. To make it causal we could, for example, use double poles at each pole location to get a causal filter. $H(z) = \frac{z^4+1}{(z^2+0.8941)^2}$

Topic 4: Tutorial Questions (Level 2)

SOLO – Level 2

5. An IIR oscillator which is initialised by an impulse is shown below. The desired impulse response is :

$y[n] = A\cos[n\theta_0]u[n]$; where θ_0 is the oscillator frequency;



Assuming θ_0 is the oscillator frequency of the digital oscillator, and by writing appropriate equations, find the values of b_1, b_2, a_0, a_1 and a_2 for sustaining the oscillation.

$A\cos[n\theta_0]u[n] \xleftrightarrow{\text{Z-transform}} A\cdot\left(\frac{1-z^{-1}\cos\theta_0}{1-2\cos\theta_0 z^{-1}+z^{-2}}\right)$

$H(z) = \frac{a_0+a_1z^{-1}+a_2z^{-2}}{1+b_1z^{-1}+b_2z^{-2}}$

both equations are the same

By equating the coefficients we get

$a_0 = A$; $a_1 = -A\cos\theta_0$; $a_2 = 0$

$b_1 = -2\cos\theta_0$; $b_2 = 1$

6. The transfer function of a digital resonator with zeros at the origin is given by

$$H(z) = \frac{b_0}{(1-re^{j\theta_0}z^{-1})(1-re^{-j\theta_0}z^{-1})}$$

where r is the pole radius and θ_0 is the resonant frequency.

Show that the gain b_0 is given by the following, when $|H(\theta_0)| = 1$,

$$b_0 = (1-r)\sqrt{1+r^2-2r\cos(2\theta_0)}$$

$H(\theta) = \frac{b_0}{(1-re^{j\theta_0}e^{-j\theta})(1-re^{-j\theta_0}e^{-j\theta})}$

$|H(\theta)|_{\theta=\theta_0} = 1 = \left|\frac{b_0}{(1-re^{j\theta_0}e^{-j\theta_0})(1-re^{-j\theta_0}e^{-j\theta_0})}\right|$

$1 = \frac{b_0}{|(1-r)(1-re^{-j2\theta_0})|} = \frac{b_0}{(1-r)|1-r\cos2\theta_0+jr\sin2\theta_0|}$

$\therefore b_0 = (1-r)\sqrt{1+r^2-2r\cos(2\theta_0)} \longleftarrow 1 = \frac{b_0}{1-r}\cdot\frac{1}{\sqrt{1+r^2-2r\cos2\theta_0}}$

Topic 4: Tutorial questions (Level 3)

SOLO – Level 3

1. For the structure shown in Figure 1, write down the appropriate difference equations and hence state the function of this structure.

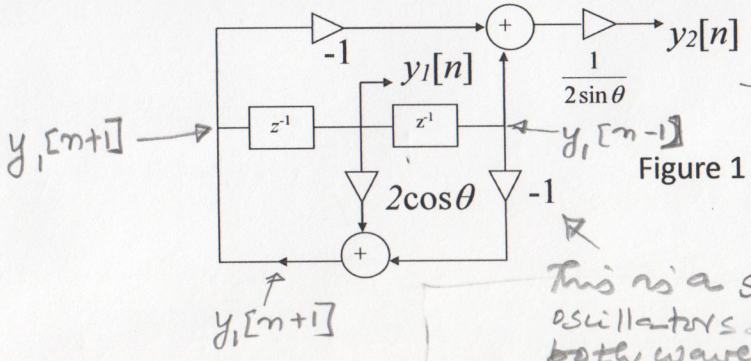


Figure 1

$y_2[n] = \frac{1}{2\sin\theta}\{y_1[n-1] - y_1[n+1]\}$ — ①

$y_1[n+1] = y_1[n]\cdot 2\cos\theta - y_1[n-1]$

$\therefore y_1[n] = \frac{1}{2\cos\theta}\{y_1[n+1] + y_1[n-1]\}$ — ②

If $y_1[n] = \cos(n\theta)$, then $y_2[n] = \sin(n\theta)$

Check: $\cos(n\theta) = \frac{1}{2\cos\theta}[\cos(n+1)\theta + \cos(n-1)\theta]$

$\therefore 2\cos\theta\cdot\cos n\theta = \cos(n+1)\theta + \cos(n-1)\theta$

This is correct

This is a sine & cosine oscillators delivering both waveforms simultaneously

Similarly it can be shown, using equation (2)

$2\sin\theta\sin(n\theta) = \cos(n-1)\theta - \cos(n+1)\theta$

2. The transfer function of a digital resonator with zeros at the origin is given by

$$H(z) = \frac{b_0(1-z^{-1})(1+z^{-1})}{(1-re^{j\theta_0}z^{-1})(1-re^{-j\theta_0}z^{-1})}$$

where r is the pole radius and θ_0 is the resonant frequency.
Determine the gain b_0 for the digital resonator so that $|H(\theta_0)| = 1$.

$$H(\theta) = \frac{b_0(1-e^{-j\theta})(1+e^{-j\theta})}{(1-re^{j\theta_0}e^{-j\theta})(1-re^{-j\theta_0}e^{-j\theta})}$$

$$|H(\theta)|_{\theta=\theta_0} = 1 = \frac{b_0|(1-e^{-j2\theta_0})|}{|(1-re^{j\theta_0}e^{-j\theta_0})(1-re^{-j\theta_0}e^{-j\theta_0})|}$$

$$1 = \frac{b_0|(1-\cos 2\theta_0) + (\sin 2\theta_0)|}{|(1-r)(1-re^{-j2\theta_0})|}$$

$$1 = \frac{b_0}{1-r}\cdot\frac{\sqrt{2-2\cos(2\theta_0)}}{\sqrt{1+r^2-2r\cos(2\theta_0)}} \qquad \sqrt{4\sin^2\theta_0} = 2|\sin\theta_0|$$

$$\therefore \boxed{b_0 = \frac{(1-r)\sqrt{1+r^2-2r\cos(2\theta_0)}}{2|\sin\theta_0|}}$$

Topic 4: Tutorial questions (Level 3)

SOLO – Level 3

3. Using the z-transform table, (a) determine the difference equations for the digital oscillators that have impulse responses $h[n] = A\sin[n\theta_0]u[n]$ and $h[n] = A\cos[n\theta_0]u[n]$ respectively; (b) Determine the structure for the coupled-form oscillator by combining the structure for the digital oscillator obtained in part (a) above.

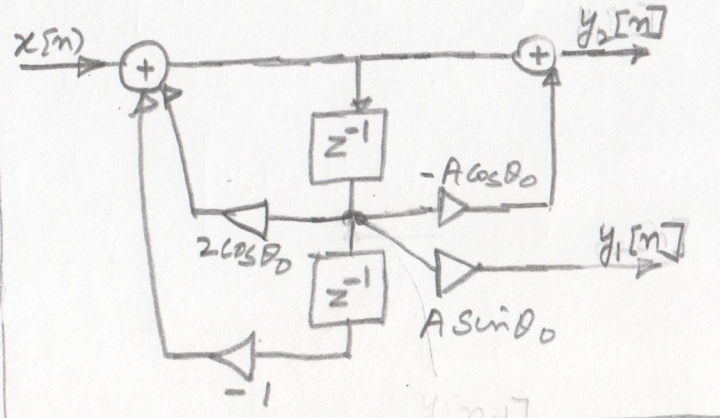
(a) $h[n] = A\sin(n\theta_0)u[n] \Rightarrow H(z) = A\dfrac{1 - z^{-1}\sin\theta_0}{1 - 2\cos\theta_0 z^{-1} + z^{-2}}$

Hence $y_1[n] = y[n] = 2\cos\theta_0\, y[n-1] - y[n-2] + A\sin\theta_0\, x[n-1]$ -①

$h[n] = A\cos(n\theta_0)u[n] \Rightarrow H(z) = A\dfrac{1 - z^{-1}\cos\theta_0}{1 - 2\cos\theta_0 z^{-1} + z^{-2}}$

Hence $y_2[n] = y[n] = 2\cos\theta_0\, y[n-1] - y[n-2] + A\,x[n] - A\cos\theta_0\, x[n-1]$ -②

(b) Combine equation ① & ② to obtain the structure.



4. Design an FIR filter that will reject a very strong 60-Hz sinusoidal interference contaminating a 200Hz useful sinusoidal signal. Determine the gain of the filter so that the useful signal does not change amplitude. The filter operates at a sampling frequency f_s = 500 samples/sec.

The rejected frequency $\theta_1 = 2\pi\frac{60}{500} = \frac{6\pi}{25}$

The frequency of useful signal $\theta_2 = 2\pi\frac{200}{500} = \frac{4\pi}{5}$

The useful signal does not change amplitude:

$\therefore |H(\frac{4\pi}{5})| = 1$ & $|H(\frac{6\pi}{25})| = 0$

$\therefore H(\theta) = k_0\,(1 - 1e^{j\frac{6\pi}{25}}e^{-j\theta})(1 - 1e^{-j\frac{6\pi}{25}}e^{-j\theta})$

filter, gain, zero at the unit circle

$\rightarrow H(\theta) = k_0\left[1 - e^{-j\theta}\left(e^{j\frac{6\pi}{25}} + e^{-j\frac{6\pi}{25}}\right) + e^{-j2\theta}\right]$

$\therefore H(\theta) = k_0\, e^{-j\theta}\left[e^{j\theta} - 2\cos\frac{6\pi}{25} + e^{-j\theta}\right]$

$H(\theta) = k_0\, e^{-j\theta}\left(2\cos\theta - 2\cos\frac{6\pi}{25}\right)$

$|H(\theta)|_{\theta=\frac{4\pi}{5}} = 1 = k\,|e^{-j\theta}|\,|2\cos\frac{4\pi}{5} - 2\cos\frac{6\pi}{25}|$ ($|e^{-j\theta}| = 1$)

$\therefore k = \dfrac{\frac{1}{2}}{\cos\frac{6\pi}{25} - \cos\frac{4\pi}{5}}$

Topic 4: Tutorial questions (Level 3)

SOLO – Level 3

5. The transfer function of a digital resonator is given by

$$H(z) = \frac{b_0}{(1 - re^{j\theta_0}z^{-1})(1 - re^{-j\theta_0}z^{-1})}$$

The impulse response of this resonator can be found by partial fraction expansion . Show that the impulse response h[n] is given by:

$$h[n] = \frac{b_0 r^n}{\sin(\theta_0)} \sin[(n+1)\theta_0]\, u[n]$$

$$H(z) = \frac{b_0 z^2}{(z - re^{j\theta_0})(z - re^{-j\theta_0})}$$

We seek the expansion of the form

$$\frac{H(z)}{z} = b_0 \left\{ \frac{z}{(z - re^{j\theta_0})(z - re^{-j\theta_0})} \right\} = \frac{A}{z - re^{j\theta_0}} + \frac{A^*}{z - re^{-j\theta_0}}$$

where

$$A = \left. \frac{z}{z - re^{-j\theta_0}} \right|_{z = re^{j\theta_0}} = \frac{re^{j\theta_0}}{re^{j\theta_0} - re^{-j\theta_0}}$$

$$= \frac{re^{j\theta_0}}{2j\, r \sin\theta_0 \, e^{j\pi/2}}$$

$$A = \frac{e^{j(\theta_0 - \pi/2)}}{2\sin\theta_0}$$

$$\therefore A^* = \frac{e^{-j(\theta_0 - \pi/2)}}{2\sin\theta_0}$$

$$\therefore H(z) = \frac{b_0}{2\sin\theta_0} \left[\frac{z\, e^{j(\theta_0 - \pi/2)}}{z - re^{j\theta_0}} + \frac{z\, e^{-j(\theta_0 - \pi/2)}}{z - re^{-j\theta_0}} \right] = \frac{b_0}{2\sin\theta_0} \left[\frac{e^{j(\theta_0 - \pi/2)}}{1 - re^{j\theta_0}z^{-1}} + \frac{e^{-j(\theta_0 - \pi/2)}}{1 - re^{-j\theta_0}z^{-1}} \right]$$

$$h[n] = \frac{b_0}{2\sin\theta_0} \left[e^{j(\theta_0 - \pi/2)} \cdot r^n e^{j\theta_0 n} + e^{-j(\theta_0 - \pi/2)} r^n \cdot e^{-j\theta_0 n} \right]$$

$$= \frac{b_0}{\sin\theta_0} \cdot r^n \left[\frac{e^{j((n+1)\theta_0 - \pi/2)} + e^{-j((n+1)\theta_0 - \pi/2)}}{2} \right] = \frac{b_0 \cdot r^n}{\sin\theta_0} \cdot \underbrace{\cos((n+1)\theta_0 - \pi/2)}_{= \sin((n+1)\theta_0)}$$

$$h[n] = \frac{b_0 r^n}{\sin\theta_0} \sin((n+1)\theta_0)\, u[n]$$

Topic 4: Tutorial questions (Level 4)

SOLO – Level 4

1. For the system shown in figure 2,

   (a) Express $y_1[n]$ in terms of $y_1[n-1]$, $y_2[n-1]$, and $x[n]$; do the same for $y_2[n]$.

   (b) You may assume $A = \cos(\theta_0)$; $B = \sin(\theta_0)$; $y_1[-1] = \cos(-\theta_0)$; $y_2(-1) = \sin(-\theta_0)$. If $x[n] = 0$, show that : $y_1[n] = \cos(n\theta_0)$ and $y_2[n] = \sin(n\theta_0)$

   (c) Obtain the system transfer function

   $H_1(z) = \frac{Y_1(z)}{X(z)}$ and $H_2(z) = \frac{Y_2(z)}{X(z)}$ where A and B arbitrary values.

   (d) If $A = B = \frac{\sqrt{2}}{2}$, sketch the pole-zero pattern for $H_1(z)$ and $H_2(z)$

   (e) If $A = B = \frac{\sqrt{2}}{2}$ and $x[n] = \delta[n]$, find the impulse response $h_1[n]$.

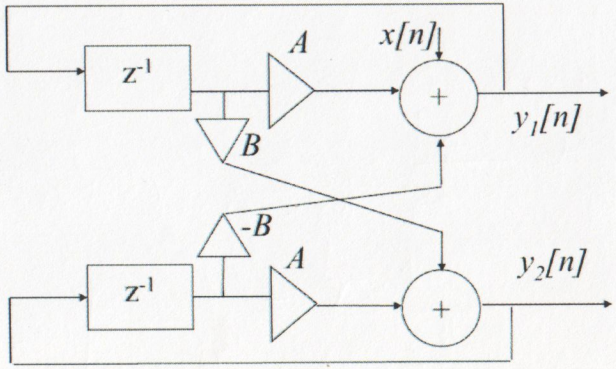


Figure 2

(a) $y_1[n] = x[n] + (-B)\, y_2[n-1] + A\, y_1[n-1]$

$y_2[n] = B\, y_1[n-1] + A\, y_2[n-1]$

(b) If $x[n] = 0$

$n=0$ $y_1[0] = \cos\theta_0\, y_1[-1] - \sin\theta_0\, y_2[-1]$
$= \cos\theta_0 \cos\theta_0 + \sin\theta_0 \sin\theta_0$
$= 1$

$n=1$ $y_1[1] = \cos\theta_0\, y_1[0] - \sin\theta_0\, y_2[0]$
$= \cos\theta$

$n=2$ $y_1[2] = \cos\theta_0\, y_1[1] - \sin\theta_0\, y_2[1]$
$= \cos\theta_0 \cos\theta_0 - \sin\theta_0 \sin\theta_0$
$= \cos^2\theta_0$

$n=n$ $\boxed{y_1[n] = \cos n\theta_0}$

$n=0$ $y_2[0] = \sin\theta_0\, y_1[-1] + \cos\theta_0\, y_2[-1]$
$= \sin\theta_0 \cos\theta_0 + \cos\theta_0 (-\sin\theta_0)$
$= 0$

$n=1$ $y_2[1] = \sin\theta_0\, y_1[0] + \cos\theta_0\, y_2[0]$
$= \sin\theta_0 \cdot 1 \quad \approx 0$

$n=2$ $y_2[2] = \sin\theta_0\, y_1[1] + \cos\theta_0\, y_2[1]$
$= \sin\theta_0 \cos\theta_0 + \cos\theta_0 \cdot \sin\theta_0$
$= \sin 2\theta_0$

$n=n$ $\boxed{y_2[n] = \sin n\theta_0}$

(c) $Y_1(z) = X(z) - BY_2(z)z^{-1} + AY_1(z)z^{-1}$ ①

$Y_2(z) = BY_1(z)z^{-1} + AY_2(z)z^{-1}$

$\therefore Y_2(z)[1 - Az^{-1}] = BY_1(z)z^{-1}$ ②

Substitute in ①

$Y_1(z) = X(z) - Bz^{-1} \cdot \frac{BY_1(z)z^{-1}}{1 - Az^{-1}} + AY_1(z)z^{-1}$

$\therefore Y_1(z)\left[1 + \frac{B^2z^{-2}}{1 - Az^{-1}} - Az^{-1}\right] = X(z)$

$\therefore H_1(z) = \frac{1 - Az^{-1}}{1 - Az^{-1} + B^2z^{-2} - Az^{-1} + A^2z^{-2}}$

$\boxed{H_1(z) = \frac{1 - Az^{-1}}{1 - 2Az^{-1} + (A^2 + B^2)z^{-2}}}$

10 (a)

Topic 4: Tutorial questions (Level 4)

SOLO – Level 4

From ②, $Y_2(z) = \frac{B z^{-1}}{1 - A z^{-1}} \cdot Y_1(z) = \frac{B z^{-1}}{1 - A z^{-1}} \cdot \frac{(1 - A z^{-1})\ X(z)}{1 - 2A z^{-1} + (A^2 + B^2) z^{-2}}$

$$\therefore \boxed{H_2(z) = \frac{Y_2(z)}{X_2(z)} = \frac{B z^{-1}}{1 - 2A z^{-1} + (B^2 + A^2) z^{-2}}}$$

(d) If $A = B = \frac{\sqrt{2}}{2}$, then $A + B = \sqrt{2}$ & $A^2 + B^2 = \frac{2}{4} + \frac{2}{4} = 1$

$$H_1(z) = \frac{1 - \frac{\sqrt{2}}{2} z^{-1}}{1 - 2 \times \frac{\sqrt{2}}{2} z^{-1} + 1 z^{-2}} = \frac{1 - \frac{\sqrt{2}}{2} z^{-1}}{1 - \sqrt{2} z^{-1} + z^{-2}}$$

$$\boxed{H_2(z) = \frac{\frac{\sqrt{2}}{2} z^{-1}}{1 - \sqrt{2} z^{-1} + z^{-2}}}$$

$$H_1(z) = \frac{z\left(z - \frac{\sqrt{2}}{2}\right)}{z^2 - \sqrt{2} z + 1} \rightarrow \text{Zeros: } z = 0;\ z = \frac{\sqrt{2}}{2}$$

$$\hookrightarrow \text{Poles: } P_{1,2} = \frac{1}{2}\sqrt{2}(1 \pm j)$$

$$H_2(z) = \frac{\frac{\sqrt{2}}{2} z}{z^2 - \sqrt{2} z + 1}$$

Zeros: $z = 0$

Poles: $P_1 = \frac{1}{2}\sqrt{2}(1 + j);\ P_2 = \frac{1}{2}\sqrt{2}(1 - j)$

(e) $H_1(z) = \frac{z^2 - z\frac{\sqrt{2}}{2}}{z^2 - \sqrt{2} z + 1}$

If $x[n] = \delta[n] \Rightarrow X(z) = 1 \Rightarrow Y_1(z) = \frac{z^2 - z\frac{\sqrt{2}}{2}}{z^2 - \sqrt{2} z + 1} \equiv \frac{z^2 - z\cos\theta_0}{z^2 - 2z\cos\theta_0 + 1}$ ← z-transform of $\cos(\theta_0 n)\, u[n]$

equate the coefficients

$\cos\theta_0 = \frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}}$

$\therefore \theta_0 = \pi/4$

$$\therefore y_1[n] = \cos\left[\frac{\pi}{4} n\right] u[n]$$ ← oscillator

10(b)

Topic 4: Tutorial questions (Level 4)

SOLO – Level 4

2. Consider a filter with transfer function

$$H(z) = \frac{b_0(1 - e^{j\theta_0}z^{-1})(1 - e^{-j\theta_0}z^{-1})}{(1 - re^{j\theta_0}z^{-1})(1 - re^{-j\theta_0}z^{-1})}$$

(a) Sketch the pole-zero pattern.

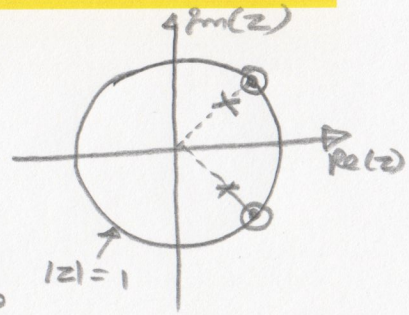
(b) Show that for $r \approx 1$, , the system is a notch filter and provide a rough sketch of its magnitude response if $\theta_0 = 60^o$.

(c) For $\theta_0 = 60^o$, choose b_0 so that the maximum value $|H(\theta)| = 1$.

(d) Determine the approximate 3-dB bandwidth of this filter.

(a) Poles : $p_{1,2} = re^{\pm j\theta_0}$

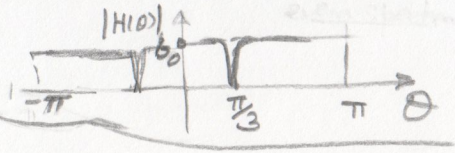
Zeros : $z_{1,2} = e^{\pm j\theta_0}$



(b) for $\theta = \theta_0$, $H(\theta_0) = 0$

If $r \approx 1$, then $H(\theta) = b_0$ for $\theta \neq \theta_0$.

i.e Poles & zeros in H(z) cancel when $r \approx 1$



(c) $|H(\theta)|^2 = b_0^2 \dfrac{|1 - e^{j\theta_0}e^{-j\theta}|^2 \, |1 - e^{-j\theta_0}e^{j\theta}|^2}{|1 - re^{j\theta_0}\cdot e^{-j\theta}|^2 \, |1 - re^{-j\theta_0}\cdot e^{-j\theta}|^2}$

$= b_0^2 \dfrac{2(1 - \cos(\theta - \theta_0))}{1 + r^2 - 2r\cos(\theta - \theta_0)} \times \dfrac{2(1 - \cos(\theta + \theta_0))}{1 + r^2 - 2r\cos(\theta + \theta_0)}$ where $\theta_0 = \pi/3$ (given)

To find the Maximum value

$\dfrac{d|H(\theta)|^2}{d\theta} = 0 \Rightarrow \theta = \pi$ [you need to do the differentiation]

$\therefore |H(\pi)|^2 = 4b_0^2 \left(\dfrac{3}{1 + r + r^2}\right)^2 = 1 \Rightarrow$ $\boxed{b_0 = \frac{1}{3}(1 + r + r^2)}$

Topic 4: Tutorial questions (Level 4)

SOLO – Level 4

(d) $|H(\theta)|^2 = b_0^2 \dfrac{|1-e^{-j\theta_0}\cdot e^{-j\theta}|^2}{|1-re^{j\theta_0}\cdot e^{-j\theta}|^2} \cdot \dfrac{|1-e^{-j\theta_0}\cdot e^{j\theta}|^2}{|1-re^{-j\theta_0}\cdot e^{-j\theta}|^2} = b_0^2 \dfrac{2[1-\cos(\theta-\theta_0)]}{1+r^2-2r\cos(\theta-\theta_0)} \times \dfrac{2[1-\cos(\theta+\theta_0)]}{[1+r^2-2r\cos(\theta+\theta_0)]}$

since the half-power frequencies are very close to θ_0, we have $\theta+\theta_0 \approx 2\theta_0$ & $r \approx 1$ and $\theta-\theta_0 \approx \frac{1}{2}\Delta\theta_0$ (↑ Bandwidth) ← assumption

$\therefore \quad \frac{1}{2} = b_0^2 \cdot \dfrac{2[1-\cos(0.5\Delta\theta_0)]}{1+r^2-2r\cos(0.5\Delta\theta_0)}$

$\therefore \cos(0.5\Delta\theta_0) = \dfrac{1+r^2-4b_0^2}{2r-4b_0^2}$

$\therefore \Delta\theta_0 = 2\cos^{-1}\left[\dfrac{1+r^2-4b_0^2}{2r-4b_0^2}\right]$ ← 3dB bandwidth

where $b_0^2 = \left[\frac{1}{3}(1+r+r^2)\right]^2$ (from part c)

Topic 4: Tutorial questions (Level 4)

SOLO – Level 4

3. By calculating the magnitude response of the following filter with a transfer function

$$H(z) = 1 + z^{-1} + z^{-2} + \ldots\ldots + z^{-8}$$

determine the frequencies of the analogue sinusoids that cannot pass through the filter. You may assume the sampling frequency f_s = 1 kHz.

$$H(z) = \frac{1 - z^{-9}}{1 - z^{-1}}$$

$$H(\theta) = \frac{1 - e^{-j9\theta}}{1 - e^{-j\theta}} = \frac{e^{-j9\theta/2}}{e^{-j\theta/2}} \cdot \frac{\left[e^{j9\theta/2} - e^{-j9\theta/2}\right]}{e^{j\theta/2} - e^{-j\theta/2}}$$

$$= e^{-j4\theta} \cdot \frac{\sin 9\theta/2}{\sin \theta/2}$$

$$|H(\theta)| = \left| \frac{\sin 9\theta/2}{\sin \theta/2} \right|$$

$H(\theta) = 0$ at $\theta = \frac{2\pi k}{9}$, $k = 1, 2, \ldots 8$

The corresponding analogue frequencies are,

$$\theta = 2\pi \frac{f_a}{f_s} = \frac{2\pi k}{9}$$

$$\therefore f_a = \frac{k f_s}{9}$$

$$f_a = \frac{k f_s}{9} \quad k = 1, 2, 3, 4 \quad (1\,kHz)$$

or

$$f_a = \frac{1}{9}\,kHz, \frac{2}{9}\,kHz, \frac{3}{9}\,kHz, \frac{4}{9}\,kHz$$

Topic 4: Tutorial questions (Level 5)

SOLO – Level 5

1. Design a digital oscillator with adjustable phase, Ø , and produces an output

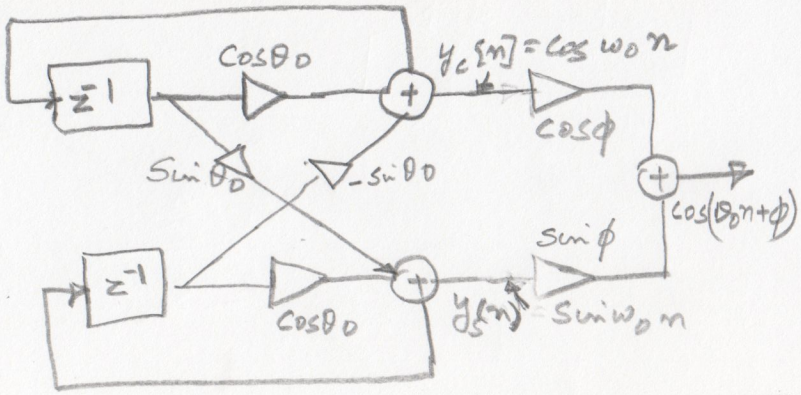
$$y[n] = \cos(n\theta_0 + \emptyset)\, u[n]$$

$h[n] = \cos(\omega_0 n + \phi)\, u[n]$

$= (\cos\omega_0 n \cos\phi - \sin\theta_0 n \cdot \sin\phi)\, u[n]$

Use the coupled-form oscillator (Page 17, Topic 4) and multiply the two outputs by $\cos\phi$ and $\sin\phi$ respectively and add the products

$y_c[n] \cdot \cos\phi + y_s[n] \sin\phi = \cos(\theta_0 n + \phi)$



2. A digital notch filter is required to remove an undesirable 60 Hz hum associated with a power supply in an ECG recording application. The sampling frequency used is fs = 500 samples/sec.

   (a) Design a second-order FIR notch filter and

   (b) A second-order pole-zero notch filter for this purpose

   In both cases choose the gain b_0 so that $|H(\theta)| = 1$ *for* $\theta = 0$.

(a) $\theta_0 = \frac{60}{500} \times 2\pi = \frac{6\pi}{50}$

$H(z) = \left(1 - e^{-j\frac{6\pi}{50}} z^{-1}\right)\left(1 - e^{-j\frac{6\pi}{50}} z^{-1}\right) \cdot b_0$ ← constant

second-order filter → $\boxed{H(z) = b_0\left(1 - 2\cos\frac{6\pi}{50} z^{-1} + z^{-2}\right)}$

$H(\theta) = b_0\left(1 - 2\cos\frac{6\pi}{50} e^{-j\theta} + e^{-j2\theta}\right)$

$H(\theta) = 2b_0 e^{-j\theta}\left(\cos\theta - \cos\frac{6\pi}{50}\right)$

$|H(\theta)|_{\theta=0} = 1 \quad \therefore 1 = 2b_0\left(1 - \cos\frac{3\pi}{25}\right)$

$\therefore \boxed{b_0 = \frac{1}{2\left(1 - \cos\frac{3\pi}{25}\right)}}$

(b) $H(z) = b_0 \dfrac{\left(1 - e^{j\frac{6\pi}{50}} z^{-1}\right)\left(1 - e^{-j\frac{6\pi}{50}} z^{-1}\right)}{\left(1 - re^{j\frac{6\pi}{50}} z^{-1}\right)\left(1 - re^{-j\frac{6\pi}{50}} z^{-1}\right)}$

$|H(\theta)|_{\theta=0} = 1 \quad = \dfrac{2b_0\left(1 - \cos\frac{6\pi}{50}\right)}{1 - 2r\cos\frac{6\pi}{50} + r^2}$

$\therefore \boxed{b_0 = \dfrac{1 - 2r\cos\frac{6\pi}{50} + r^2}{2\left(1 - \cos\frac{6\pi}{50}\right)}}$

Topic 4: Tutorial questions (Level 5)

SOLO – Level 5

3. A comb filter may be viewed as a notch filter in which the spectral nulls occur periodically across the frequency band. Comb filters find applications in a wide range of practical systems, for example, in the rejection of power-line harmonics. A comb filter can be described using a moving average FIR filter with the difference equation

$$y[n] = \frac{1}{M+1}\sum_{k=0}^{M} x[n-k]$$

(a) Show that the frequency response of the above filter is given by $H(\theta) = \frac{e^{-j\theta M/2}}{M+1}\frac{\sin\theta(\frac{M+1}{2})}{\sin(\frac{\theta}{2})}$

(b) Sketch the magnitude characteristic of the comb filter (for M= 10) that you obtained in part (a) above, but clearly illustrating the existence of the periodically spaced zeros at $\theta_k = \frac{2\pi k}{M+1}$ $for\ k = 1,2,\ldots.M.$

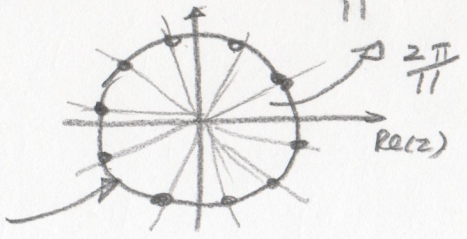
(a) $y[n] = \frac{1}{M+1}\left[x[n] + x[n-1] + x[n-2] + \cdots + x[n-M]\right]$

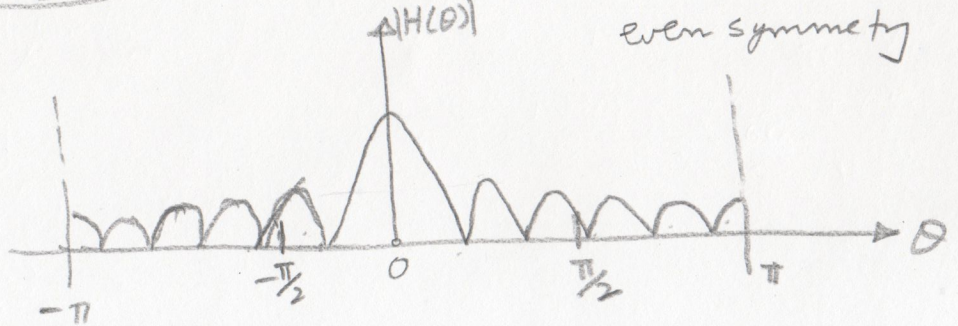
$H(z) = \frac{1}{M+1}\left[1 + z^{-1} + z^{-2} + \cdots + z^{-M}\right] = \frac{1}{M+1}\cdot\frac{1-z^{-(M+1)}}{1-z}$

$H(\theta) = \frac{1}{M+1}\cdot\frac{1-e^{-j(M+1)\theta}}{1-e^{-j\theta}} = \frac{1}{M+1}\cdot\frac{e^{-j(\frac{M+1}{2})\theta}}{e^{-j\frac{\theta}{2}}}\cdot\frac{e^{+j(\frac{M+1}{2})\theta} - e^{-j(\frac{M+1}{2})\theta}}{e^{j\frac{\theta}{2}} - e^{-j\frac{\theta}{2}}}$

$$\boxed{H(\theta) = \frac{e^{-j\frac{M\theta}{2}}}{M+1}\cdot\frac{\sin(\frac{M+1}{2}\theta)}{\sin\frac{\theta}{2}}} \Rightarrow H(\theta) = 0 \text{ at } \theta = \frac{2\pi k}{M+1} \text{ for } k = 1,2,\ldots M$$

(b) If $M = 10$ $\therefore \theta = \frac{2\pi}{11} \times k$





ELEC3104: Digital Signal Processing

Topic 5: Digital filter Design

Model Answers to Tutorial Questions

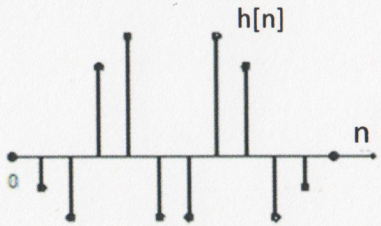
Prof. Eliathamby Ambikairajah, School of EE&T

Term 3, 2021

Topic 5: Tutorial Questions (SOLO -Level 1)

SOLO – Level 1

1. The impulse response h[n] of a digital filter is shown below. What can be said about this filter?



I) It is an FIR filter; ✓
II) It is an IIR filter;
III) It has linear phase; ✓
IV) It is a low-pass filter.

Tick the correct answer:

a) I
✓b) I, III
c) II, III
d) I, III, IV
e) None of the above

- The coefficients are symmetrical ⇒ linear phase
- Also finitie number of coefficients

2. One of the major problem that you may encounter when designing digital IIR filters using impulse invariant transformation is: (Tick the correct answer)

a) Mapping the analogue impulse response h(t) to its sampled version h[n];
✓b) Effects of Aliasing ; Increase the sampling rate to avoid aliasing
c) Inability to map the poles of the analogue fitters to the corresponding digital filters;
d) None of the above.

Topic 5: Tutorial Questions (SOLO -Level 1)

SOLO – Level 1

3. Window functions are used to truncate a signal to produce a signal of finite duration. The truncation process: (Tick the correct answers)

   ✓a) Causes frequency distortion in the output spectrum;
   ✓b) Creates ripples in the output spectrum;
   ✓c) Convolves the input spectrum with the window function spectrum;
   d) None of the above.

} see lecture notes Page 28 (Topic 5)

4. The bilinear transform is used for: (Tick the correct answers)

   ✓a) Mapping the left half of s plane into the unit circle in the z-plane thus avoiding aliasing of frequency components;
   ✓b) Transforming a stable analogue filter into a stable digital filter;
   c) Avoiding non-linear mapping;
   d) None of the above.

} see lecture notes Page 13 (Topic 5)

5. Which digital filter design method avoids aliasing but requires 'pre-warping' of the filter specifications ?

   a) Frequency sampling;
   b) Windowing;
   ✓c) Bi-linear transform;
   d) Impulse invariant method.

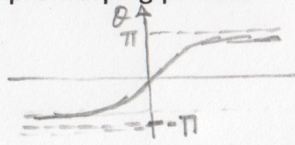
} (See Lecture notes Page 16 (Topic 5)

Topic 5: Tutorial Questions (SOLO-Level 2)

SOLO – Level 2

1. (a) When using the bilinear transformation to design a recursive digital filter from an analogue filter, pre-warping of the analogue filter's frequency response is required. State the reason for this pre-warping process.

$\omega_c = \frac{2}{T}\tan\left(\frac{\theta}{2}\right)$ ← analogue frequency



This mapping is highly non-linear therefore pre-warping of the analogue filter's response is required

(b) Use the bilinear z-transform to design an IIR digital lowpass filter based on the prototype function G(s): $G(s) = \frac{1}{s^2+s+1}$; The digital filter is to be designed to operate at a sampling rate of 8 kHz with a 3dB cut-off frequency at 2 kHz. Show that G(z) reduces to $G(z) = \frac{(z+1)^2}{(3z^2+1)}$.

We must prewarp the cut-off frequency ω_c

$\omega_c' = \frac{2}{T}\tan\left(\frac{\theta}{2}\right) = \frac{2}{T}\tan\left(\frac{1}{2}\left(2\pi\,\frac{2}{8}\right)\right) = \frac{2}{T}$ ← θ

$G(s) = \frac{1}{s^2+s+1}$; The denormalised analogue transfer function is:

$G_1(s) = G(s)\Big|_{s=\frac{s}{\omega_c'}} = \frac{1}{\left(\frac{s}{\omega_c'}\right)^2+\left(\frac{s}{\omega_c'}\right)+1} = \frac{(\omega_c')^2}{s^2+\omega_c' s+(\omega_c')^2}$

$\therefore G(z) = G_1(s)\Big|_{z=\frac{2}{T}\left(\frac{z-1}{z+1}\right)} = \frac{\left(\frac{2}{T}\right)^2}{\left[\frac{2}{T}\frac{z-1}{z+1}\right]^2+\frac{2}{T}\left(\frac{2}{T}\frac{z-1}{z+1}\right)+\left(\frac{2}{T}\right)^2}$

$\therefore G(z) = \frac{1}{\left(\frac{z-1}{z+1}\right)^2+\left(\frac{z-1}{z+1}\right)+1} = \frac{(z+1)^2}{(z^2-2z+1)+(z^2-1)+z^2+2z+1} = \frac{(z+1)^2}{3z^2+1}$

$\therefore G(z) = \frac{(z+1)^2}{3z^2+1}$

2. Use the bilinear z-transform to design an IIR digital lowpass filter based on the prototype function $G(s) = \frac{2}{s+2}$. The digital filter is to be designed to operate at a sampling rate of 8 kHz with a 3 dB cut-off frequency at 2kHz. Determine the transfer function G(z).

We must Prewarp the cut-off frequency ω_c.

$\omega_c' = \frac{2}{T}\tan\left(\frac{\theta}{2}\right) = \frac{2}{T}\tan\left(\frac{1}{2}\,2\pi\,\frac{f_a}{f_s}\right) = \frac{2}{T}\tan\left(\frac{1}{2}\times 2\pi\times\frac{2}{8}\right) = \frac{2}{T}$

$G(s) = \frac{2}{s+2}$ let $\omega_c = 2$ $\therefore G_1(s) = \frac{\omega_c'}{s+\omega_c'}$

$G(z) = G(s)\Big|_{s=\frac{2}{T}\left(\frac{1-z^{-1}}{1+z^{-1}}\right)} = \frac{\omega_c'}{\frac{2}{T}\left(\frac{1-z^{-1}}{1+z^{-1}}\right)+\omega_c'} = \frac{T\omega_c'(1+z^{-1})}{2-2z^{-1}+T\omega_c'+T\omega_c' z^{-1}}$

$G(z) = \frac{T\cdot\omega_c'}{2+T\omega_c'}\cdot\frac{1+z^{-1}}{1-\frac{2-T\omega_c'}{2+T\omega_c'}z^{-1}}$

$= \frac{T\cdot\frac{2}{T}}{2+T\cdot\frac{2}{T}}\cdot\frac{1+z^{-1}}{1+\left(\frac{2-T\cdot\frac{2}{T}}{2+T\cdot\frac{2}{T}}\right)z^{-1}} = \frac{1}{2}\cdot\frac{1+z^{-1}}{1+0} = \frac{1}{2}\cdot(1+z^{-1})$

$G_1(z) = \frac{1}{2}\cdot(1+z^{-1})$

SOLO – Level 2

3. Convert $H(s) = \frac{4s+7}{s^2+5s+4}$ to $H(z)$ using impulse invariant transformation, where the sampling frequency is 2Hz.

   [**Hint:** Find the partial fraction for H(s) first, before applying impulse invariant transformation]

$H(s) = \frac{4s+7}{(s+1)(s+4)} = \frac{A}{s+4} + \frac{B}{s+1} = \frac{3}{s+4} + \frac{1}{s+1}$

$T = \frac{1}{2} = 0.5s$

Impulse invariant transformation:

$\frac{1}{s+a} \rightarrow \frac{1}{1-e^{-aT}z^{-1}}$

$H(z) = \frac{3}{1-e^{-4T}z^{-1}} + \frac{1}{1-e^{-T}z^{-1}} = \frac{3}{1-e^{-2}z^{-1}} + \frac{1}{1-e^{-0.5}z^{-1}}$

$H(z) = \frac{3(1-e^{-0.5}z^{-1}) + (1-e^{-2}z^{-1})}{(1-e^{-2}z^{-1})(1-e^{-0.5}z^{-1})} = \frac{4 - 1.9549z^{-1}}{1 - 0.7419z^{-1} + 0.0821z^{-2}}$

4. (a) A second-order analogue band pass filter can be designed from a first-order normalised analogue low pass Butterworth filter $H(s) = \frac{1}{s+1}$ by performing the following low pass to band pass transformation:

$$s \rightarrow \frac{s^2 + {\omega_0}^2}{B\,s}; \quad B = \omega_u - \omega_l; \quad {\omega_0}^2 = \omega_u \omega_l$$

where ω_u is the upper cut-off frequency and ω_l is the lower cut-off frequency. Show that the analogue bandpass filter obtained using the above transformation is given by

$$G(s) = \frac{Bs}{s^2 + Bs + {\omega_0}^2}$$

(b) An IIR digital bandpass filter G(z) is to be designed to approximate the above analogue transfer function G(s). The digital bandpass filter is to operate at a sampling rate of 2 kHz with a lower cut-off frequency (f_l)at 200 Hz and the upper cut-off frequency (f_u) at 300 Hz. Using the bilinear transformation method, obtain the transfer function G(z) of the digital filter and show that G(z) reduces to

$$G(z) = 0.1367\frac{(1 - z^{-2})}{1 - 1.2362z^{-1} + 0.7265z^{-2}}$$

(a) $H(s) = \frac{1}{s+1} \Rightarrow G(s) = H(s)\Big|_{s \rightarrow \frac{s^2+\omega_0^2}{Bs}} = \frac{1}{\frac{s^2+\omega_0^2}{Bs} + 1} = \frac{Bs}{s^2 + Bs + \omega_0^2}$

(b) $f_u = 300$ Hz, $f_L = 200$ Hz

The prewarped passband edge frequencies are

$\omega_L' = \frac{2}{T}\tan\left(\frac{\omega_L T}{2}\right) = \frac{2}{T}\tan\left(2\pi\frac{200}{2000}\times\frac{1}{2}\right) = \frac{2}{T}\times 0.3249$

Similarly $\omega_u' = \frac{2}{T}\tan\left(2\pi\frac{300}{2000}\times\frac{1}{2}\right) = \frac{2}{T}\times 0.5095$

$\therefore \omega_0^2 = \omega_L' \cdot \omega_u' = \left(\frac{2}{T}\right)^2 \times 0.116$; $B = \omega_u' - \omega_L' = \frac{2}{T}(0.1846)$

(b) $G(z) = G(s)\Big|_{s=\frac{2}{T}\frac{z-1}{z+1}}$

$$= \frac{\frac{2}{T}(0.1846) \times \frac{2}{T}\frac{z-1}{z+1}}{\left(\frac{2}{T}\frac{z-1}{z+1}\right)^2 + \frac{2}{T}0.1846\left(\frac{2}{T}\frac{z-1}{z+1}\right) + \left(\frac{2}{T}\right)^2 \cdot 0.1655}$$

$$= \frac{0.1846\,\frac{z-1}{z+1}}{\left(\frac{z-1}{z+1}\right)^2 + 0.1846\left(\frac{z-1}{z+1}\right) + 0.1655}$$

$$= \frac{0.1846\,(z+1)(z-1)}{(z^2 - 2z + 1) + 0.1846z^2 - 0.1846 + 0.1655(z^2 + 2z + 1)}$$

$G(z) = \frac{0.1367(1 - z^{-2})}{1 - 1.2362z^{-1} + 0.7265z^{-2}}$

Topic 5: Tutorial Questions (SOLO-Level 2)

SOLO – Level 2

5. You are required to design a linear phase FIR low pass filter. The length of the impulse response is set to N = 7 and the coefficients are to be determined using a windowing technique. An ideal low pass filter $H_d(\theta)$, with linear phase can be characterised in the frequency domain by

$$H_d(\theta) = \begin{cases} e^{-j\beta\theta} & 0 \le |\theta| \le \frac{\pi}{2} \\ 0 & \text{otherwise} \end{cases}$$

(i) Determine β .

(ii) Determine the impulse response $h_d[n]$.

(i) $\beta = \frac{N-1}{2} = \frac{7-1}{2} = 3$

For a linear phase filter symmetrical coefficients centred around 3.

(ii) $h_d[n] = \frac{1}{2\pi} \int_{-\pi/2}^{\pi/2} e^{-j3\theta} e^{jn\theta} d\theta$

$= \frac{1}{2\pi} \left[\frac{e^{j(n-3)\theta}}{(n-3)j} \right]_{-\pi/2}^{\pi/2}$

$h_d[n] = \frac{1}{2\pi j(n-3)} \left[e^{j(n-3)\pi/2} - e^{-j(n-3)\pi/2} \right]$

$= \frac{1}{\pi(n-3)} \left(\frac{e^{j(n-3)\pi/2} - e^{-j(n-3)\pi/2}}{2j} \right)$

$= \sin(n-3)\frac{\pi}{2}$

$\therefore h_d[n] = \frac{\sin(n-3)\frac{\pi}{2}}{\pi(n-3)}$

$h_d[3] = \frac{1}{2}$ [for $n=3$]

Topic 5: Tutorial questions (SOLO - Level 3)

SOLO – Level 3

1. (a) A second-order analogue band pass filter with an s-domain transfer function is given by

$$H(s) = \frac{b_p s}{s^2 + b_p s + \omega_p^2} \qquad (1)$$

Where ω_p and b_p are the centre frequency and bandwidth of the filter respectively, both expressed in rad/s. By applying the bilinear transformation to equation (1) a digital filter with the following transfer function can be obtained:

$$H(z) = \frac{a_0 - a_1 z^{-2}}{1 + b_1 z^{-1} + b_2 z^{-2}} \qquad (2)$$

Show that the digital filter coefficients are given by

$$a_0 = a_1 = \frac{2b_pT}{4 + 2b_pT + \omega_p^2 T^2}; \quad b_1 = \frac{2\omega_p^2 T^2 - 8}{4 + 2b_pT + \omega_p^2 T^2}; \quad b_2 = \frac{4 + \omega_p^2 T^2 - 2b_pT}{4 + 2b_pT + \omega_p^2 T^2}$$

$$H(s) = \frac{b_p s}{s^2 + b_p s + \omega_p^2}$$

$$\therefore H(z) = \frac{b_p \cdot \frac{2}{T}\left(\frac{1 - z^{-1}}{1 + z^{-1}}\right)}{\left(\frac{2}{T}\right)^2\left(\frac{1 - z^{-1}}{1 + z^{-1}}\right)^2 + b_p \cdot \frac{2}{T}\left(\frac{1 - z^{-1}}{1 + z^{-1}}\right) + \omega_p^2}$$

$$= \frac{2b_pT\,(1 - z^{-1})(1 + z^{-1})}{4(1 - z^{-1})^2 + 2b_pT(1 - z^{-1})(1 + z^{-1}) + \omega_p^2 T^2 (1 + z^{-1})^2}$$

$$= \frac{2b_pT - 2b_pT z^{-2}}{(4 + 2b_pT + \omega_p^2 T^2) + (2\omega_p^2 T^2 - 8) z^{-1} + (4 - 2b_pT + \omega_p^2 T^2) z^{-2}}$$

$$H(z) = \frac{a_0 - a_1 z^{-2}}{1 + b_1 z^{-1} + b_2 z^{-2}}$$

$$\therefore a_0 = a_1 = \frac{2b_pT}{4 + 2b_pT + \omega_p^2 T^2}; \quad b_1 = \frac{2\omega_p^2 T^2 - 8}{4 + 2b_pT + \omega_p^2 T^2}; \quad b_2 = \frac{4 - 2b_pT + \omega_p^2 T^2}{4 + 2b_pT + \omega_p^2 T^2}$$

(b) A digital filter with a centre frequency of 1000 Hz and a bandwidth of 150 Hz is required. Assuming a sampling frequency of 10kHz, compute the digital filter coefficients a_0, a_1, b_1 & b_2 and show that $a_0 = a_1 = 0.04409115$; $b_1 = -1.551$ and $b_2 = 0.918176$.

$\omega_p = 2\pi \times 1000 = 2000\pi$ rad/s

$b_p = 2\pi \times 150 = 300\pi$ rad/s

In order to compensate for non-linear nature of the bilinear transform, it is necessary to prewarp these frequencies:

$$\omega_p' = \frac{2}{T}\tan\left(\omega_p \frac{T}{2}\right) = \frac{2}{10^{-4}}\tan\left(2000\pi \times \frac{10^{-4}}{2}\right) = 6498.4 \text{ rad/s}$$

$$b_p' = \frac{2}{10^{-4}}\tan\left(300\pi \times \frac{10^{-4}}{2}\right) = 943.2 \text{ rad/s}$$

(c) Comment on the stability of the digital filter H(z) (see equation 2 above) which you have obtained .

$$\therefore 4 + 2b_pT + \omega_p^2 T^2 = 4 + (2 \times 943.2 \times 10^{-4}) + (6498.4 \times 10^{-4}) = 4.610926$$

$$a_0 = a_1 = \frac{2 \times 943.2 \times 10^{-4}}{4.610926} = 0.04409115$$

$$b_1 = \frac{2 \times (6498.4 \times 10^{-4})^2 - 8}{4.610926} = -1.551$$

$$b_2 = \frac{4.233646}{4.610926} = 0.918176$$

$-2 \leq b_1 = -1.551 \leq 2$

$-1 \leq b_2 = 0.918176 \leq 1$

Stable filter (see stability triangle)

Topic 5: Tutorial questions (SOLO - Level 3)

SOLO – Level 3

2. Using the second order normalised Butterworth lowpass filter G(s), design an IIR digital filter G(z) to approximate the analogue transfer function G(s). The digital filter is to be designed to operate at a sampling rate of 8 kHz with a 3 dB cut-off frequency at 2 kHz. Using the bilinear transformation method, obtain the transfer function G(z) of the digital filter and show that G(z):

$$G(z) = \frac{(z+1)^2}{(2+\sqrt{2})\left(z^2+\frac{2-\sqrt{2}}{2+\sqrt{2}}\right)}$$

$H(s) = \frac{1}{s^2+\sqrt{2}s+1}$

$f_c = 2\,kHz$, $f_s = 8\,kHz$

$\theta_c = 2\pi\left(\frac{2000}{8000}\right) = \pi/2$

Prewarp the frequency f_c (ie θ_c)

$\omega_c' = \frac{2}{T}\tan\left(\frac{\theta_c}{2}\right) = \frac{2}{T}\tan\left(\frac{\pi/2}{2}\right) = \frac{2}{T}$ rad/sec

$\therefore$ Denormalised transfer function

$\therefore H(s)\Big|_{s \to \frac{s}{\omega_c'}} = \frac{1}{\left(\frac{s}{\omega_c'}\right)^2+\sqrt{2}\left(\frac{s}{\omega_c'}\right)+1}$

$H(s)\Big|_{\text{denormalised}} = \frac{(\omega_c')^2}{s^2+\sqrt{2}\,\omega_c' s+(\omega_c')^2}$

$\omega_c' = \frac{2}{T}$

$\therefore H(s) = \frac{\left(\frac{2}{T}\right)^2}{s^2+\frac{2}{T}\sqrt{2}s+\left(\frac{2}{T}\right)^2}$

$\therefore H(z) = \frac{\left(\frac{2}{T}\right)^2}{\left(\frac{2}{T}\right)^2\left(\frac{z-1}{z+1}\right)^2+\frac{2}{T}\sqrt{2}\left(\frac{2}{T}\frac{z-1}{z+1}\right)+\left(\frac{2}{T}\right)^2}$

$H(z) = \frac{1}{\left(\frac{z-1}{z+1}\right)^2+\sqrt{2}\left(\frac{z-1}{z+1}\right)+1}$

$H(z) = \frac{(z+1)^2}{z^2(2+\sqrt{2})+2-\sqrt{2}}$

$\therefore H(z) = \frac{(z+1)^2}{(2+\sqrt{2})\left(z^2+\frac{2-\sqrt{2}}{2+\sqrt{2}}\right)}$

Topic 5: Tutorial questions (SOLO - Level 3)

SOLO – Level 3

3. (a) Determine the impulse response $h_d[n]$ of the bandpass filter whose frequency response is given by

$$H_d(\theta) = \begin{cases} e^{-j3\theta} & \frac{\pi}{4} \le |\theta| \le \frac{3\pi}{4} \\ 0 & \text{otherwise} \end{cases}$$

(b) To obtain a finite impulse response from $h_d[n]$ a Bartlett window of length N = 7 is used. Compute the coefficients of the FIR filter with this impulse response.

Note: The Bartlett Window function is given by

$$w_B[n] = \begin{cases} \frac{2n}{N-1} & 0 \le n < \frac{(N-1)}{2} \\ 2 - \frac{2n}{N-1} & \frac{N-1}{2} \le n \le N-1 \\ 0 & \text{otherwise} \end{cases}$$

$$h_d[n] = \frac{\sin(n-3)\frac{3\pi}{4}}{\pi(n-3)} - \frac{\sin(n-3)\frac{\pi}{4}}{\pi(n-3)} \qquad n \neq 3$$

$$= \frac{1}{2} \quad \text{when } n = 3$$

(b) $N = 7$

$W_B[0] = 0$, $W_B[1] = \frac{1}{3}$, $W_B[2] = \frac{2}{3}$, $W_B[3] = 1$

$W_B[4] = \frac{2}{3}$, $W_B[5] = \frac{1}{3}$, $W_B[6] = 0$ } symmetrical

$\therefore h[n] = h_d[n] \cdot w[n]$

$h[0] = h_d[0] \cdot W_B[0] = 0$

$h[1] = h_d[1] \cdot W_B[1] = -\frac{1}{3\pi}$

$h[2] = h_d[2] \cdot W_B[2] = 0$

$h[3] = h_d[3] \cdot W_B[3] = \frac{1}{2}$ ($h_d[3] = \frac{1}{2}$, $W_B[3] = 1$)

$h[4] = h[2] = 0$

$h[5] = h[1] = -\frac{1}{3\pi}$

$h[6] = h[0] = 0$

$$h_d[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} H_d(\theta) e^{jn\theta} d\theta$$

$$= \frac{1}{2\pi}\int_{-3\pi/4}^{-\pi/4} e^{-j3\theta} \cdot e^{jn\theta} d\theta + \frac{1}{2\pi}\int_{\pi/4}^{3\pi/4} e^{-j3\theta} \cdot e^{jn\theta} d\theta$$

$$= \frac{1}{2\pi}\left[\frac{e^{j(n-3)\theta}}{j(n-3)}\right]_{-3\pi/4}^{-\pi/4} + \frac{1}{2\pi}\left[\frac{e^{j(n-3)\theta}}{j(n-3)}\right]_{\pi/4}^{3\pi/4}$$

$$= \frac{1}{2\pi j(n-3)}\left[e^{-j(n-3)\frac{\pi}{4}} - e^{-j(n-3)\frac{3\pi}{4}} + e^{j(n-3)\frac{3\pi}{4}} - e^{j(n-3)\frac{\pi}{4}}\right]$$

Topic 5: Tutorial questions (SOLO - Level 3)

SOLO – Level 3

4. Using impulse invariant method design a digital filter H(z) to approximate the following normalized analogue transfer function:

$$H(s) = 1/(s^2 + \sqrt{2}\,s + 1)$$

Assuming that the 3dB cut-off frequency of the digital filter is 150Hz and the sampling frequency is 1.28 kHz, Show that H(z) is given by:

$$H(z) = \frac{393.9264z^{-1}}{1 - 1.0308z^{-1} + 0.3530z^{-2}}$$

Before applying the impulse invariant method, we need to denormalise the transfer function.

$$H_1(s) = H(s)\Big|_{s \to \frac{s}{\omega_c}} = \frac{1}{\left(\frac{s}{\omega_c}\right)^2 + \sqrt{2}\left(\frac{s}{\omega_c}\right) + 1}$$

$$H_1(s) = \frac{\omega_c^2}{s^2 + \sqrt{2}\,\omega_c s + \omega_c^2} \quad \text{where } \omega_c = 2\pi \times 150 = 942.4778$$

$$H_1(s) = \frac{\omega_c^2}{s^2 + \sqrt{2}\omega_c s + \omega_c^2} = \frac{A}{s - p_1} + \frac{B}{s - p_2}$$

$$p_1, p_2 = \frac{-\sqrt{2}\omega_c \pm \sqrt{2\omega_c^2 + 4\omega_c^2}}{2}$$

$$= -\sqrt{2}\omega_c\left(\frac{1 \pm j}{2}\right)$$

$$\therefore p_1 = -\frac{\sqrt{2}\omega_c}{2}(1+j) = -666.4324(1+j)$$

$$p_2 = -\frac{\sqrt{2}\omega_c}{2}(1-j) = -666.4324(1-j)$$

Find A & B

$$\Rightarrow A = -\frac{\omega_c}{\sqrt{2}}j$$

$$B = \frac{\omega_c}{\sqrt{2}}j$$

Use impulse invariant transformation

$$H_1(z) = \frac{A}{1 - e^{p_1 T} z^{-1}} + \frac{B}{1 - e^{p_2 T} z^{-1}} \qquad T = \frac{1}{1.28\,kHz}$$

$$= \frac{A(1 - e^{p_2 T} z^{-1}) + B(1 - e^{p_1 T} z^{-1})}{(1 - e^{p_1 T} z^{-1})(1 - e^{p_2 T} z^{-1})}$$

$$= \frac{(A+B) - (A e^{p_2 T} + B e^{p_1 T}) z^{-1}}{1 - (e^{p_1 T} + e^{p_2 T}) z^{-1} + e^{p_1 T} z^{-1}}$$

By substituting for A, B, T, p_1 & p_2

$$H_1(z) = \frac{393.9264\,z^{-1}}{1 - 1.0308\,z^{-1} + 0.3530\,z^{-2}}$$

Topic 5: Tutorial questions (SOLO-Level 4)

SOLO – Level 4

1. Convert an analogue filter $H(s) = \frac{s+0.1}{(s+0.1)^2+16}$ into a digital IIR filter by means of the bilinear transformation. The digital filter is to have a resonant frequency $\theta_0 = \frac{\pi}{2}$.

The analogue filter has a resonant frequency $\omega_0 = 4$ rad/sec

We have $\omega_0 = \frac{2}{T}\tan\frac{\theta_0}{2} \Rightarrow 4 = \frac{2}{T}\tan\left(\frac{\pi/2}{2}\right) \Rightarrow T = \frac{1}{2}$ in order to have $\theta_0 = \pi/2$ rad/sec

Thus the desired mapping is

$$s = \frac{2}{T}\left(\frac{1-z^{-1}}{1+z^{-1}}\right) = 4\left(\frac{1-z^{-1}}{1+z^{-1}}\right)$$

$$H(z) = H(s)\Big|_{s=4\left(\frac{1-z^{-1}}{1+z^{-1}}\right)} = \frac{4\left(\frac{1-z^{-1}}{1+z^{-1}}\right)+0.1}{\left[4\left(\frac{1-z^{-1}}{1+z^{-1}}\right)\right]^2 + 0.2\times 4\left(\frac{1-z^{-1}}{1+z^{-1}}\right) + 16.00}$$

$$H(z) = \frac{0.128 + 0.0006z^{-1} - 0.122z^{-2}}{1 + 0.0006z^{-1} + 0.975z^{-2}} = \frac{0.128 + 0.0006z^{-1} - 0.122z^{-2}}{1 + 0.975z^{-2}}$$

This filter has Poles at:

$p_1 = 0.987\, e^{j(\pi/2)}$ ← resonant frequency

$p_2 = 0.987\, e^{-j(\pi/2)}$ ← resonant frequency

Topic 5: Tutorial questions (SOLO-Level 4)

SOLO – Level 4

2. Consider the following analogue system with a transfer function $H(s) = \frac{\alpha}{s+\alpha} \Rightarrow h(t) = e^{-\alpha t}$; where α (= 10^4 rad/sec) is the analogue cut-off frequency and the sampling period T is 100 μs.

   (a) Using bilinear transformation, show that the transfer function H(z) is: $H(z) = \frac{1-b}{2}\left[1 + \frac{(1+b)z^{-1}}{1-bz^{-1}}\right]$; where b= $\frac{2-\alpha T}{2+\alpha T}$

   (b) (i) What is the dc gain of H(z)?

   (ii) At what frequency is the H(θ) equal to zero? (θ - digital frequency).

   (iii) Calculate the impulse response h[n].

   (iv) Assuming that the impulse response decays to 1/e of its initial value at n = N samples, show that: N =(ln{b/(b+1)}-1)/(ln(b))

$H(s) = \frac{\alpha}{s+\alpha}$; $H(z) = H(s)\Big|_{s=\frac{2}{T}\frac{1-z^{-1}}{1+z^{-1}}}$

$\therefore H(z) = \frac{\alpha}{\frac{2}{T}\frac{1-z^{-1}}{1+z^{-1}}+\alpha} = \frac{\alpha T(1+z^{-1})}{2(1-z^{-1})+\alpha T(1+z^{-1})}$

$H(z) = \frac{\alpha T(1+z^{-1})}{(2+\alpha T)+(\alpha T-2)z^{-1}} = \frac{\alpha T}{2+\alpha T}\left[\frac{1+z^{-1}}{1-\left(\frac{2-\alpha T}{2+\alpha T}\right)z^{-1}}\right]$

note : Let $b = \frac{2-\alpha T}{2+\alpha T} \Rightarrow 1-b = \frac{2+\alpha T-2+\alpha T}{2+\alpha T}$

$1-b = \frac{2\alpha T}{2+\alpha T}$

$\therefore \frac{1-b}{2} = \frac{\alpha T}{2+\alpha T}$

$H(z) = \frac{1-b}{2}\left[\frac{1+z^{-1}}{1-bz^{-1}}\right] = \frac{1-b}{2}\left[1 + \frac{(1+b)z^{-1}}{1-bz^{-1}}\right]$

$b = \frac{2-10^4\times 100\times 10^{-6}}{2+10^4\times 100\times 10^{-6}} = \frac{1}{3}) \therefore \frac{1-b}{2} = \frac{1}{3}$

$1+b = 4/3$

$H(z) = \frac{1}{3}\left[1 + \frac{\frac{4}{3}z^{-1}}{1-\frac{1}{3}z^{-1}}\right]$

$H(\theta) = \frac{1}{3}\left[1 + \frac{\frac{4}{3}e^{-j\theta}}{1-\frac{1}{3}e^{-j\theta}}\right]$

(i) $H(0) = 1 + \left(\frac{\frac{4}{3}}{1-\frac{1}{3}}\right) = 1$; (ii) $H(\theta)\Big|_{\theta=\pi} = \frac{1}{3}\left[1 + \frac{4e^{-j\pi}}{3-e^{-j\pi}}\right] = 0$

DC gain

$\therefore \theta = \pi$

(iii) Inverse z transform

$h[n] = \frac{1-b}{2}\left[\delta[n] + (1+b)\,b^{n-1}u[n-1]\right]$

(iv) $h[n]\Big|_{n=N} = \frac{1-b}{2}\left[\underbrace{\delta[N]}_{0} + (1+b)\,b^{N-1}\underbrace{u[N-1]}_{=1}\right]$

$h[n] = \frac{1-b}{2}(1+b)\,b^{N-1}$

We have now $\frac{1}{e}h[0] = \frac{1-b}{2}(1+b)\,b^{N-1}$

$\frac{1}{e}\left(\frac{1-b}{2}\right) = \left(\frac{1-b}{2}\right)\cdot(1+b)\,b^{N-1} \Rightarrow \frac{1}{e} = \frac{1+b}{b}\,b^{N}$

$\therefore \ln\left(\frac{1}{e}\right) = \ln\left(\frac{1+b}{b}\right) + N\ln(b)$

$\therefore N\ln(b) = -\ln\left(\frac{1+b}{b}\right) + \ln\left(\frac{1}{e}\right) = \ln\left(\frac{b}{1+b}\right) - 1$

$\therefore N = \frac{\ln\left(\frac{b}{1+b}\right)-1}{\ln(b)}$

Topic 5: Tutorial questions (SOLO-Level 4)

SOLO – Level 4

3. A second-order resonant system is given by $H(s) = \frac{1}{s^2+b_p s+\omega_p^2}$ and $b_p = \frac{\omega_p}{Q_p}$ {ω_p is the resonant frequency, Q_p is the Q - factor associated with the pole , b_p is the bandwidth}. By applying the impulse invariant transformation to the above equation, a digital filter with the transfer function $H(z)$ can be obtained: $H(z) = \frac{k_o z^{-1}}{1+b_1 z^{-1}+b_2 z^{-2}}$. Show that the filter coefficients b_1 and b_2 are given by,

$$b_1 = -2e^{-p_2 T}\cos(q_2 T);\ b_2 = e^{-2p_2 T} \quad \text{where} \quad p_2 = \frac{\omega_p}{2Q_p};\ q_2 = p_2\sqrt{4Q_p^2 - 1}$$

Also show that $k_o = -4q_2 e^{-p_2 T}\sin(q_2 T)$ { T is the sampling period}

$$H(s) = \frac{1}{s^2 + \frac{\omega_p}{Q_p}s + \omega_p^2} = \frac{1}{s^2 + b_p s + \omega_p^2}$$

$$s_{1,2} = \frac{-b_p \pm \sqrt{b_p^2 - 4\omega_p^2}}{2} = -\frac{b_p}{2} \pm j\sqrt{4\omega_p^2 - b_p^2}$$

$$= \frac{-\omega_p}{2Q_p} \pm j\frac{\omega_p}{2Q_p}\sqrt{4Q_p^2 - 1}$$

$$s_{1,2} = -p_2 \pm j q_2 \ ; \quad p_2 = \frac{-\omega_p}{2Q_p}; \quad q_2 = p_2\sqrt{4Q_p^2 - 1}$$

$$H(s) = \frac{1}{(s-s_1)(s-s_2)} = \frac{A}{s-s_1} + \frac{B}{s-s_2} = \frac{\frac{1}{s_1-s_2}}{s-s_1} + \frac{\frac{-1}{s_1-s_2}}{s-s_2}$$

$$H(s) = \frac{1}{s_1-s_2}\left[\frac{1}{s-s_1} - \frac{1}{s-s_2}\right]$$

↓ impulse invariant transform

$$H(z) = \frac{1}{s_1-s_2}\left[\frac{1}{1-e^{-p_2 T + j q_2 T}z^{-1}} - \frac{1}{1-e^{-p_2 T - j q_2 T}z^{-1}}\right]$$

$$H(z) = \frac{1}{s_1-s_2}\cdot\frac{z^{-1}\left(e^{jq_2T} - e^{-jq_2T}\right)e^{-p_2T}}{1 - 2e^{-p_2T}z^{-1}\left(\frac{e^{jq_2T}+e^{-jq_2T}}{2}\right) + e^{-2p_2T}z^{-2}}$$

($\cos(q_2T)$)

$$H(z) = 2jq_2 \cdot \frac{z^{-1}(\ )e^{-p_2T}}{1 + b_1 z^{-1} + b_2 z^{-2}}$$

($\sin(q_2T)$)

$$= \frac{-4q_2 e^{-p_2T} z^{-1}\left(\frac{e^{+jq_2T} - e^{-jq_2T}}{2j}\right)}{1 + b_1 z^{-1} + b_2 z^{-2}}$$

$$= \frac{K_0 z^{-1}}{1 + b_1 z^{-1} + b_2 z^{-2}}$$

$$K_0 = -4q_2 e^{-p_2T}\sin(q_2T)$$

$$b_1 = -2e^{-p_2T}\cos(q_2T)$$

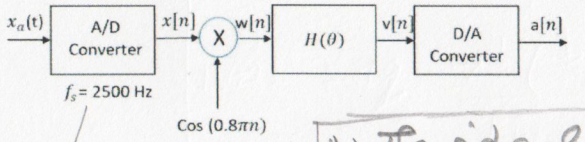
$$b_2 = e^{-2p_2T}$$

Topic 5: Tutorial questions (SOLO-Level 5)

SOLO – Level 5

1. An analogue signal of the form $x_a(t) = a(t) \cos 2000\pi t$ is bandlimited to the range $900 \le f \le 1100\ Hz$. It is used as an input to the system shown below:
   (a) Determine and sketch the spectra for the signal x[n] and w[n].
   (b) Use a Hamming window of length M= 7 to design a lowpass linear phase FIR filter $H(\theta)$ that passes a[n]

$f_c = 1000\ Hz = \frac{1000}{2500} = 0.4 \Rightarrow \theta_c = 0.8\pi$



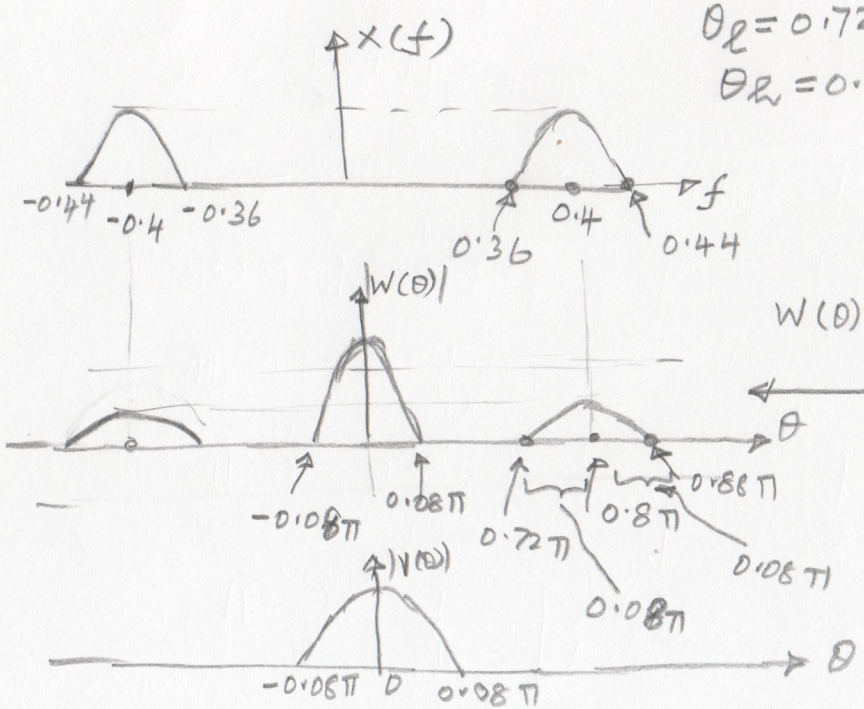
$W_H[n] = 0.54 - 0.46 \cos \frac{2\pi(n-3)}{6}$ [given]

$x_a(t) = a(t) \cos 2000\pi t$

$f_l = \frac{900}{2500} = 0.36$; $f_h = \frac{1100}{2500} = 0.44$

$\theta_l = 0.72\pi$

$\theta_h = 0.88\pi$



$W(\theta) = \frac{1}{2}\left[X(\theta-\theta_0) + X(\theta-\theta_0)\right]$

see Page 21 of lecture notes

(b) linear phase filter:

$M = 7$

Symmetry around $n = 3$

(b) The ideal lowpass filter has a passband of $-0.08\pi \le \theta \le 0.08\pi$,

Hence

$$H_d(\omega) = \begin{cases} 1 \cdot e^{-j(3\theta)} & -0.08\pi \le \theta \le 0.08\pi \\ 0 & \text{otherwise} \end{cases}$$

$$h_d[n] = \frac{1}{2\pi} \int_{-0.08\pi}^{0.08\pi} e^{-j3\theta} \cdot e^{jn\theta}\, d\theta$$

$$= \frac{1}{2\pi} \int_{-0.08\pi}^{0.08\pi} e^{j\theta(n-3)}\, d\theta$$

$$\boxed{h_d[n] = 2\,\frac{\sin 0.08\pi(n-3)}{(n-3)}}$$

$\therefore h[n] = h_d[n] \cdot W_H[n]$

$\left[W_H[n] = 0.54 - 0.46\left\{\cos\frac{2\pi(n-3)}{6}\right\}\right]$ given

$h[n]$ is the impulse response of the lowpass filter $H(\theta)$.

Topic 5: Tutorial questions (SOLO-Level 5)

SOLO – Level 5

2. A continuous time system is called an integrator if the response of the system $y_a(t)$ to an input $x_a(t)$ is :

$$y_a(t) = \int_{-\infty}^{t} x_a(\tau)d\tau$$

The system function for an integrator is $H_a(s) = \frac{1}{s}$

(a) Design a discrete –time integrator using bilinear transformation and find the difference equation relating the input x[n] to the output y[n] of the discrete-time system.

(b) Find the frequency response of the discrete-time integrator found in part (a) , and determine whether or not this system is a good approximation to the continuous-time system.

Bilinear Transform :

$s = \frac{2}{T}\frac{1-z^{-1}}{1+z^{-1}}$.

with a system function $H_a(s) = \frac{1}{s}$)

the bilinear transform produces a discrete-time filter with a system function

$H(z) = \frac{T}{2}\frac{1+z^{-1}}{1-z^{-1}}$

$\frac{Y(z)}{X(z)} = \frac{T}{2}\frac{1+z^{-1}}{1-z^{-1}}$

$Y(z)[1-z^{-1}] = \frac{T}{2}(1+z^{-1})X(z)$

$\therefore y[n] = \frac{T}{2}\{x[n]+x[n-1]\}+y[n-1]$

(b) Because the frequency response of the continuous-time system $H_a(j\omega) = \frac{1}{j\omega}$ is related to the discrete-time filter through the mapping $\omega = \frac{2}{T}\tan(\theta/2)$

the frequency response of the discrete-time system is

$H(\theta) = H_a(j\omega)\Big|_{\omega=\frac{2}{T}\tan\theta/2} = \frac{1}{j\cdot\frac{2}{T}\tan\theta/2}$

$H(\theta) = \frac{T}{2j}\cot\theta/2$ & $H_a(j\omega) = \frac{1}{j\omega}$

$H(\theta) = \frac{T}{2j}\frac{\cos\theta/2}{\sin\theta/2}$

for small θ, $H(\theta) = \frac{T}{2j}\cdot\frac{1}{\theta/2}$ ($\theta \approx 0$ small θ)

$= \frac{T}{2j}\cdot\frac{1}{\omega T/2}$

$= \frac{1}{\omega j}$

$\therefore H(\theta) \approx H_a(j\omega)$

ELEC3104: Digital Signal Processing

Topic 6: Multirate Systems

Model Answers

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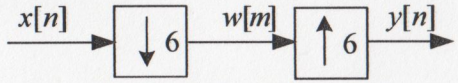
Term 3, 2021

Topic 6: Tutorial Questions (SOLO -Level 1)

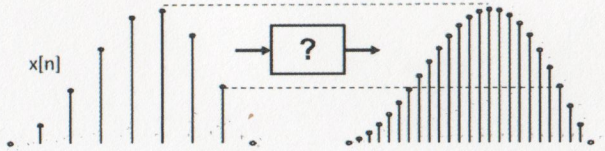
SOLO – Level 1

1. For the diagram shown below, tick the correct answer(s).



✓ a) w[m] = x[6n] ← Down sampling
b) w[m] = x[n/6]
c) y[n] = x[n/6]
✓ d) y[n] = w[m/6] ← upsampling
e) None of the above

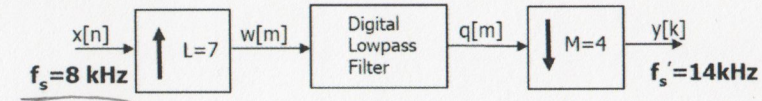
2. What operation is performed on the input sequence x[n] below? (Tick the correct answer)



a) Decimation by 2
b) Decimation by 3
c) Interpolation by 2
d) Interpolation by 3

SOLO – Level 1

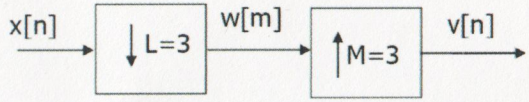
3. Figure below shows sampling rate conversion by non-integer factors. The minimum cut-off frequency of the digital lowpass filter is: (Tick the correct answer)



Bandwidth: 0 to 2 kHz

a) 4 kHz
b) 7 kHz
✓ c) 2 kHz
d) None of the above.

4. For the diagram shown below, tick the correct answer:



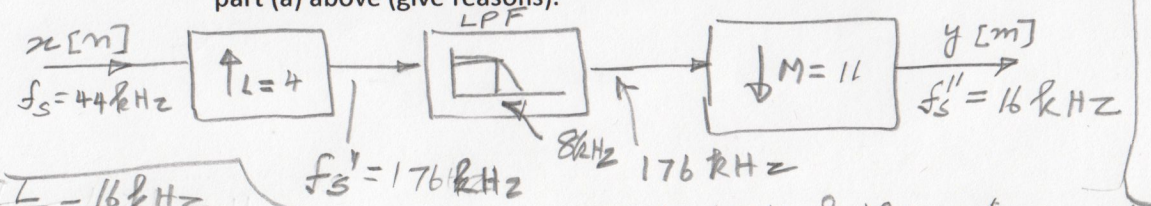
a) w[m] = x[n]
b) v[n]= w[m]
c) v[n]=x[n/3]
✓ d) None of the above

Topic 6: Tutorial Questions (SOLO-Level 2)

SOLO – Level 2

1. A signal has been sampled at 44 kHz. For a particular application, this signal is required to be used at a reduced sampling rate of 16 kHz.
   (a) Draw a block diagram to show how you would achieve this.
   (b) Indicate the type of digital filter which you would choose for part (a) above (give reasons).

(a)

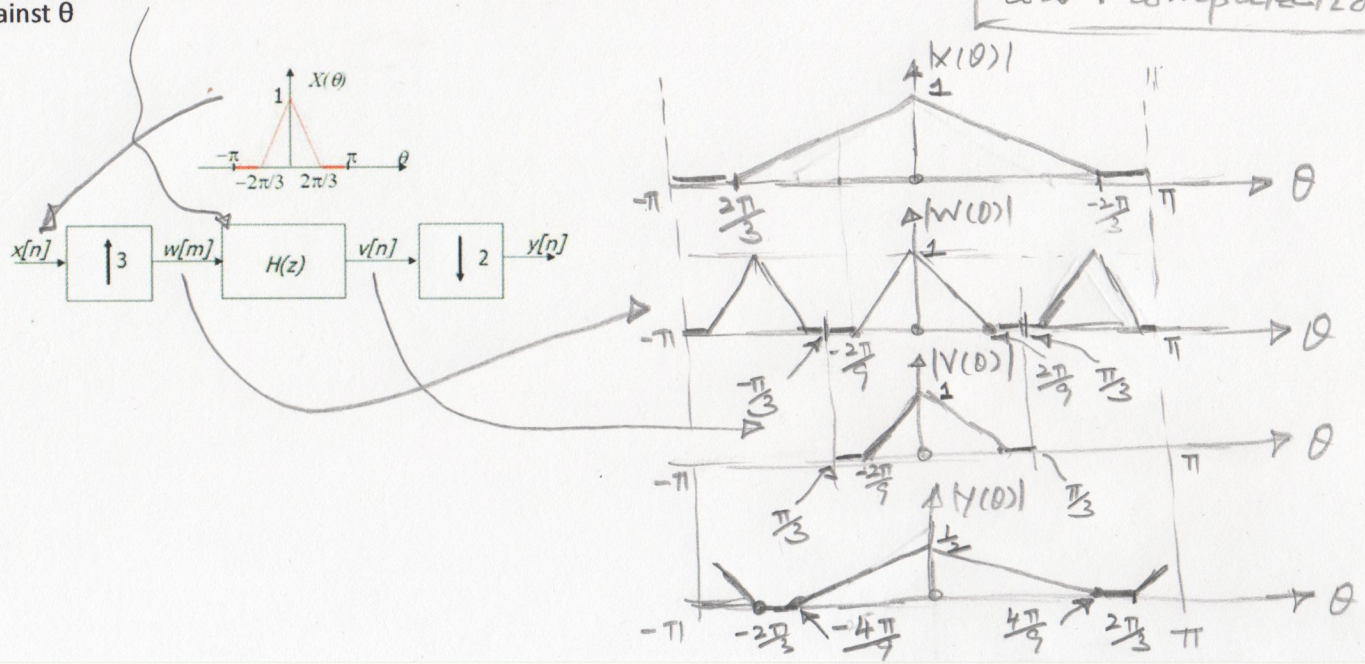


$\frac{L}{M} = \frac{16\,kHz}{44\,kHz} = \frac{4\times4}{4\times11} = \frac{4}{11}$ ⇒ up sample by L = 4 & then down sample by 11

→ The cut-off frequency of LPF ≈ $\frac{f_s''}{2}$ = 8 kHz.

(b) Using an IIR filter in this case has an obvious shortcoming. We cannot take advantage of the fact that we have to compute only every M^{th} output, since previous outputs are required to compute M^{th} output. Thus no saving realised. Using an FIR filter implies we can do our computations at f_s'/M since previous outputs are not required to compute the M^{th} output. This will lead to a significantly lower computation rate

2. A signal x[n] has a spectrum X(θ) as shown below. The signal is applied to the system shown below. The ideal lowpass filter H(z) has a gain factor of 1 in the passband and a cut-off frequency θ=π/3. Sketch |X(θ)|, |W(θ)|, |V(θ)|, |Y(θ)| against θ

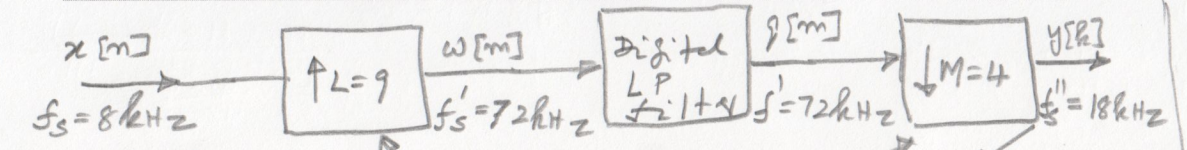


Topic 6: Tutorial Questions (SOLO-Level 2)

SOLO – Level 2

3. A signal (with a bandwidth of 0 to 3kHz) has been sampled at 8 kHz and for a particular application this signal is required to be used at an increased sampling rate of 18 kHz:
   (a) Draw a block diagram to show how you would efficiently implement a system to achieve this , indicating the type of digital filter you would use (give reasons);
   (b) Calculate the minimum cut-off frequency of the digital lowpass filter in terms of rad/sec.

(a)



$f_s = 8kHz$, $BW = 0$ to $3kHz$

$f_s' = 72kHz$, $f_s' = 72kHz$, $f_s'' = 18kHz$

$$\frac{L}{M} = \frac{18}{8} = \frac{2\times 9}{2\times 4} = \frac{9}{4}$$

$$f_s'' = \frac{L}{M} f_s = \frac{9}{4} \times 8kHz = 18kHz$$

(b) * The cut off frequency (min) of the lowpass filter must be 3kHz (because the input signal is band limited to 3kHz)

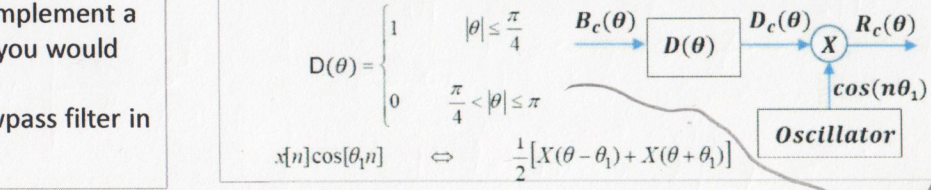
Type of digital filter is FIR

See Question 1 (b) answer Page 3

SOLO – Level 2

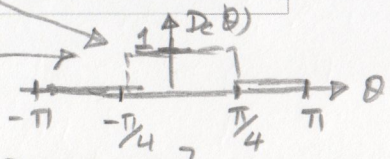
4. A section of a digital modulator is shown below. The input $B_C(\theta) = 1$ for $-\pi \le \theta \le \pi$, By writing appropriate equations for $R_c(\theta)$ and sketch $|R_c(\theta)|$ vs θ, and where $\theta_1 = 3\pi/4$.

$$D(\theta) = \begin{cases} 1 & |\theta| \le \frac{\pi}{4} \\ 0 & \frac{\pi}{4} < |\theta| \le \pi \end{cases}$$



$$x[n]\cos[\theta_1 n] \Leftrightarrow \frac{1}{2}[X(\theta-\theta_1)+X(\theta+\theta_1)]$$

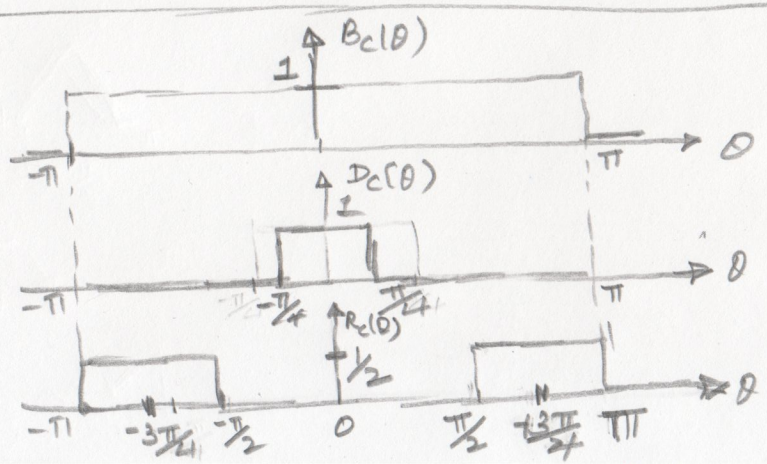
$B_c(\theta) = 1$; $D_c(\theta) = 1$



$$R_c(\theta) = \frac{1}{2}[D_c(\theta-\theta_1) + D_c(\theta+\theta_1)]$$

$$= \frac{1}{2}[D_c(\theta-\tfrac{3\pi}{4}) + D_c(\theta+\tfrac{3\pi}{4})]$$

$$R_c(\theta) = \begin{cases} \frac{1}{2} & -\frac{\pi}{4} \le (\theta-\frac{3\pi}{4}) \le \frac{\pi}{4} \Rightarrow \frac{\pi}{2} \le \theta \le \pi \\ \frac{1}{2} & -\frac{\pi}{4} \le (\theta+\frac{3\pi}{4}) \le \frac{\pi}{4} \Rightarrow -\pi \le \theta \le -\frac{\pi}{2} \\ 0 & \text{elsewhere} \end{cases}$$



Topic 6: Tutorial questions (SOLO - Level 3)

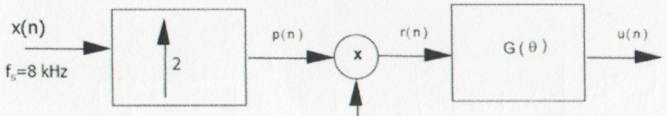
SOLO – Level 3

1. A section of a digital modulator is shown below. If the input signal x[n] is

$$x[n] = \frac{1}{2}\delta[n+1] + \delta[n] + \frac{1}{2}\delta[n-1]$$

(a) Show $P(\theta) = 1+\cos(2\theta)$ for $-\pi \le \theta \le \pi$,

(b) By writing appropriate equations for $R(\theta)$ and $U(\theta)$, sketch $|R(\theta)|$ and $|U(\theta)|$ vs θ, and where $\theta_1 = \pi/4$.



$$G(\theta) = \begin{cases} 1/2 & 0 \le |\theta| \le \frac{\pi}{4} \\ 0 & \frac{\pi}{4} < |\theta| \le \pi \end{cases}$$

(a) $X(z) = \frac{1}{2}z^{1} + 1 + \frac{1}{2}z^{-1} \quad \therefore X(\theta) = \frac{1}{2}e^{j\theta} + 1 + \frac{1}{2}e^{-j\theta}$

$\therefore X(\theta) = 1+\cos\theta \quad -\pi \le \theta \le \pi$

$\therefore P(\theta) = X(2\theta) = 1+\cos 2\theta$

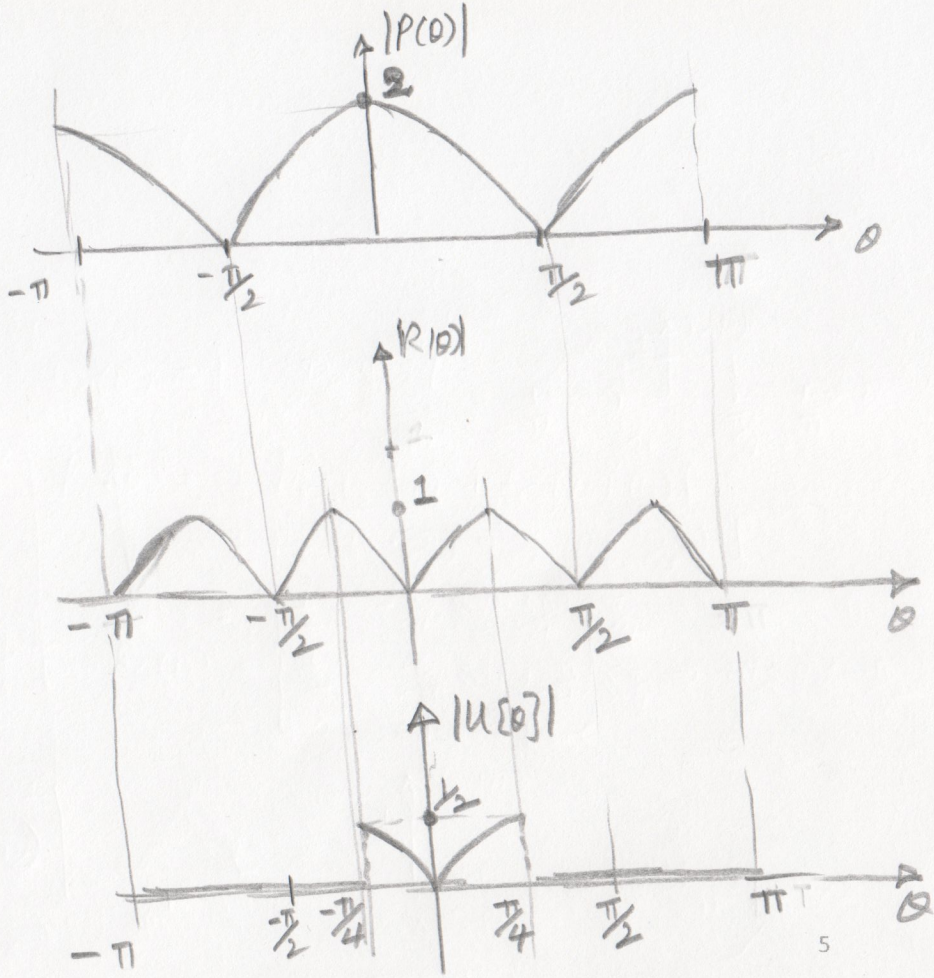
(b) $R(\theta) = \frac{1}{2j}\cdot[P(\theta-\theta_1) - P(\theta+\theta_1)]$

$= \frac{1}{2j}\cdot[1+\cos 2(\theta-\frac{\pi}{4}) - \cos 2(\theta+\frac{\pi}{4}) - 1]$

$= \frac{1}{2j}\cdot[\sin 2\theta + \sin 2\theta] = \frac{1}{j}\sin 2\theta$

$U[\theta] = R[\theta]\cdot G_1[\theta] = \begin{cases} \frac{1}{2j}\cdot\sin 2\theta & -\frac{\pi}{4} \le \theta \le \frac{\pi}{4} \\ 0 & \frac{\pi}{4} \le |\theta| \le \pi \end{cases}$

Given



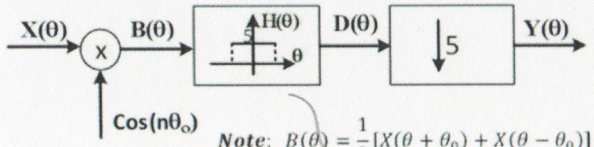
Topic 6: Tutorial questions (SOLO - Level 3)

SOLO – Level 3

2. The input $X(\theta)$ for the digital modulator shown below is given by:

$$X(\theta) = \begin{cases} 1 & 0.4\pi \le |\theta| \le 0.8\pi \\ 0 & elsewhere \end{cases}$$

The modulator signal frequency is $\theta_o = 0.6\pi$. $H(\theta)$ is an ideal lowpass filter with a cut-off frequency of 0.2π and a gain of 5. Sketch, one above another, $|X(\theta)|$, $|B(\theta)|$, $|D(\theta)|$ and $|Y(\theta)|$ against θ.

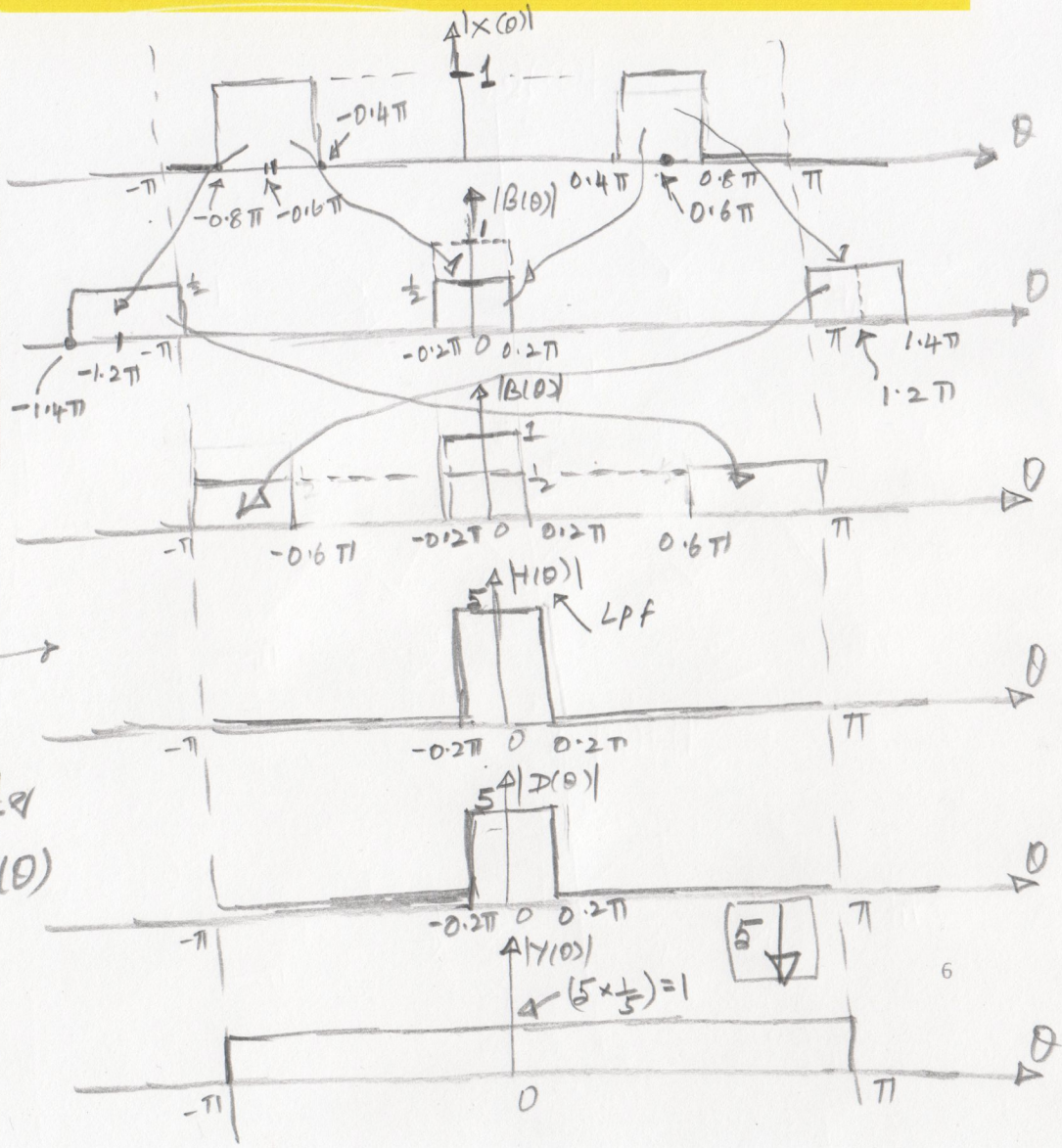


Note: $B(\theta) = \frac{1}{2}[X(\theta + \theta_0) + X(\theta - \theta_0)]$

Lowpass filter

After lowpass filter

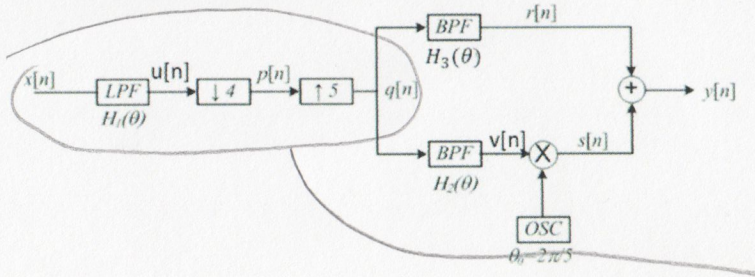
$D(\theta) = B(\theta) \cdot H(\theta)$



Topic 6: Tutorial questions (SOLO – Level 4)

SOLO – Level 4

3. A multi-rate system is given below:

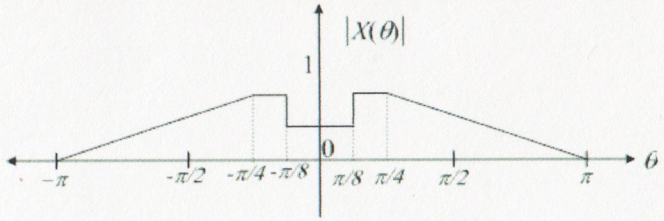


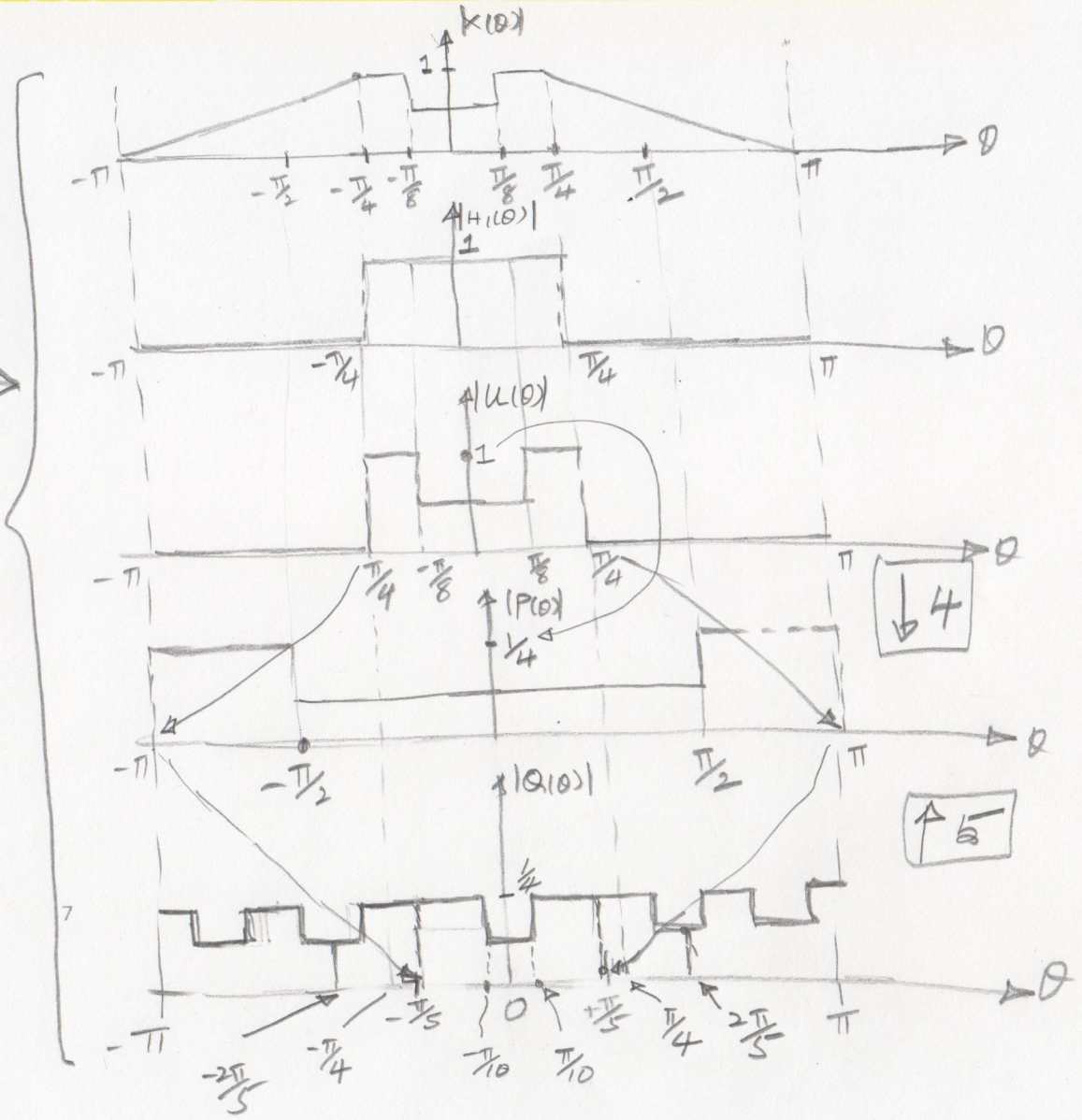
where

$$H_1(\theta) = \begin{cases} 1 & |\theta| \le \pi/4 \\ 0 & otherwise \end{cases} \qquad H_2(\theta) = \begin{cases} 1 & \pi/5 \le |\theta| \le 2\pi/5 \\ 0 & otherwise \end{cases}$$

$$H_3(\theta) = \begin{cases} 1 & 2\pi/5 \le |\theta| \le 3\pi/5 \\ 0 & otherwise \end{cases}$$

If the frequency spectrum of $x[n]$, $|X(\theta)|$ is given as follows, sketch $|P(\theta)|$, $|Q(\theta)|$, $|R(\theta)|$, $|S(\theta)|$ and $|Y(\theta)|$.

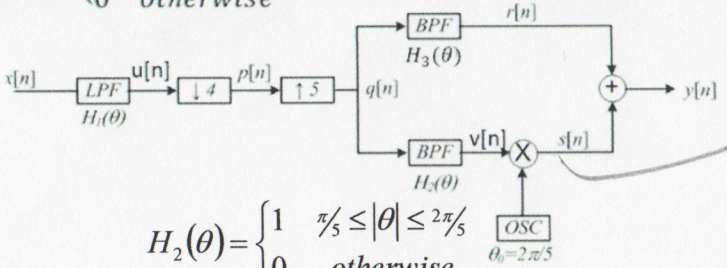




SOLO – Level 4

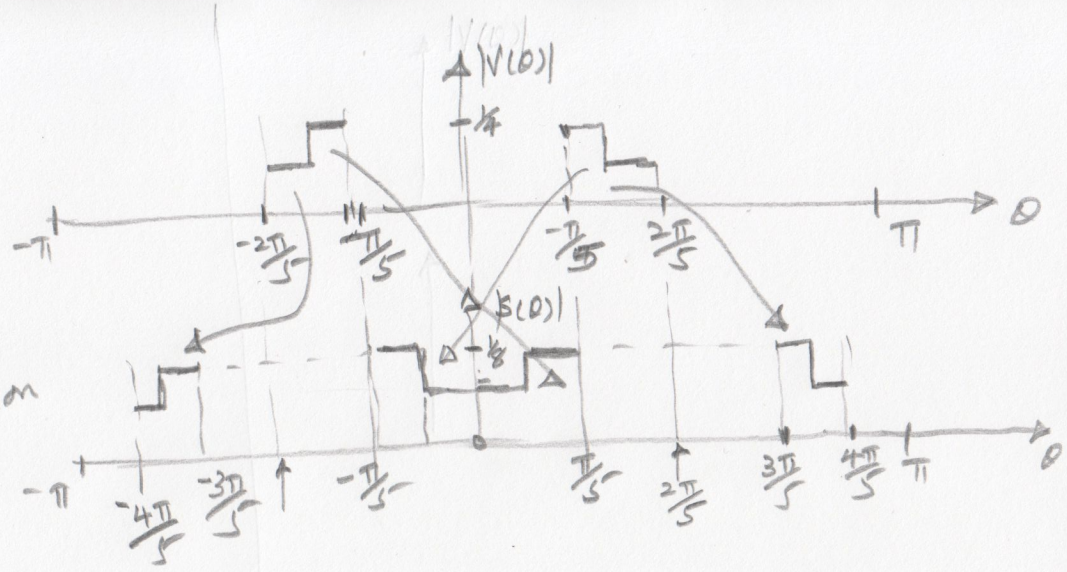
$$H_1(\theta)=\begin{cases}1 & |\theta|\le \pi/4\\ 0 & otherwise\end{cases}$$

$$H_3(\theta)=\begin{cases}1 & 2\pi/5\le|\theta|\le 3\pi/5\\ 0 & otherwise\end{cases}$$



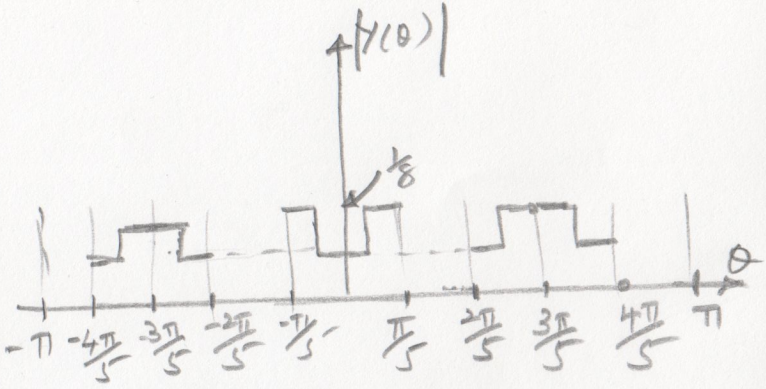
$$H_2(\theta)=\begin{cases}1 & \pi/5\le|\theta|\le 2\pi/5\\ 0 & otherwise\end{cases}$$

after Modulation



If we now add R(θ) and S(θ), we get

$V(\theta) = S(\theta) + R(\theta)$



Topic 6: Tutorial questions (SOLO - Level 4)

SOLO – Level 4

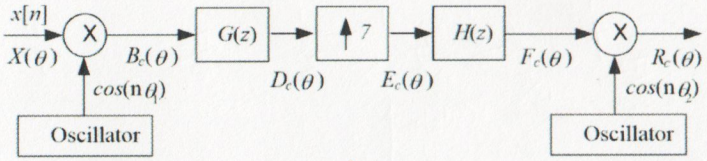
2. The diagram below shows a section of a digital modulator. The sampling period T of the input signal x[n] is 125 μs . The first oscillator generates a carrier of frequency θ_1. The second oscillator generates a carrier of frequency θ_2.

$$\theta_1 = \omega_1 \frac{T}{2} \quad where \omega_1 = 2\pi \times 2 \times 10^3 rad/\sec; \qquad \theta_2 = (\omega_1 + \omega_c)\frac{T}{2}$$

$$where, \omega_c = 2\pi \times 4 \times 10^3 rad/\sec$$

The low-pass filters G(z) and H(z) have the following characteristics:

$$G(\theta) = \begin{cases} 1 & 0 \le |\theta| \le \frac{\pi}{4} \\ 0 & \frac{\pi}{4} < |\theta| \le \pi \end{cases} \qquad H(\theta) = \begin{cases} 1 & 0 \le |\theta| \le \frac{\pi}{28} \\ 0 & \frac{\pi}{28} < |\theta| \le \pi \end{cases}$$



(a) If $X(\theta) = 1 + \cos(2\theta)$, by writing appropriate equations, show that

$$D_c(\theta) = \begin{cases} 1 & -\pi/4 \le \theta \le \pi/4 \\ 0 & elsewhere \end{cases}$$

(b) Sketch, one above another, $|E_c(\theta)|$, $|F_c(\theta)|$ and $|R_c(\theta)|$ against θ in the fundamental frequency region $-\pi \le \theta \le \pi$.

$\theta_1 = \omega_1 T/2 = \omega_1 \times \frac{1}{2f_s} = 2\pi \times 2\times10^3 \times \frac{1}{2\times8\times10^3} = \pi/4$

$\theta_2 = (\omega_1+\omega_c)T/2 = \frac{\omega_1+\omega_c}{2f_s} = \frac{2\pi\times2\times10^3 + 2\pi\times4\times10^3}{2\times8\times10^3} = 3\pi/4$

given $\rightarrow X(\theta) = 1+\cos 2\theta$

$B_c(\theta) = \frac{1}{2}[X(\theta-\theta_1) + X(\theta+\theta_1)]$

$= \frac{1}{2}[X(\theta-\pi/4) + X(\theta+\pi/4)]$

$= \frac{1}{2}[\cos 2(\theta-\pi/4) + \cos(2(\theta+\pi/4))]$

$= \frac{1}{2}[1+\sin 2\theta + 1 - \sin 2\theta] = 1$

$D_c(\theta) = B_c(\theta) \cdot G(\theta) = \begin{cases} 1 & -\pi/4 \le \theta \le \pi/4 \\ 0 & elsewhere. \end{cases}$

$= 1$ (given)

without taking equation you can sketch the spectrum.

$F_c[\theta] = D_c(7\theta) \circ H(\theta) = \begin{cases} \begin{cases} 1 & -\pi/4 \le 7\theta \le \pi/4 \\ 0 & \pi/4 \le |7\theta| \le \pi \end{cases} \\ 0 & \frac{\pi}{7} \le |\theta| \le \pi \end{cases}$

$= \begin{cases} 1 & -\frac{\pi}{28} \le \theta \le \pi/28 \\ 0 & \pi/28 \le |\theta| \le \pi \end{cases}$

$R_c(\theta) = \frac{1}{2}[F_c(\theta-\theta_2) + F_c(\theta+\theta_2)]$

$= \frac{1}{2}[F_c(\theta-3\pi/4) + F_c(\theta+3\pi/4)]$

$= \begin{cases} 1/2 & -\pi/8 \le \theta - 3\pi/4 \le \pi/28 \\ 1/2 & -\pi/28 \le \theta + 3\pi/4 \le \pi/28 \\ 0 & elsewhere \end{cases} = \begin{cases} 1/2 & 10\pi/14 \le \theta \le 11\pi/14 \\ 1/2 & -11\pi/14 \le \theta \le -10\pi/14 \\ 0 & elsewhere \end{cases}$

Topic 6: Tutorial questions (SOLO - Level 4)

SOLO – Level 4

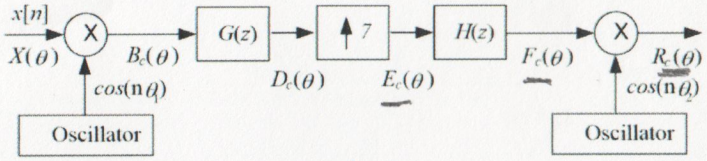
2. The diagram below shows a section of a digital modulator. The sampling period T of the input signal x[n] is 125 μs . The first oscillator generates a carrier of frequency θ_1. The second oscillator generates a carrier of frequency θ_2.

$$\theta_1 = \omega_1 \frac{T}{2} \quad where \omega_1 = 2\pi \times 2 \times 10^3 rad/\sec; \qquad \theta_2 = (\omega_1 + \omega_c)\frac{T}{2}$$

$$where, \omega_c = 2\pi \times 4 \times 10^3 rad/\sec$$

The low-pass filters G(z) and H(z) have the following characteristics:

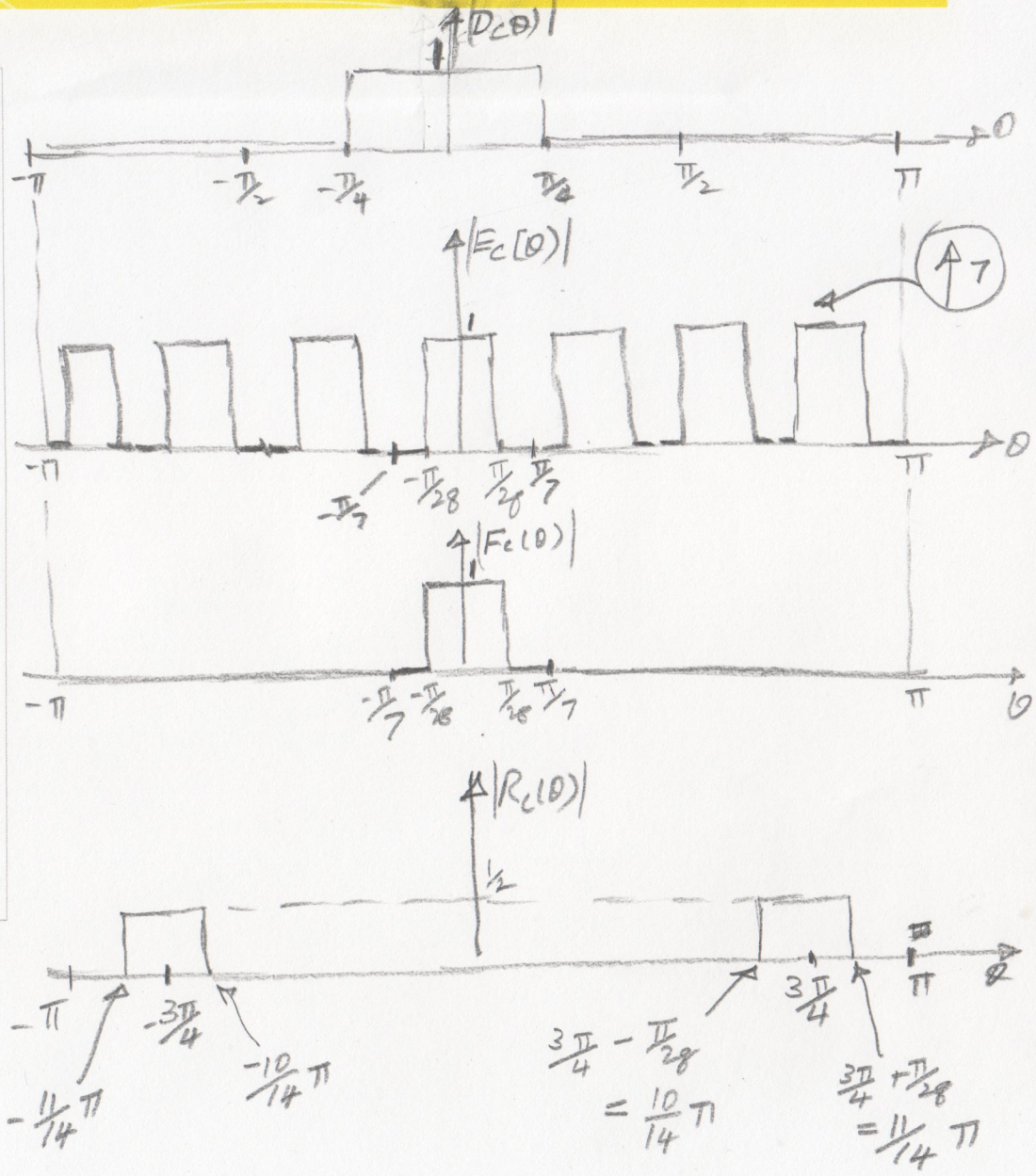
$$G(\theta) = \begin{cases} 1 & 0 \le |\theta| \le \frac{\pi}{4} \\ 0 & \frac{\pi}{4} < |\theta| \le \pi \end{cases} \qquad H(\theta) = \begin{cases} 1 & 0 \le |\theta| \le \frac{\pi}{28} \\ 0 & \frac{\pi}{28} < |\theta| \le \pi \end{cases}$$



(a) If $X(\theta) = 1 + \cos(2\theta)$, by writing appropriate equations, show that

$$D_c(\theta) = \begin{cases} 1 & -\pi/4 \le \theta \le \pi/4 \\ 0 & elsewhere \end{cases} \quad \checkmark$$

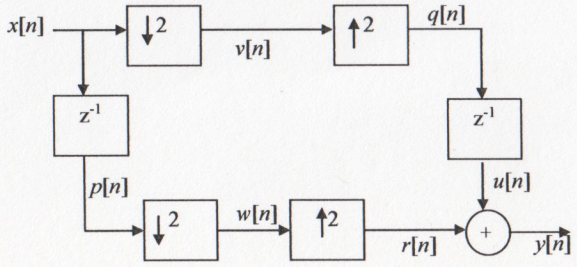
(b) Sketch, one above another, $|E_c(\theta)|$, $|F_c(\theta)|$ and $|R_{c(\theta)}|$ against θ in the fundamental frequency region $-\pi \le \theta \le \pi$.



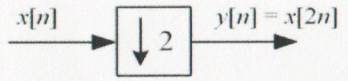
Topic 6: Tutorial questions (SOLO - Level 5)

SOLO – Level 5

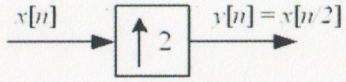
1. Express the output y[n] of the figure below as a function of the input x[n]. By simplifying the expression derived, show that y[n] = x[n-1].



Note:



$Y(z) = \frac{1}{2}\left[X(z^{\frac{1}{2}}) + X(-z^{\frac{1}{2}})\right]$



$Y(z) = X(z^2)$

$p[n] = x[n-1] \Rightarrow P(z) = X(z)\,z^{-1}$

$P(z^{\frac{1}{2}}) = X(z^{\frac{1}{2}}) \cdot z^{-\frac{1}{2}} \Rightarrow P(-z^{\frac{1}{2}}) = X(-z^{\frac{1}{2}})(-1)\,z^{-\frac{1}{2}}$

$W[n] = p[2n] \Rightarrow W(z) = \frac{1}{2}\left[P(z^{\frac{1}{2}}) + P(-z^{\frac{1}{2}})\right]$

$\therefore W(z) = \frac{1}{2}\left[X(z^{\frac{1}{2}}) \cdot z^{-\frac{1}{2}} - X(-z^{\frac{1}{2}})\,z^{-\frac{1}{2}}\right]$ — (1)

$V[n] = x[2n] \Rightarrow V(z) = \frac{1}{2}\left[X(z^{\frac{1}{2}}) + X(-z^{\frac{1}{2}})\right]$ — (2)

$q[n] = V[\frac{n}{2}] \Rightarrow Q(z) = V(z^2)$

$U(z) = Q(z) \cdot z^{-1}$

Similarly $R(z) = W(z^2)$

We have:

$Q(z) = V(z^2) = \frac{1}{2}\left[X(z) + X(-z)\right]$ (from 2)

$R(z) = W(z^2) = \frac{z^{-1}}{2} X(z) - \frac{z^{-1}}{2} X(-z)$ (from 1)

$U(z) = z^{-1} Q(z) = \frac{1}{2} z^{-1}\left[X(z) + X(-z)\right]$

$\therefore Y(z) = U(z) + R(z)$

$= \frac{z^{-1}}{2} X(z) - \frac{z^{-1}}{2} X(-z) + \frac{1}{2} z^{-1}\left[X(z) + X(-z)\right]$

$\therefore Y(z) = X(z)\,z^{-1}$

$y[n] = x[n-1]$

↑ The original signal x[n] is reconstructed. The above structure is a perfect reconstruction system.